

**POWER GENERATION,
INDUSTRIAL CONTROL & DCS
KNOWLEDGE COMPENDIUM**

BY ALEXANDER KUESEL

TRADITIONAL POWER

Glossary

Saturday, June 17, 2017 1:59 PM

MFT = Master Fuel Trip [boiler]
TPS = Turbine Protection System
EHC = Electro-Hydraulic Control
TSI = Turbine Supervisory Instrumentation
EFPH = Equivalent full power hours [for measuring outage costs]
GTB = Generator transformer High Voltage breaker
GCB = Generator Circuit Breaker
UAT = Unit Auxiliary Transformer
TP = Throttle Pressure (pressure of steam at inlet of turbine)
CV = control valves (GE turbines)
GV = governor valves (Westinghouse turbines)
BTG = Boiler - Turbine - Generator Unit
UCC = Unit Coordinated Control
LDC = Load Demand Control
HRH = Hot Reheat Turbine Bypass
ACV = Automatic Valve Control
MRE = Manual Reject

Runback

When a critical piece of equipment fails, and the unit can no longer maintain Load Demand, a runback is triggered. A Runback will trigger a rapid Load change to remain the unit online. Ex. FD Fans can runback to about 42% (250 MW for a 600 MW plant).

Rundown

Many times, one subsystem may have a hard time keeping up with other subsystems to meet demand. A rundown will allow for that subsystem to catch up. IN the case in which a block inc/dec isn't enough,

Block Increase/Decrease

Will prevent an increasing load ramp from causing a trip.

Surge

Loss of flow through turbine. Causes a stall.

Excursion Event

Cross-limiting

In control design, air and fuel flow are limited by comparing each others values.

Once-Through boiler:

Steam is not collected in a steam drum prior to superheat sections.

Selected

In the context of multiple transmitters, the one being used as the actual value is known as "Selected"

Boiler Drum Level Control

Thursday, January 27, 2022 12:31 PM

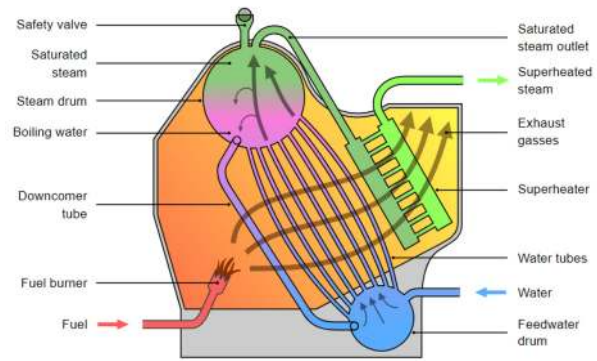
If level is too high, steam purification equipment is flooded
 If level is too low, reduction in efficiency, overheating and pressure starts to build up

Water level control is performed via single, two-element or three-element control

What affects water level?

- Bubble volume constantly misrepresenting true water level
- Drum pressure. Bubbles contract/expand depending on pressure.

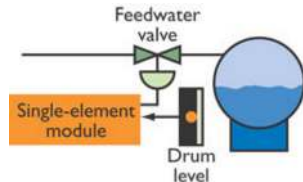
High steam demand = lower drum pressure = bubble expansion = higher apparent level = feedwater input valve closes



System Classification	Load Change Capabilities		Typical Applications
	Magnitude	Rate	
Single-element	Moderate	Slow	Institutional and industrial heating plants. Steady process conditions
Two-element	Moderate	Moderate	Industrial plants with essentially continuous type processes and good feedwater pressure regulation
Three-element	Wide	Fast	Combination of batch and continuous type operations such that plant steam load characteristics varies continuously and usually unpredictably. Most industrial power applications fall into this category.

Single-element

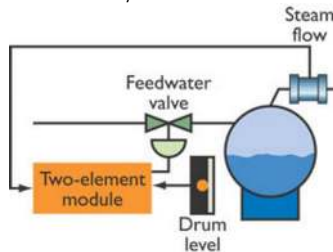
No relationship between drum level and steam/feedwater flow, only applied to single feed pump with a stable load.



Proportional only controller
 One AI, One AO

Two-element

Single drum boilers where feedwater is at constant pressure
 Adds steam-flow element to correct for density in mass flow-rate

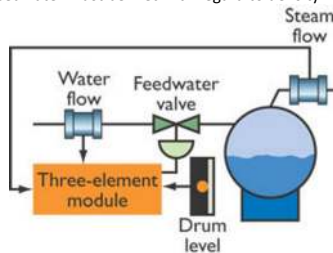


- Level: PV coming from drum level transmitter.
- Steam Flow: mass flow rate signal that controls feedwater flow corrections in response to load changes. Imbalances can be:
 - o Blowdown variations due to changes in water makeup
 - o Variations in feedwater supply pressure
 - o Steam circuit leaks

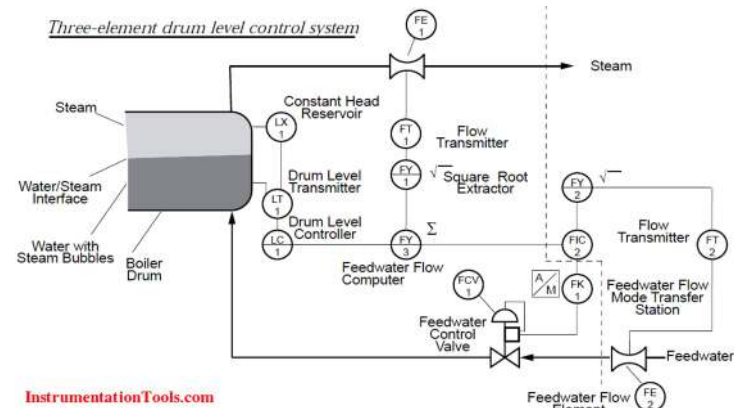
Three-element (feed-forward control)

Used in coal power plants

Correct flow for steam and feedwater must be met with regard to density



- Level & Steam Flow: corrects for boiler blowdown (water blown out of boiler to avoid impurities) and leaks
- Feedwater Flow: variations in feedwater demand: steam flow rate feedforward or feedwater pressure/flow fluctuations
- When adding water and steam flow measurement, a disparity will override bubble issues



Boilers

Friday, June 16, 2017 10:43 AM

Furnace Pressure Control

Furnace pressure is controlled by induced draft (ID) fans and inlet dampers. Typically, a PID will drive an ID fan. A feedforward from a forced draft (FD) fan is also used. Very negative pressure will prompt an inlet damper closing and decrease in fan speed. Very high pressure will prompt a blocked of speed decrease, that is, only increase of ID speed.

Master Fuel Trip

If the Burner Management System (BMS) detects a dangerous condition, it shut down the fuel flow, and the flame in the furnace collapses. MFT kicker logic is implemented to reduce fans output. Turbine trips via a lock-out relay (solenoid). Generator is not tripped immediately, so as not to drive turbine to overspeed

Airflow and O₂ trim

There must always be higher airflow than fuel going into the furnace, to avoid explosive conditions. During decreasing load, the air demand follows a lag function, while fuel demand follows the boiler demand. During increasing load, the air demand follows boiler demand, while fuel lags. This ensures fuel will never ride above air. Air and Fuel flow mass flow rate must be compensated/scaled so that the percentage equals Unit Load. For example, if Unit load is at 50%, air and fuel should be at 50%.

Drum Level and Feedwater

Feedwater is fed into the drum via constant pressure pumps. High levels can cause entrainment, low levels can cause overheating. There are two modes of automatic operation: Single element and three element. Single element uses only the difference between drum level and drum level setpoint. Three-element uses drum level, steam flow and feedwater flow. Drum level: cascaded controller. Steam flow: feedforward. If drum level goes out of range, it will lock into manual.

Superheat Temperature

Steam leaves the drum and enters a superheater, then a desuperheater. The outlet temp is measured at the exit of the superheater. This will enter as a PV, and the output of this PID will be the set point for the inlet temp control

Boiler Runback

Boiler running at fraction of base load, tuning parameters are latched to avoid erratic behaviour

Relations:

Boiler demand and superheater temperature
Throttle pressure and over/underfiring of boiler
Ratio of fuel flow to top of mill vs. other levels is a good indication of dynamics.
Top levels of boiler affect temp more than pressure, while low levels affect pressure more than temp
Reheater temp control affect superheater temp
Superheater spray adds efficiency. Reheater spray decreases efficiency.

Most boilers operate at 1800, 2200, 2400 psi, the latter being the most common

Lowest rate of combustion is defined by coal mill. 40% is standard lowest capability

Fuel/Air Ratio

Low air flow: may cause unburned fuel
High air flow: excess heat carried out stack
Logic ensures minimum air flow present at low loads.

Air Flow Master

Regulates flow through FD Fans.

Draft: difference between atm pressure and pressure in the furnace.

Furnace Pressure

Sunday, June 25, 2017 10:44 AM

Combustion air in boiler is supplied by Primary Air and Forced Draft fans

Low pressure:

After a long enough bout of low pressure, an implosion occurs, due to excessive force on the outer walls.

Causes:

- ID fan trip/failure
- Load throw-off, large fuel cutting
- Reduction in air flow
- Tripping of FD fan

Response:

- Change ID fan control to manual
- Change SP of ID fan vane control
- Restart tripped FD fans
- Correct air flow leakage

High Pressure

Can lead to an explosion. Is a direct indication of combustion disturbance

Causes:

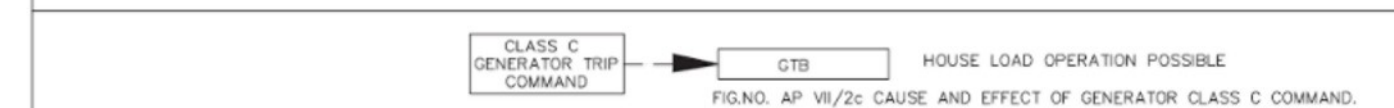
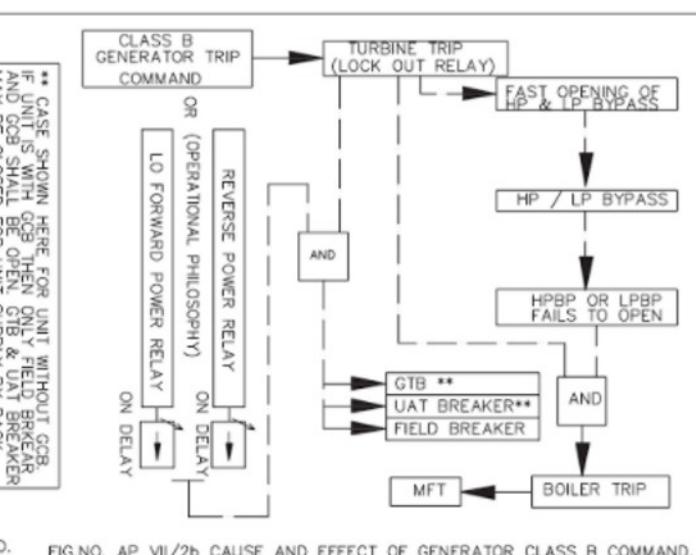
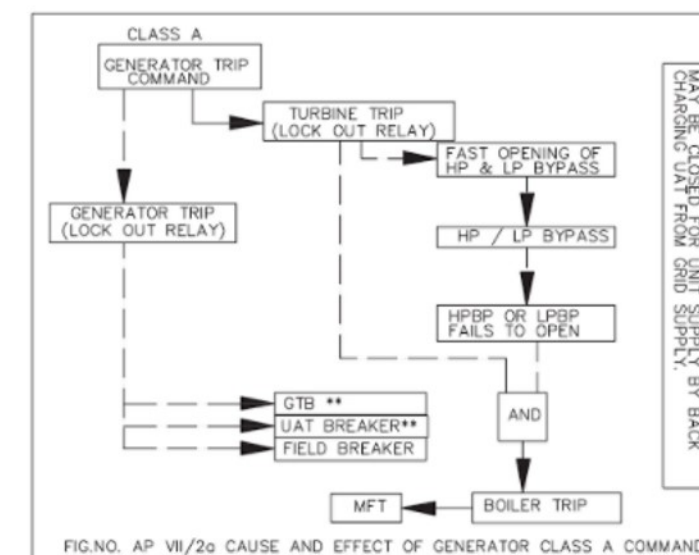
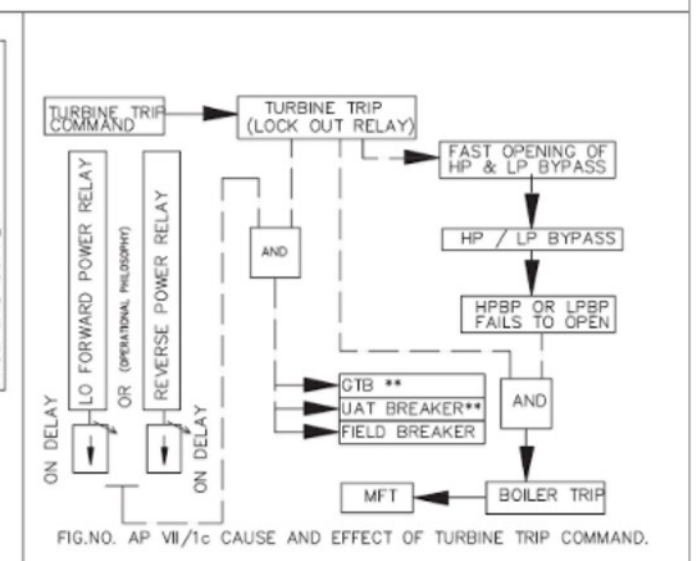
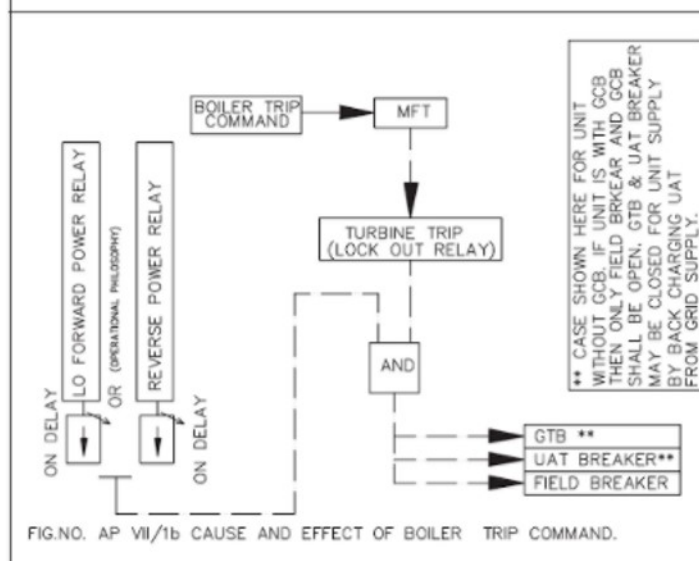
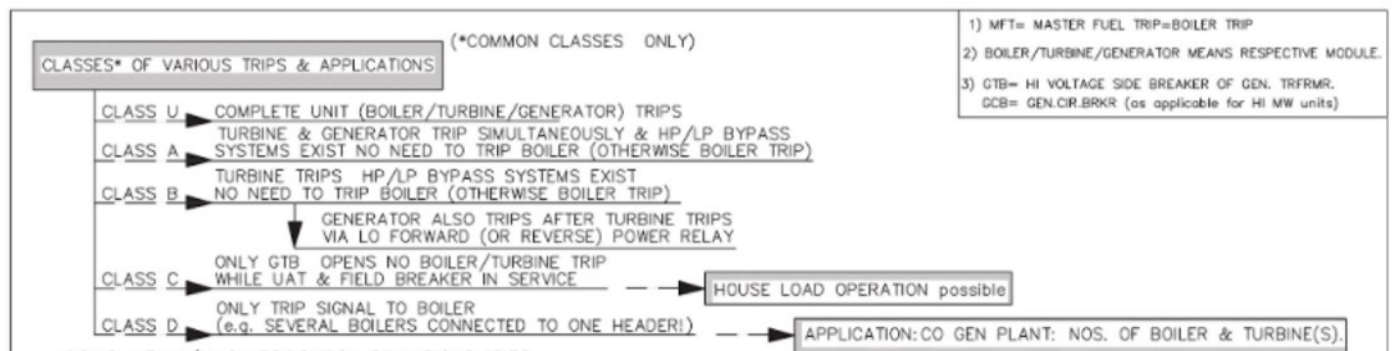
- ID fan trip
- Mal-operation of regulating vanes
- Unstable flame
 - Low wind box pressure
 - Improper burner operation
 - Sudden start of mills with fuel
 - Sudden fuel input into furnace
 - Loss of ignition energy
- Fouling buildup in pre-heaters
- Air pre-heater trip
- Furnace water seal broken
- High excess air levels

Response

- If ID fan tripped, reduce load.

Trips

Saturday, June 17, 2017 12:57 PM



IMPORTANT POINT DISCUSSIONS: **CLASS U** TRIP CAUSES **TRIPPING OF BOILER TURBINE AND GENERATOR IMMEDIATELY**. THEORETICALLY WITH HP LP BY PASS SUCH SITUATION IS VERY IMPORTANT POINT DISCUSSIONS: **CLASS U** TRIP CAUSES TRIPPING OF BOILER TURBINE AND GENERATOR IMMEDIATELY. **THEORETICALLY WITH HP LP BY PASS SUCH SITUATION IS VERY RARE**. HOWEVER IN PRACTICE FOR CERTAIN CASES, EXTREME BIG ELECTRICAL FAULT OR SAY LUBE OIL FAILURE OF TURBINE (BEARING IS DAMAGED , IN SUCH SITUATION RUNNING HP/LP BYPASS MAY NOT BE POSSIBLE SO IN THOSE CASES, UNIT TRIPPING MAY BE DONE. THERE ARE VARIATIONS HERE, SOME DIRECTLY CLUSTER

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FIGURE APVII/2 Classification cause and effect of various trips, Scheme 2.

Turbine Shutdown

Wednesday, July 15, 2020 9:49 PM

Shutdown

- Decrease output:
 - o Close control valves
 - o Decrease inlet steam parameters
- When minimum load reached
 - o Close valves completely.
- Cooling done via
 - o Decrease live and reheat steam temp
 - o Throttle control valves
- Rotor will continue to spin from 3000 to 40 rpm

Turbine Startup

Wednesday, July 15, 2020 4:50 PM

Three primary stages for a successful Turbine Startup:

1. Start-up prelim preparation
2. Run-up to idle and synchronization
3. Loading phase

1. Start-up prep

Start up initial conditions:

- a. Cold Start: High Pressure (HP) and Intermediate Pressure (IP) inner casing temp lower or equal to 170 C. usually occurs after 6 day outage. Target rotational speed set to 3000 rpm.
- b. Warm Start: HP and IP inner casing temp between 170 C and 430 C
- c. Hot Start: HP and IP inner casing temp greater than 430 C. Run up acceleration of 250 - 300 rev/min².

2. Run up to idle and sync

- Turning gear will be used to roll axle from standstill (40 rpm). The conditions for turning gear motor to start are:
 - o Lube oil pumps On
 - o Bearing oil present
- Condensation system started.
 - o Cooling water
 - o Air ejection
 - o Condensate pumps
 - o Steam pipelines drained and heated
- At 3000 rpm (sub-synchronous speed)
 - o the auxiliary oil pump is switched OFF
 - o Turbine governor ready to sync Turbine and grid
 - o Once sync'd the governor loads turbine with a percentage of nominal load

3. Loading phase

- Load increase
 - o Increase live steam pressure
 - o Gradually open control valves

Turbine run-up rate must be increased during critical speeds and must never be stopped at a critical speed. Fast speed is considered 600 - 800 rev/min.

Stops are made to reduce excessive vibration, for example at 1000 rpm.

Critical frequencies:

- 1st harmonic
- generator rotor 940 rpm,
 - LP rotor 1490 rpm,
 - HP rotor 1900 rpm,
 - IP rotor 2230 rpm,
- 2nd harmonic
- generator rotor 2580 rpm.

Thermal stress limits:

- Relative stress in HP and IP must be under 80%, at this point the load must be decreased.
- Rotor differential expansion must be monitored to avoid rotor rubbing.
- Temperature differentials are highest in HP and IP inlet sections
- The thicker the section, the higher the temp difference.

Thermal Stress Controllers (TSC) supervise unsteady thermomechanical operation of turbine components and ensures safe operation.

Controls turbine startup and shutdown by manipulating speed and load changes so as to avoid thermal stress.

Turbine loading = positive expansion

Turbine unloading = negative expansion

Startup curve: relationship between rotational speed, power output and steam params.

Means of start-up	Characteristics	Advantages	Drawbacks
Constant pressure operation with throttle control	Constant live steam pressure in the entire load range Throttle control Typically two control valves No control stage	Load step changes Low cost	High throttling losses in partial load operation
Constant pressure operation with nozzle group control	Constant live steam pressure in the entire load range Nozzle group control Typically four control valves Control stage	Load step changes High efficiency at partial load operation	Expensive control stage
Pure sliding pressure operation	Live steam pressure changes proportionally to live steam flow rate No throttling in control valves Typically two control valve No control stage	Low cost	Quick load step changes for frequency control not possible
Modified sliding pressure operation	Load changes according to grid requirements Valves not wide open Typically two control valves No control stage	Quick load step changes possible in the range of throttling	High throttling losses in the entire load range
Modified sliding pressure/constant pressure operation	Sliding pressure at low loads constant pressure at high loads Typically four control valves Control stage	Quick load step changes possible	Expensive control stage

ANSI Relay Device Numbers

Thursday, February 6, 2020 2:08 PM

ANSI/IEEE Standard Device Numbers

In North America protective relays are generally referred to by standard device numbers. Letters are sometimes added to specify the application (IEEE Standard C37.2-2008).

Device Numbers (the more commonly used ones are in bold)

1 - Master Element	53 - Field Excitation Relay
2 - Time Delay Starting or Closing Relay	55 - Power Factor Relay
3 - Checking or Interlocking Relay	56 - Field Application Relay
4 - Master Contactor	59 - Overvoltage Relay
5 - Stopping Device	60 - Voltage or Current Balance Relay
6 - Starting Circuit Breaker	62 - Time-Delay Stopping or Opening Relay
7 - Rate of Change Relay	63 - Pressure Switch
8 - Control Power Disconnecting Device	64 - Ground Detector Relay
9 - Reversing Device	65 - Governor
10 - Unit Sequence Switch	66 - Notching or jogging device
11 - Multifunction Device	67 - AC Directional Overcurrent Relay
12 - Overspeed Device	68 - Blocking or "out of step" Relay
13 - Synchronous-speed Device	69 - Permissive Control Device
14 - Underspeed Device	74 - Alarm Relay
15 - Speed or Frequency-Matching Device	75 - Position Changing Mechanism
16 - Data Communications Device	76 - DC Overcurrent Relay
20 - Elect. operated valve (solenoid valve)	78 - Phase-Angle Measuring Relay
21 - Distance Relay	79 - AC-Reclosing Relay
23 - Temperature Control Device	81 - Frequency Relay
24 - Volts per Hertz Relay	83 - Automatic Selective Control or Transfer Relay
25 - Synchronizing or Synchronism-Check Device	84 - Operating Mechanism
26 - Apparatus Thermal Device	85 - Pilot Communications, Carrier or Pilot-Wire Relay
27 - Undervoltage Relay	86 - Lockout Relay
30 - Annunciator Relay	87 - Differential Protective Relay
32 - Directional Power Relay	89 - Line Switch
36 - Polarity or Polarizing Voltage Devices	90 - Regulating Device
37 - Undercurrent or Underpower Relay	91 - Voltage Directional Relay
38 - Bearing Protective Device	92 - Voltage and Power Directional Relay
39 - Mechanical Condition Monitor	94 - Tripping or Trip-Free Relay
40 - Field (over/under excitation) Relay	
41 - Field Circuit Breaker	
42 - Running Circuit Breaker	
43 - Manual Transfer or Selector Device	
46 - Rev. phase or Phase-Bal. Current Relay	
47 - Phase-Seq. or Phase-Bal. Voltage Relay	
48 - Incomplete-Sequence Relay	
49 - Machine or Transformer Thermal Relay	
50 - Instantaneous Overcurrent	
51 - AC Inverse Time Overcurrent Relay	
52 - AC Circuit Breaker	

Suffixes indicating zone of protection

B—Bus

G—Ground or generator

L—Line

N—Neutral

T—Transformer

U—Unit

Auxiliary Air

Wednesday, December 16, 2020 4:10 PM

A damper is used to direct air or gas in the desired direction. It is useful for air dynamics within a boiler. It will always have a minimum opening of 30%.

The dampers modulate according to coal input in three different ways as:

When coal is fed into a particular(own) elevation ,the corresponding air damper starts modulating based on the fuel input to let the secondary air required for the combustion of given input.

Rather when coal is fed into an (auxillary) elevation adjacent to a particular damper, the damper modulates to maintain the windbox to furnace differential pressure, thereby controlling the constant negative pressure essential to create swirling fireballs of combustion in tangentially fired boilers.

When the condition of tripping the whole unit arises, all the six coal elevation dampers are opened to 100 % (maximum).

When there is no fuel input in any of the above elevations then the dampers will be in minimum opening position. (0%)

The opening of dampers depends on the load and coal input. The above table I illustrate the percentage of coal input in each mill for given percentage of load. Here 30 Tonnes/Hr of coal is the 100% input and 10 Kg/cm² of oil is the 100% input of oil.

When unit load is 50%, then coal mills A, B are in full operation and C mill operates 10% with AB elevation having oil.

For a load of 60%, coal mills A, B, C will operate till 30% and mill D has 10% input. AB oil elevation is fed with full percentage of oil.

If the load is 70%, then there is no need for oil but coal mills A, B, C, and D are fed with maximum input.

For 90% ,there is no need for oil but the coal mills

A, B, C, D is fed with maximum input and E with 20% of fuel input.

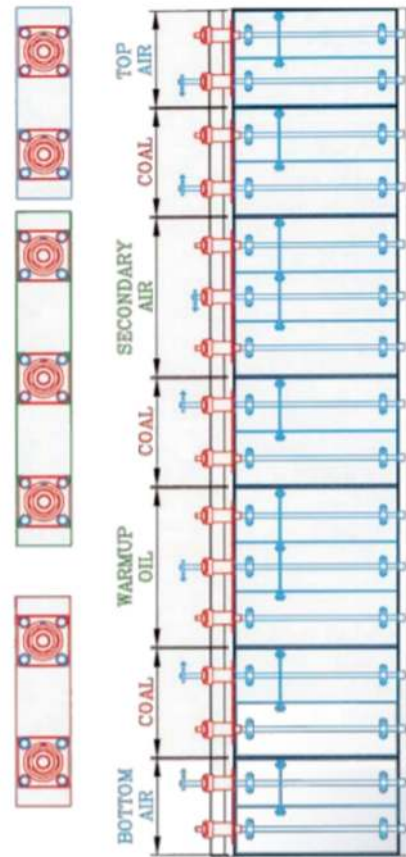
For 100% load, all the coal elevations are fed with maximum input.

During the furnace purge period and initial operations up to 30% boiler loading all elevations of auxiliary and end air dampers modulate to maintain a predetermined (approx. 40 mmWC) set point differential pressure between windbox to furnace.

When the unit load exceeds 30% MCR, the differential pressure set point is changed and ramps to a higher setting (approx. 100 mmWC). Simultaneously, the auxillary air dampers associated with coal or oil elevations not in service close in timed sequence starting with the upper elevation of dampers and progressing to the lowest elevation.

When the unit loading is reduced below 30% loading, the auxillary air dampers open in a timed sequence starting with the lowest elevation of dampers. Simultaneously, the differential set point changes to its lowest setting.

All the auxillary air dampers maintain the status quotient upon a boiler trip and will open fully when both the FD fans are off. The bottom end air damper is normally kept open to a fixed predetermined position to reduce unburnt coal dust fall out.



B. Fuel (oil) air dampers:

Elevation AB, CD and EF dampers modulate according to the oil header pressure when heavy oil is being fired. Otherwise they open and close with the rest of the aux air dampers. Elevation AB dampers are kept at a fixed opening when light oil is being fired.

C. Fuel (coal) air dampers:

Their operations are independent of Boiler load. All coal air dampers are normally closed. They open fifty seconds after the associated feeder is started, and modulate as a function of feeder speed. Fifty seconds after the coal feeders are removed from service, the associated fuel air dampers close. The fuel (coal) air dampers open fully, when both FD fans are off or when the boiler trips on master fuel trip.

Secondary air for combustion enters through windbox. A windbox is a tall structure containing 6 coal nozzles and 5 air nozzles. Coal nozzles are designation A,B,C,D,E,F from bottom to top. Air elevations are AA and FF. Aux air elevations are AB, BC, CD, DE, EF. Furnace corners are 1, 2, 3, 4 in clockwise direction.

Combustion Dynamics

Monday, October 26, 2020 4:47 PM

Refers to what happens within the combustors in the combustion chamber.

Improper combustion dynamics can cause combustion damage that can leak downstream to the rest of the turbine.

Combustion dynamics are caused by flame instabilities on premix-type combustors.

They are pressure waves caused by acoustic oscillations related to heat release.

Frequency + amplitude are key to damage evaluation

The closer to natural frequency, the more damage.

These dynamics were introduced with Nox injection systems designed to reduce emissions.

Diluent injection: steam / water injected to reduce flame temp

Premix combustion: mix air with fuel prior to combustion. Also known as Dry Low Nox (DLN), Dry Low Emissions (DLE), LowEV, Premix

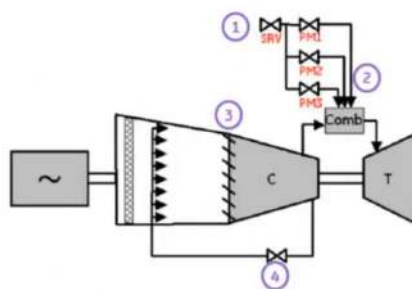
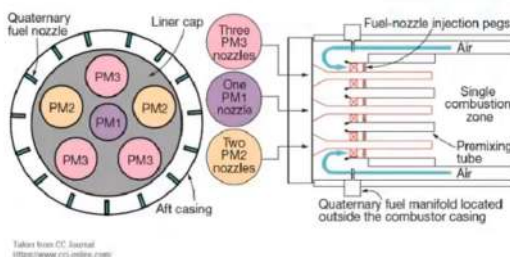
Stoichiometric mix is the optimum mix of fuel/air: gas turbines run at 0.5 stoichiometric ratio, running lean all the time.

- NOx produced at high flame temp.
- CO produced at low flame temp
- Dry Low NO combustor has a narrow equivalence ratio operating ratio.
- Combustors operate close to Lean Blowout limit

DLN/Premix include additional controls:

- Airflow controls
- Multiple fuel manifolds to introduce fuel differently and scheduled
- Tuning used manual or automated tuning system.

GE F-Class Turbine



1. Total Fuel
2. Fuel Splits
3. IGV Angle
4. Inlet Bleed Heat (IBH) Flow

GE F-Class with Dry Low NOx controls

Measure

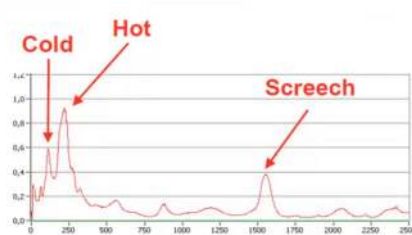
- Barometric Pressure
- Ambient Temp
- Inlet Temperature
- Fuel pressure
- Comp Discharge Press
- Comp Discharge Temp
- Exhaust Temp
- Megawatts
- RPMs

Calculate

- Inlet air flow
- Firing Temperature

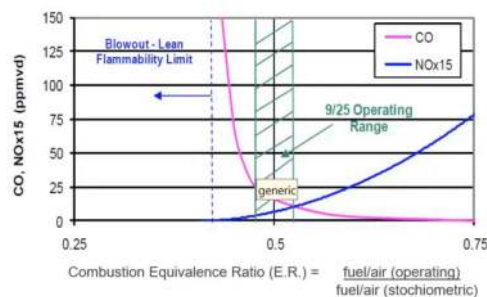
Control

- Fuel flow PM1, PM2, PM3, PM4 (1,2)
- Air flow – IGV (3)
- Air flow – IBH (4)

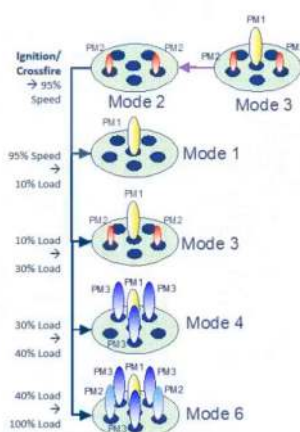


Dynamics for GE Gas Turbine

- ✓ Cold tones
 - ✓ 10 – 100HZ
 - ✓ Lean conditions, impending blowout
 - ✓ Cold flame condition
- ✓ Hot tones
 - ✓ 100 – 300HZ
 - ✓ Hot flame condition
- ✓ Screech
 - ✓ 500 – 2500HZ
 - ✓ Very destructive



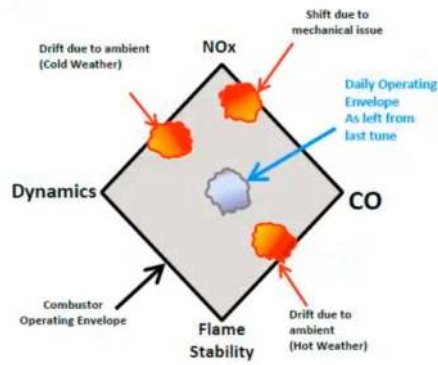
Control system has to control fuel flow through different stages. Mode 6 is normal loaded operation range.



Operating tuning envelope. There are different limits, in which increasing fuel or air ratio can cause issues with emissions. The goal is to tune into the middle of the

envelope.

Tuning is done at the beginning of winter and then at the beginning of summer to account for significant temperature (and as such air density) changes.



The envelope quadrants are:

NOx = Emissions of nitrogen oxides

CO = Carbon Monoxide

FS = Flame stability

Dynamics = Combustion pressure fluctuations and pulsation frequency

Practical events that produce high dynamics

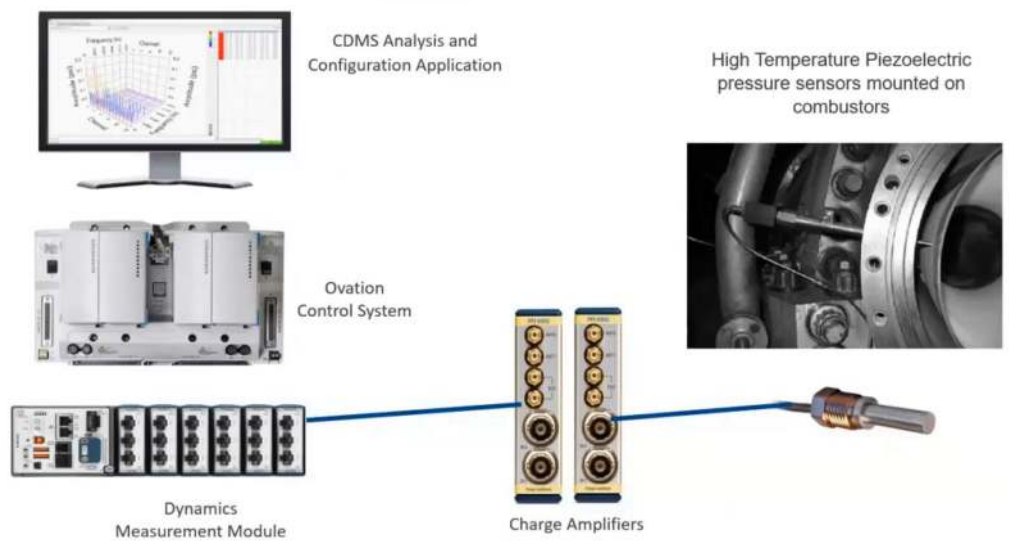
- Low load (turndown) the further from base load, the further to the edges of the envelope
- High Load (peak) same as above
- Different fuel composition (Wobbe index)
- Ambient temperature changes
- Instrumentation out of calibration
- Combustion emission monitoring system (CEMS) feedback to the control system

Ovation Combustion Dynamics Monitoring System

- High temp piezoelectric pressure sensors mounted on combustors.
- MHM module will replace the dynamics measurement module
- CDMS connected via ethernet.

Autotune technology

- Eliminates need for seasonal tuning.
- Integrated into Ovation
- Improves turndown and peak firing.
- Takes all data from turbine and moves the parameters around to follow conditions.

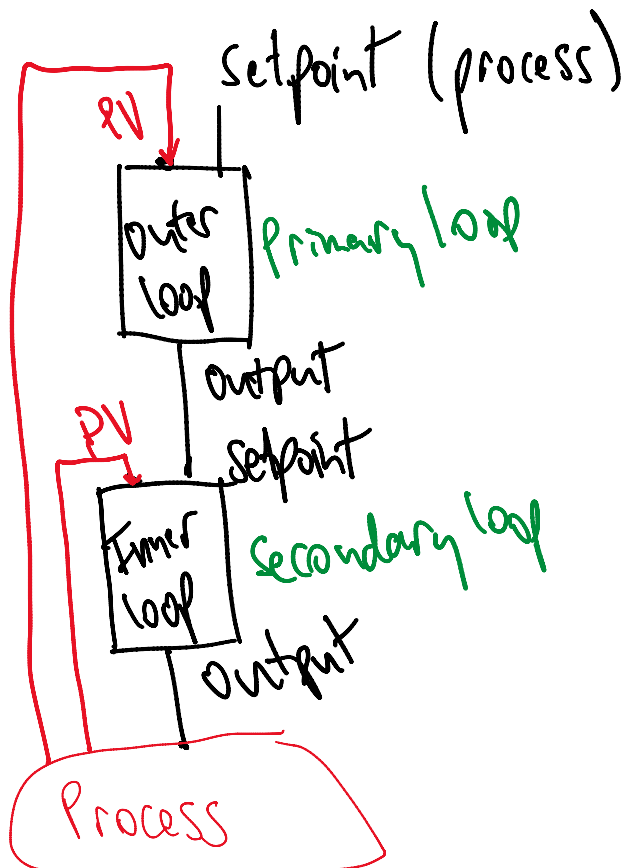


Cascade Control

Wednesday, April 14, 2021 15:33

A cascade control loop is one in which a feedback loop uses the output of an outer loop as its setpoint. In essence you are using an intermediate variable within the process (from the outer loop) to help reach the inner setpoint. This splits the lag dynamics into two parts. Cascading control is beneficial for processes with long lags, such as temperature and volume control. A few general guidelines are:

- The secondary loop is the one with the most volatile disturbances, and that requires quicker response time.
- The secondary loop should only have the min of lags and should be 3 times faster than the outer loop.
- The secondary variable should have variables related to the outer loop
- The secondary controller should operate at the highest possible gain



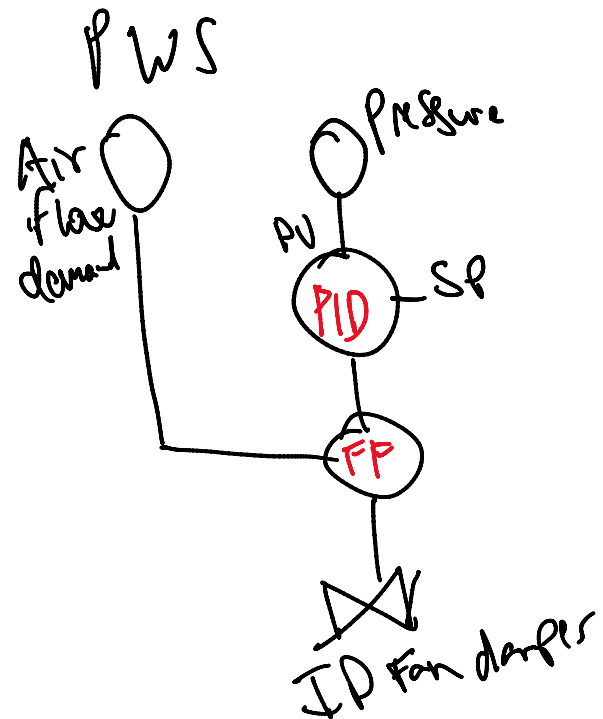
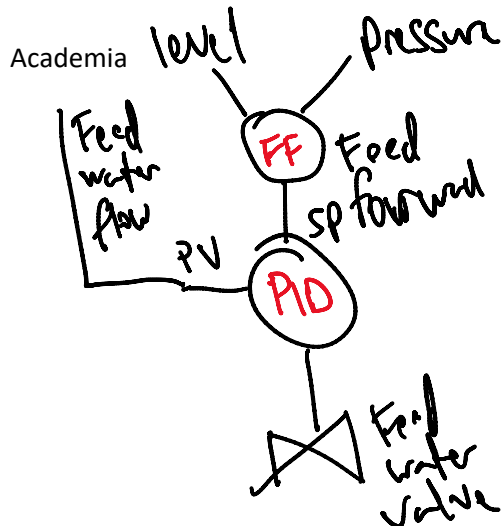
In certain cases, the outer loop will track a slower magnitude such as temperature, while the secondary loop will handle a faster magnitude such as flow.

Feed Forward Control

Wednesday, April 14, 2021 16:12

PWS definition differs from the standard definition:

- Academia: A control system in which the control output is calculated from measurements of the process input using predictive methods
- PWS: A control strategy that anticipates a process upset or disturbance so the deviation to the controlled variable is minimized



In the PWS variant, the feedforward action is applied directly to the final control element

UCC

Monday, June 19, 2017 11:35 AM

Made up of three main systems:

- BMS: Burner management system
 - o Controls intro if fuel to boiler furnace via igniters/burners
 - o Monitors air flow, furnace pressure, drum level
 - o In case of dangerous levels, initiates MFT
- BCS: Boiler control system
 - o Regulates flow of liquid/fuel through boiler to maintain proper pressure/temp
 - o Pressure at inlet of turbine is known as throttle pressure
- TCS: Turbine Control System
 - o Valve modulation to regulate steam flow via Control and Governor valves

Boiler Master Control Station

- Flow of water regulated to maintain Steam Drum level
- Final output of UCC to BCS comes from Boiler Master
- Load demand increases, Boiler Master Demand must increase.
- Constant Turbine GV position, increase in BM demand will increase TP

Turbine Master

- Local: Operator uses control screen to position TGV:
 - o Manual: operator sets control station position
 - o Auto: TCS regulates initial speed ramp
 - o If boiler is in manual, GV opening will remove more steam from boiler.
 - o In Local, Turbine Master tracks GV position for transfer to remote. TM output does not control GV.
 - o Boiler follow mode, the UCC is tracking inferred steam flow (GV pos and TP demand)
- Remote: Turbine on-line above min load, TCS receives position demand from Turbine Master Station.
 - o Operator can manually change demand to GV using TM station.
 - o If valves are opened at constant TP, ammount of steam will increase and more load is generated.
 - o When both BM and TM are in auto, load ramps are controlled from UCC. Local TCS is disabled.

Boiler Follow Mode

firing is arranged to automatically keep the steam pressure constant and the turbine is controlled to suit the load required by the governor.
Traditional way

Turbine Follow Mode

Turbine governor valves are arranged to keep the steam pressure constant and the boiler varies with the load or fuel supply variations. Throttle pressure is controlled by modulating gov valves.
Waste heat boilers.

Sliding Pressure mode

The turbine valves are set fully open and the pressure is left to balance with the boiler output similar to turbine follow but with no valve position change. no throttling
Avoid throttling losses on large turbines

Balanced Mode (Coordinated)

The turbine valves and the boiler firing are adjusted at the same time by a computer model so that the pressure never changes but the load is varied as required. Boiler master looks at throttle pressure, turbine looks at megawatt error.
Modern way

Mode	Boiler master	Turbine master
Base load	Manual	Manual
Boiler follow	Auto	Manual
Turbine follow	Manual	Auto
Coordinated	Auto	Auto

Base Mode:

BM and TM in Manual.

- BCS sub loops for pressure / air / fuel / feedwater are controlled independently by Op.
- BM regulates fuel flow, BM output tracks total fuel flow.
- If BM wants to change fuel flow in Man: Furnace Press, Air Flow, Fuel Master must be in Auto.
- TCS in local Auto, load can be changed by operator, but not TP.
- TCS in local, UCC TM is rejected to Manual and its demand tracks turbine GV position.
- TCS in Remote, TM stays in Manual until switched.
- If more load required, TM demand and BM demand must be increased coordinated.
- Lowest level of control for both is the TP control. TP setpoint tracks actual pressure.

Boiler Follow 1:

BM Auto / TM Manual

- BM controls TP, operator controls load by positioning TM/local TCS.
- UCC demand tracks inferred steam flow

Turbine Follow 1

TM Auto, BM Manual

- TM controls TP, operator controls load manually with BM demand.
- UCC tracks BM demand.
- No way to enter MW setpoint in UCC.

Boiler Follow 2:

BM and TM in Auto

- BM follows UCC feed-forward demand.
- TP is regulated by PID.
- TM only responds to UCC demand.
- UCC demand is sent in parallel to both BM and TM.

Turbine Follow 2

BM Auto / TM Auto

- TM follows UCC feed-forward demand while TP is PID'd.
- BM only responds to UCC.
- Parallel UCC demand

In these modes, operator can change UCC demand to BCS/TCS to obtain desired load. MW trim may not equal actual MW load. Coordinated control must be used:

Coordinated UCC modes use a PID to compare actual MW to target setpoint.

Coordinated Boiler Follow (CBF)

TM/BM Auto

- TM Load demand is sum of UCC demand trimmed by out of TM

Coordinated Turbine Follow Mode (CTF)

- Etc etc

Remote Dispatch Mode

Remote Load Dispatch system is used to match power gen to customer load demand

Automatic Generation Control (AGC) or Automatic Dispatch System (ADS).

UCC must be placed in remote mode to act on dispatch setpoint

Sheet design

Scaling at the top allows feed-forward action that minimizes need for controller response to load changes. Scaling changes MW target into percentage
Tracking avoids bumps.

Valve characterization must be programmed in as well.

Frequency control is required in case of a desync with the grid. If several units on the same grid fail, a supply/demand imbalance will cause a frequency droop, subsequently causing a drop in output.

Boiler Master Station

Forced into manual by:

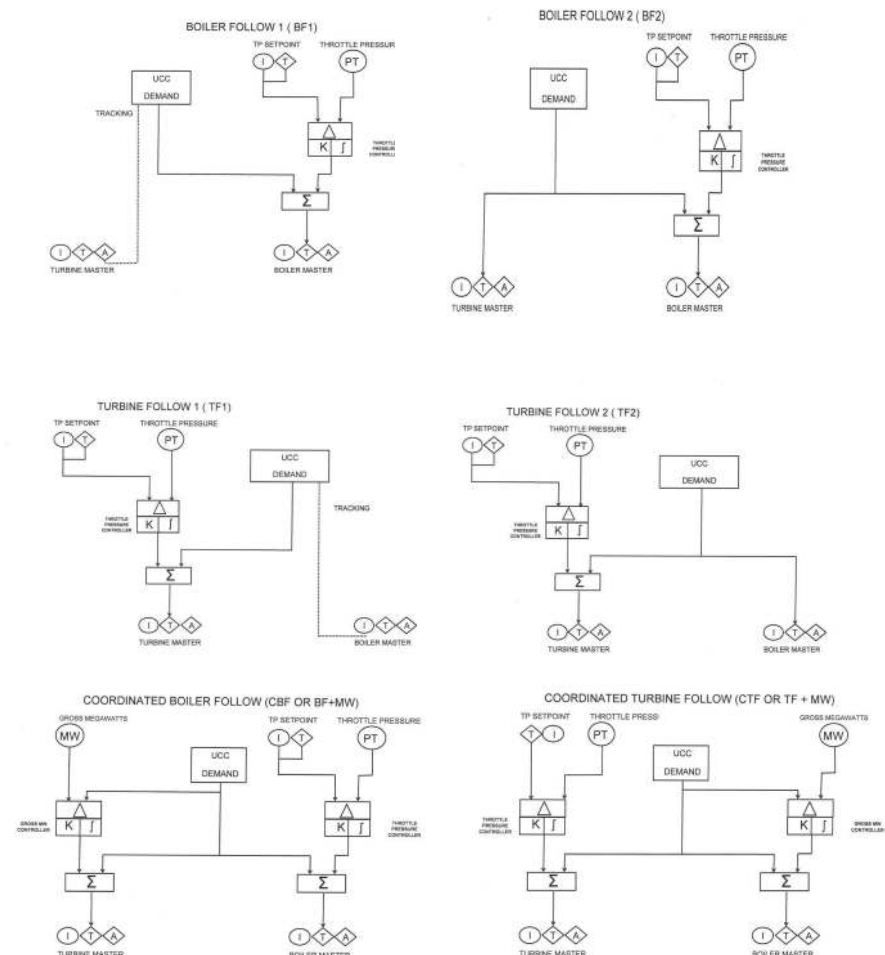
- Fuel Master Not in Auto
- Master Fuel Trip
- Throttle Pressure Bad Quality

Inhibits:

- Fuel Master at max (raise) or min (lower)
- Air Flow Master or FD Fans at max (raise) or min (lower)

Fuel Master Mode	TCS Mode	BM Track Signal	TM Track Signal
Man	Local	Fuel Flow	GV Position
Auto	Local	None	GV Position
Man	Remote	Fuel Flow	None
Auto	Remote	None	None

Table 1 - Boiler and Turbine Master Trackin



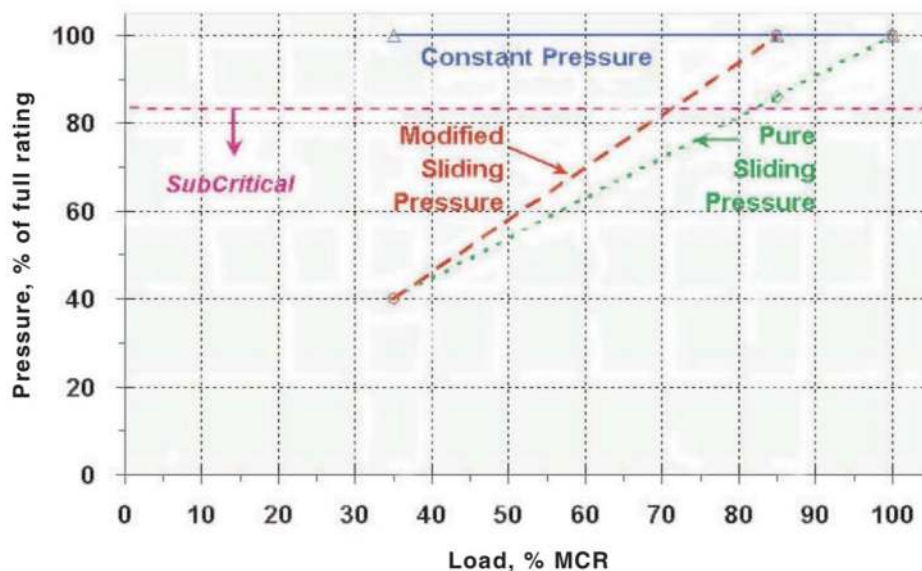
Constant vs Sliding Pressure

Saturday, January 29, 2022 14:39

According to theory, a fixed, rotating turbine will require less pressure as flow rate is reduced. If following load (lower load = lower flow rate), pressure will lower with load as well.

Sliding pressure is a startup/shutdown method that brings pressure up/down for all components in plant in unison (steam lines, turbines, boiler, etc). This is down via pressure control within the boiler. This reduces the thermal stress to which components are subjected to.

Constant pressure on the other hand will have a large load reserve regardless of load reduction. Constant pressure responds to load changes via the control valve of the turbine, boiler pressure remains the same. Sudden reduction of pressure in the steam generator prompts an instant expulsion of steam mass due to increase of system volume. Advantageous for base-load.



NUCLEAR POWER

Acronyms and Glossary

Saturday, July 11, 2020 11:25 AM

CRCS: Control Rod Control System

CEDCMS: Control Element Drive Mechanism Control System

ASTS: Automatic Seismic Trip System

SPV: Single Point of Vulnerability

Demin: demineralizer (water purification)

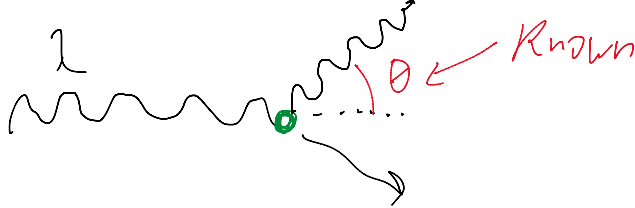
NPP location is determined by proximity to load (end users), water source (cooling). Fuel source is not an important factor as refueling frequency is extremely broad (years compared to constant refueling for coal/gas)

Neutron Physics

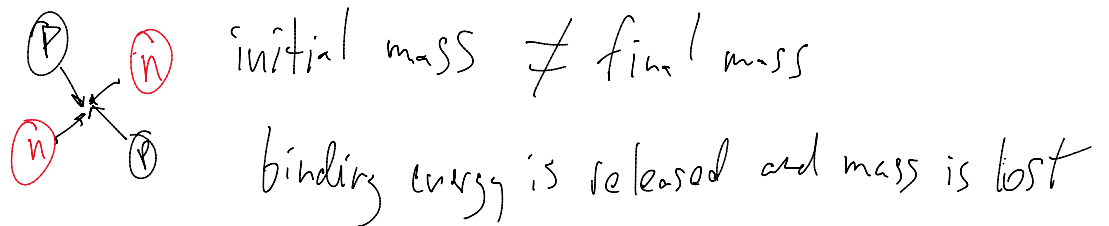
Wednesday, January 19, 2022 09:30

Neutrons are the basis of nuclear physics. Chadwick's article "Possible existence of a neutron" was the first report.

Compton effect: Photon strikes electron, picking up energy and transferring energy to electron.



Mass defect:

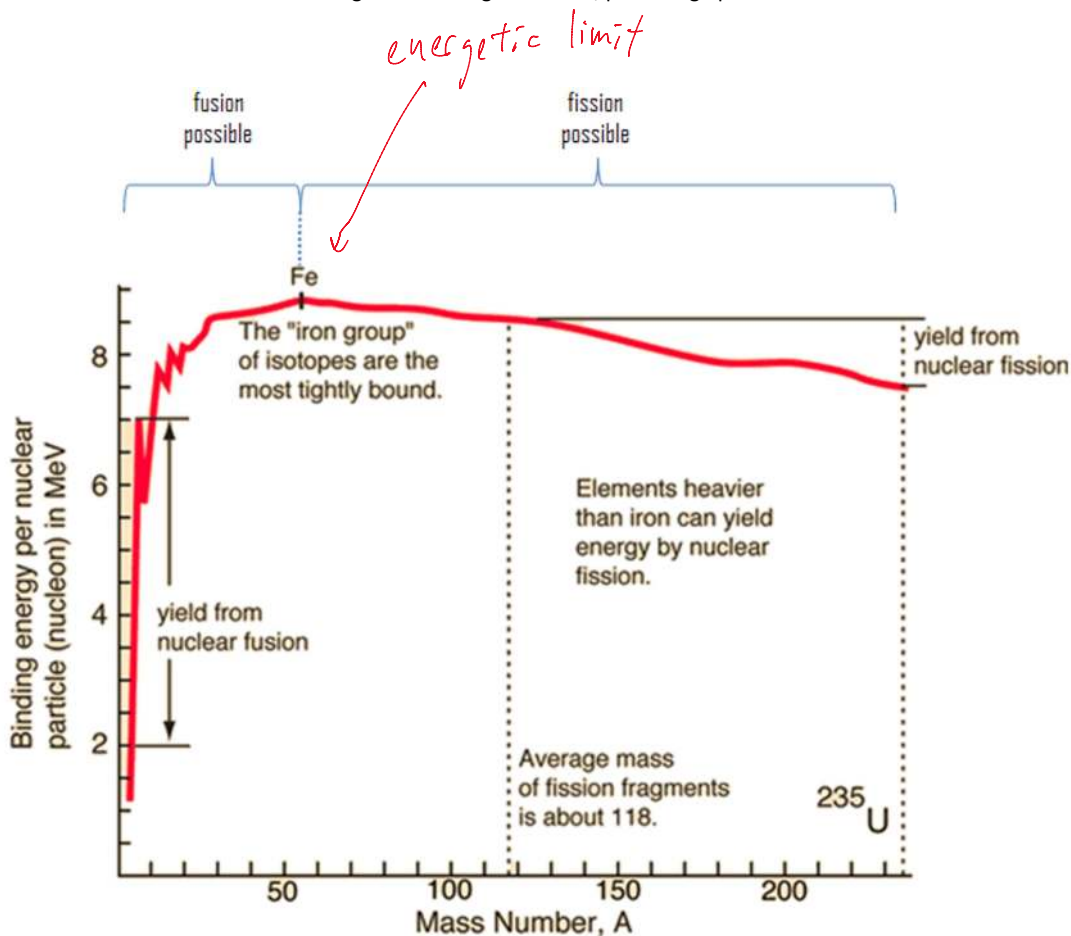


Mega Electron Volt (MeV): binding energy

Neutrino: speed of light particle that doesn't interact with mass.

The probability that a nucleus can capture a neutron depends on the energy of the neutron.

Cherenkov Radiation: electrons moving faster than light in water, producing optical shockwaves



Notation:

→ atomic mass (neutrons)

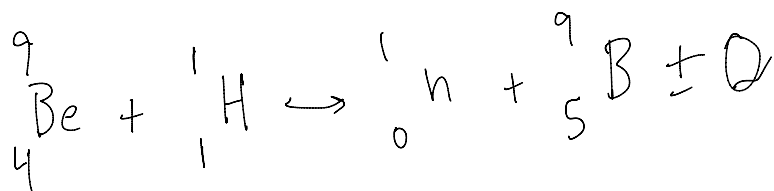
Notation:

A $\rightarrow$ atomic mass (neutrons)

$Name \pm q$ $\rightarrow$ charge (zero if not ion)

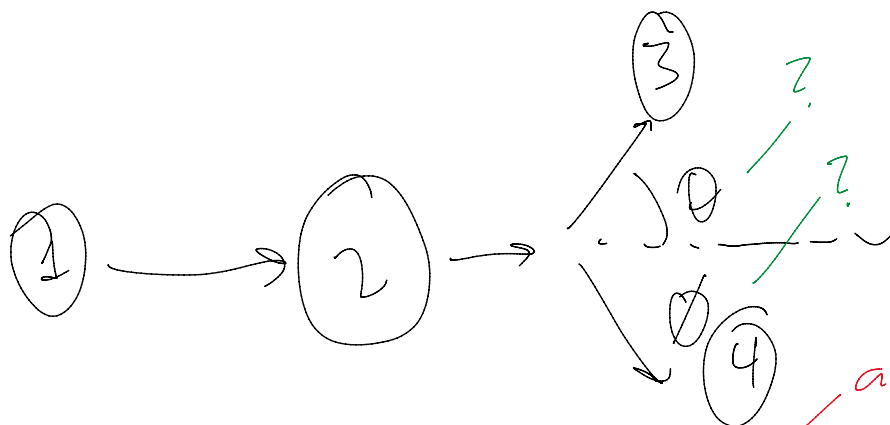
Z $\rightarrow$ atomic number (protons)

Beryllium Reaction



Beryllium is the lightest structural material, used in satellites.

JANIS site
KAERI Table



$M = \text{mass}$
 $T = \text{Kinetic}$

$$M_1 c^2 + T_1 + M_2 c^2 + \cancel{T_2} = M_3 c^2 + T_3 + M_4 c^2 + \cancel{T_4}$$

$$Q = [M_1 + M_2 - M_3 - M_4] c^2 = T_3 + T_4 - T_1 - \cancel{T_2}$$

Neutron Flux

Sunday, January 16, 2022 20:59

Used for:

- Reactivity measurement and control;
- Reactor power control (integral and three dimensional);
- Reactor emergency shutdown;
- Core diagnostics

Can be measured by ion chambers, fission chambers and self-powered detectors. Can be in core (in-core) or exterior to core (ex-core):

- In-core: used for localized flux measurements,
- Ex-core: provide global measurements as function of reactor power

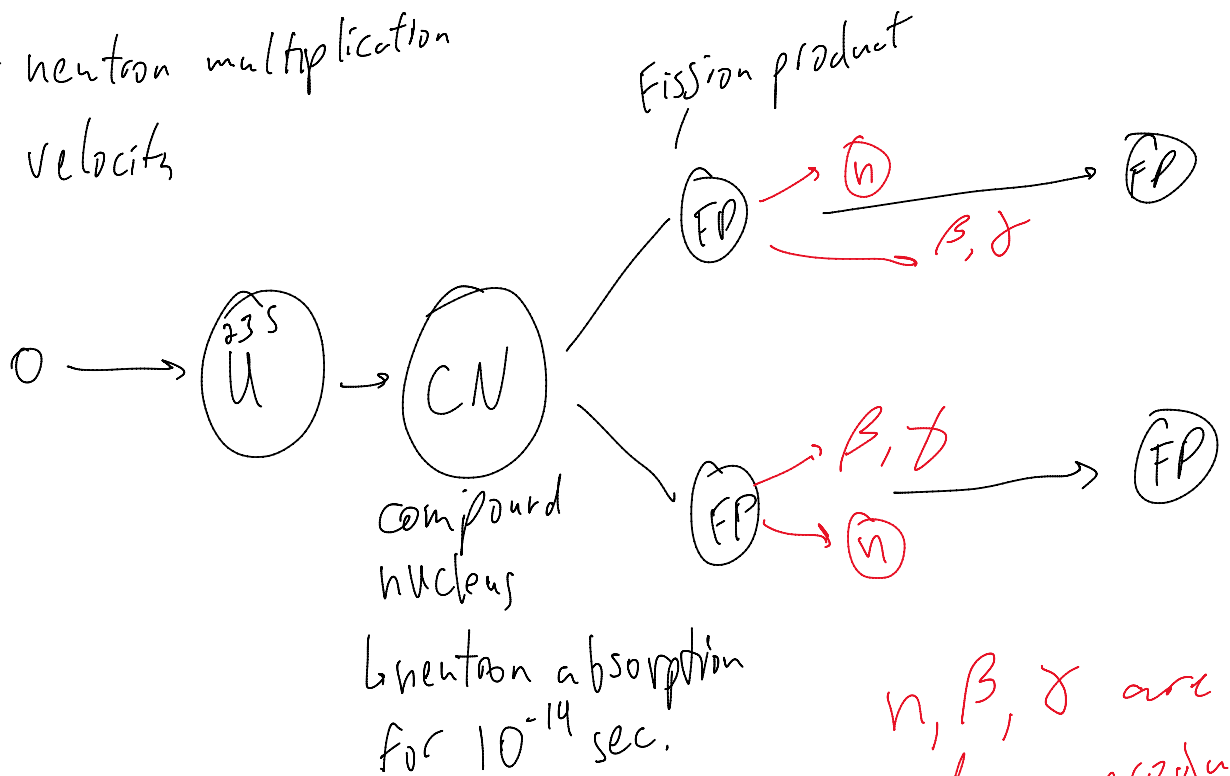
Three range approach to cover entire operating envelope:

- Source: high sensitivity, measure electrical charge build up produced by ions during start-up
- Intermediate: compensated ion radiation produced by incident gamma radiation
- Power: neutron to gamma ratio is high enough to not require compensation

Nuclear Reactions

Wednesday, January 26, 2022 08:59

λ = neutron multiplication
 V = velocity



n, β, γ are decay products of the fission product

Almost all fission energy is within thermal regions

$$n(\vec{r}, E, \vec{\Omega}, t)$$

$\vec{r} \rightarrow (x, y, z)$
 $E \rightarrow \text{energy}$
 $\vec{\Omega} \rightarrow (\theta, \phi) \rightarrow \text{angle of vector}$
 $t \rightarrow \text{time}$

$\chi(E)$ = "fission birth spectrum"

Almost all reactions with U-235 are fission, making it very attractive for reactors.

Nuclear Reactor Physics

Tuesday, December 15, 2020 12:00 AM

Current technology is based around nuclear fission. A neutron is fired at an atom, which fissions into two atoms and additional neutrons. These released neutrons then hit other atoms, which fission more atoms, therefore creating a chain reaction.

The probability of nuclear reaction is defined by the cross section with units of barns.

Criticality is reached once the neutron population remains steady from one generation to the next. This means that as many new neutrons are being created as being lost during the fission reaction. The "six-factor formula" ($\rightarrow$) will determine the criticality.

- $K = 0$ means critical (stable)
- $K < 0$ means subcritical, more neutrons are being lost than produced and as such losing power
- $K > 0$ means supercritical, more neutrons are being created than lost, increasing power output

Control rods are used to absorb neutrons, therefore slowing down the chain reaction.

Many reactors contain a starter source of neutrons to facilitate chain reactions. A common source for fresh cores is a mixture of Americium-241 and beryllium-9. An operational core may use antimony and beryllium-9.

Four fundamental forces

- **Gravitational:** holds even down to sub-atomic level
- **Electromagnetic:** hold electrons in balance with protons
- **Strong:** binds quarks that keep together protons and neutrons in the nucleus
- **Weak:** form and function of quarks

Radioation: wave and particle energy that decays off a radioactive particle. The **half-life** determines the time at which 50% of material will be decayed.

Radioactive material basics

- Stable isotopes tend to have nearly equal proportions of proton to neutrons.
- Heavier neutrons require more neutrons than protons to bind the nucleus
- Elements with more than 83 protons are radioactive.
- Nuclear decay occurs to bring the nucleus to a less excited state, less neutrons.

Types of radioactive particles:

Alpha

- Emitted from uranium and plutonium. Transuranic emission.
- Short range (few centimeters in air)
- Unable to penetrate skin. If ingested or inhaled causes severe damage. Unlikely to due to being very heavy.
- Charged particles of helium nuclei emitted at high energy.
- Slowed down by coulombic reactions with protons, eventually gaining electrons and becoming helium.

Beta

- High energy electrons emitted from fission. Strontium for example.
- Can cause damage to outside of skin with increased exposure.
- Used in medical applications
- Range of a few feet.
- Slowed down less easily than alpha.
- Blocked by more dense solids, aluminum.

Gamma ray

- High energy photons that displace electrons (causing ionization).
- Materials with a high Z number (proton count) stop gamma rays more easily.
- Travel long distances at speed of light
- Lead is a good blocker.

Neutron

- Does not have a charge, but interact with atoms they go through, causing them to become radioactive.
- More energy is lost in a collision with a small atom than a large one.
- Travel long distances and require very thick Hydrogen containing materials to block (concrete or water).
- Primarily occurs within reactor, which contains water shielding.

Fertile vs Fissile

The Z number distinguishes an atom's chemical properties from other elements.

The neutron number distinguishes the atom's nuclear properties from other elements.

A fissile isotope (odd neutron number) will fission with slow (thermal) neutrons (U-233, U-235, Pu-239, Pu-241)

A fertile isotope (even neutron number) can be converted into a fissile isotope by capture and subsequent decay-chain from the capture reaction daughter. Fertile isotopes can fission with high enough energy.

Neutrons are born from fission with a mean energy of 1.2 MeV. This is above the energy threshold for fast fission in most fertile isotopes. In HTGRs/LWRs the neutrons lose energy via collisions with the moderator.

Fissionable: material that can undergo fission following absorption of a more energetic neutron.

Breeder Reactors

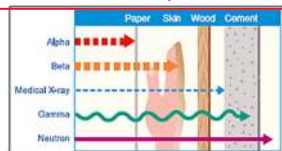
Add extra U-238 and Th-232 to the reactor.
Higher fuel efficiency as they create their own fuel.

Cross-section

Probability that a nuclear reaction will occur

$$\text{six-factor formula: } k = \eta f p \epsilon P_{FNL} P_{TNL}^{[1]}$$

Symbol	Name	Meaning
η	Thermal Fission Factor (Eta)	The number of fission neutrons produced per absorption in the fuel.
f	The thermal utilization factor	Probability that a neutron that gets absorbed does so in the fuel material.
p	The resonance escape probability	Fraction of fission neutrons that manage to slow down from fission to thermal energies without being absorbed.
ϵ	The fast fission factor (Epsilon)	$\frac{\text{total number of fission neutrons}}{\text{number of fission neutrons from just thermal fissions}}$
P_{FNL}	The fast non-leakage probability	The probability that a fast neutron will not leak out of the system.
P_{TNL}	The thermal non-leakage probability	The probability that a thermal neutron will not leak out of the system.



Reactor History

Saturday, December 12, 2020 11:36 AM

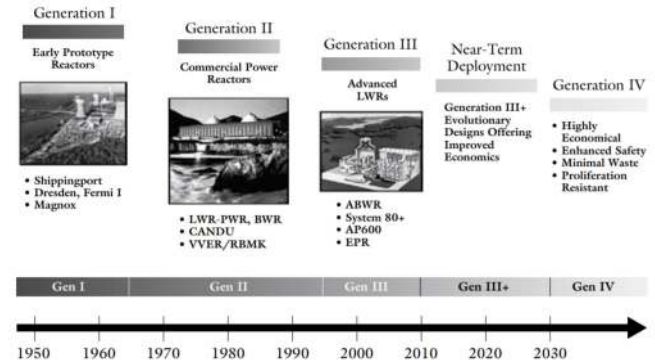
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Six factors define nuclear reactor development and deployment:

1. Cost-effectiveness: competition with other energy sources, how much does kw/h cost compared to coal?
2. Safety: Improve upon safety measures, passive (inherent) over active (require human intervention)
3. Security / Non-proliferation (weaponization): mitigation of availability of nuclear fuel for nuclear weapons
4. Grid Appropriateness: is the output of the reactor appropriate for the grid's capacity?
5. Commercialization: Can production improvements streamline commercialization?
6. Fuel Cycle Management:
 - a. Front-end: Reactors require continuous re-fueling, which must be sourced internationally
 - b. Back-end: reduce the amount and toxicity of nuclear waste

Generation classifications: I, II, III, III+, IV

- I. First reactors built in the 1950s/60s. Prototype and proof of concept
- II. Designed for 40 year operation. Light water reactors (LWR): PWR, CANDU, BWR, AGR, VVER. Use traditional active safety features that are initiated automatically or by operators. Most reactors are built by either Westinghouse, Framatome or General Electric. Require relatively large grids, produce significant amount of used fuel. Primary improvements on safety for these would be: 1) Containment structures 2) common-mode emergency core cooling 3) Mixed oxide fuel usage 4) critical safety analysis in pool design
- III. Started with WEC APR-600. GE made the ABWR. Only 4 exist in operation today. Gen II with improved: fuel tech, thermal efficiency, modular construction, passive safety, standardized design. 60 years operation.
- III+. AP1000, ESBWR, Advanced CANDU. Passive safety systems that rely on gravity and/or convection. Achieve higher fuel burnup
- IV. Currently under development. Sodium fast reactor and molten salt reactor. Focus on minimized nuclear waste duration (centuries vs millennia), increased yield (100-300x) for same amount of fuel, closed fuel cycle (ability to consume nuclear waste to produce electricity). Improvement on safety would include minimization of pressurized operation, automatic passive (uncommanded, unpowered) reactor shutdown, avoidance of water cooling in design.



Pressurizer usage is undesirable now due to the added safety / cost / maintenance component.

- Pressurizers require robust tanks that are expensive.

Pressurizer (PWR)

Thursday, November 12, 2020 10:39 PM

An essential part of a reactor coolant system, especially in Pressurized Water Reactor (PWR) designs. Pressure stability is critical in this closed loop system, and any deviation requires immediate correction. The pressurizer is in charge of maintaining pressure stability. It contains spray nozzles that spray cold water into the steam to condense it into water upon pressure increase. If pressure continues to increase, the pressurizer relief valves will open and dump more water into the tank.

The pressurized water system is also used to spray the spent fuel pool if necessary.

During steady-state operating conditions, approximately 60 percent of the pressurizer volume is occupied by water and 40 percent by steam.

A reduction in electrical load will create an increase in coolant temperature and by relation volume. This will then increase the water level within in the pressurizer which then raises the pressure. The rise in pressure will result in the valves to open on the spray lines, opening up and condensing steam. This quenches further pressure increases.

An increase in electrical will have the reverse effect, eventually closing the spray valves, causing water in the pressurizer to turn to steam and increase pressure.

Transient load conditions can create a change in pressurizer water level. This is due to the fact that water volume changes in proportion to reactor coolant average temp. "

By programming the pressurizer level setpoint as a function of reactor coolant average temperature almost no level control system action will be required because of external load changes.

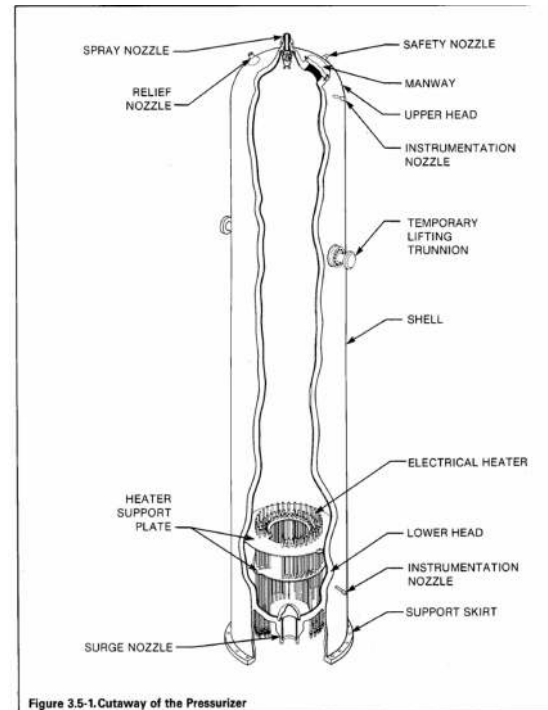
Under steady-state conditions, the level controller will only take control action to maintain the level at the programmed setpoint. This will serve to correct for changes in water inventory in the RCS. Then, when the level does deviate from the setpoint, the level control system will re-establish the in and out water flow conditions and maintain the required water inventory by restoring the level back to the setpoint.

The pressurizer level control system regulates the positive displacement pump speed and/or the centrifugal charging pump flow to drive the pressurizer level to its setpoint value.

The pressurizer pressure setpoint is fixed. A controller is used to maintain at setpoint. Deviation of pressure to setpoint is used as PV. The output drives proportional spray controllers, backup heaters and power relief valves.

Some heaters provide heat input to compensate for heat losses and reach water / steam equilibrium.

- Small Pressure variation: Proportional heaters will increase/decrease heat to increase/decrease steam levels and return to setpoint.
- Large pressure increases: steam is compressed causing pressure to be above setpoint. Heaters are turned off and water is sprayed to condense steam.
- Large pressure decrease: heaters are set to max to convert water to steam.



Reactor types

Sunday, December 27, 2020 10:47 AM

Components that are common to all reactors and may differentiate them

- **Fuel:** most commonly Uranium. A new reactor requires a neutron source to jump start the reaction. This is usually beryllium mixed with polonium.
- **Moderator:** Material to slow down neutrons released by fission. Can be water, graphite or heavy water
- **Control Rods/Blades:** made of neutron absorbing material such as cadmium, hafnium or boron. Inserted or withdrawn from core to control reaction rate or halt it if necessary. PWR control rods are inserted from the top down, BWR blades bottom up.
- **Coolant:** Fluid surrounds the core to extract heat. Light water reactors the water moderator is the primary coolant.
- **Pressure Vessel/Tubes:** A robust steel vessel containing reactor core and moderator/coolant. Can also be a series of tubes.
- **Steam generator:** PWRs use the coolant from the reactor to make steam.
- **Containment:** Structure enveloping reactor vessel and steam generators. Keeps radiation in if there is a leak. RBMK famously does not have containment as part of the design.

Heavy water contains two deuterium atoms instead of two protium atoms as with normal (light) water. Heavy water absorbs less neutrons than light water.

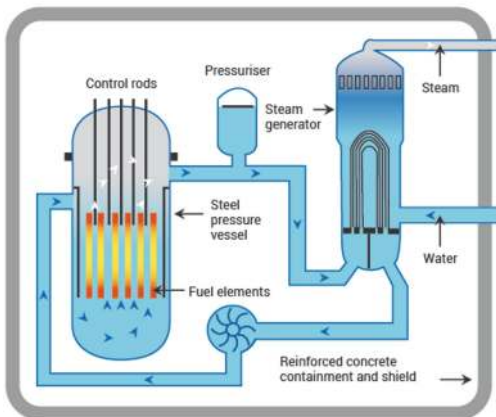
* In fission, most of the neutrons are released promptly, but some are delayed. These are crucial in enabling a chain reacting system (or reactor) to be controllable and to be able to be held precisely critical.

Commercial Reactors

Reactor type	Main countries	Number	GWe	Fuel	Coolant	Moderator
Pressurised water reactor (PWR)	USA, France, Japan, Russia, China, South Korea	301	286	enriched UO_2	water	water
Boiling water reactor (BWR)	USA, Japan, Sweden	64	65	enriched UO_2	water	water
Pressurised heavy water reactor (PHWR)	Canada, India	48	24	natural UO_2	heavy water	heavy water
Advanced gas-cooled reactor (AGR)	UK	14	8	natural U (metal), enriched UO_2	CO_2	graphite
Light water graphite reactor (LWGR)	Russia	12	8.4	enriched UO_2	water	graphite
Fast neutron reactor (FBR)	Russia	2	1.4	PuO_2 and UO_2	liquid sodium	none
TOTAL		441	393			

	Fuel	Moderator	Coolant			Spent Fuel Reprocessing	Steam Cycle Efficiency	Main Economic and Safety Characteristics
			Heat extraction	Outlet temp.	Pressure			
Magnox	Natural uranium metal (0.7% U^{235}) Magnesium alloy cladding	Graphite	Carbon dioxide gas heated by fuel raises steam in steam generator	360°C	300 psia	Typically within one year, for operational reasons	31%	Safety benefit that coolant cannot undergo a change of phase. Also ability to refuel whilst running gives potential for high availability
AGR	Uranium dioxide enriched to 2.3% U^{235} Stainless steel cladding	Graphite	Carbon dioxide gas heated by fuel raises steam in steam generator	650°C	600 psia	Can be stored under water for tens of years, but storage could be longer in dry atmosphere	42%	Same operational and safety advantages as Magnox but with higher operating temperatures and pressures, leading to reduced capital costs and higher steam cycle efficiencies
PWR	Uranium dioxide enriched to 3.2% U^{235} Zirconium alloy cladding	Light Water	Pressurised light water pumped to steam generator which raises steam in a separate circuit	317°C	2235 psia	Can be stored for long periods under water giving flexibility in waste management	32%	Low construction costs resulting from design being amenable to fabrication in factory-built sub-assemblies. Wealth of operating experience now accumulated world wide. Off load refuelling necessary
BWR	Uranium dioxide enriched to 2.4% U^{235} Zirconium alloy cladding	Light Water	Pressurised light water boiling in the pressure vessel produces steam which directly drives a turbine	286°C	1050 psia	As for PWR	32%	Similar construction cost advantages to PWR enhanced by design not requiring a heat exchanger, but offset by need for some shielding of steam circuit and turbine. Off load refuelling necessary
CANDU	Unenriched uranium dioxide (0.7% U^{235}) Zirconium alloy cladding	Heavy water	Heavy water pumped at pressure over the fuel raises steam via a steam generator in a separate circuit.	305°C	1285 psia	As for PWR	30%	Good operational record but requires infrastructure to provide significant quantities of heavy water at reasonable costs.
RBMK	Uranium dioxide enriched to 1.8% U^{235}	Graphite	Light water boiled at pressure, steam used to drive a turbine directly	284°C	1000 psia	Information not available	31%	Information not available but operated in considerable numbers in the former USSR. Believed in the West to be inherently less safe

A Pressurized Water Reactor (PWR)



Fuel assemblies of 200-300 rods each, arranged vertically. 150-250 fuel assemblies with around 90 tonnes of Uranium.

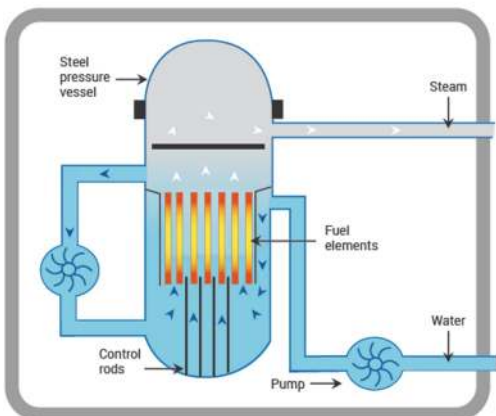
Water in the reactor core reaches 325 C, meaning it must be pressurized to 150 x atm to prevent boiling. The pressurizer is in charge of this (see Pressurizer Section)

Water acts as moderator so if any of it flashes to steam, the fission will slow down. This is a negative feedback safety loop.

The secondary shutdown system adds boron to the primary circuit.

The Secondary loop is under less pressure and water boils in the heat exchangers.

A Boiling Water Reactor (BWR)



When a neutron hits water, it thermalizes completely with the hydrogen atom.

Fuel assemblies of 90-100 fuel rods. 750 assemblies with 140 tonnes of U.

Similar to PWR, only that there is a single circuit. Water is at 75 x atm (boils at 285 C)

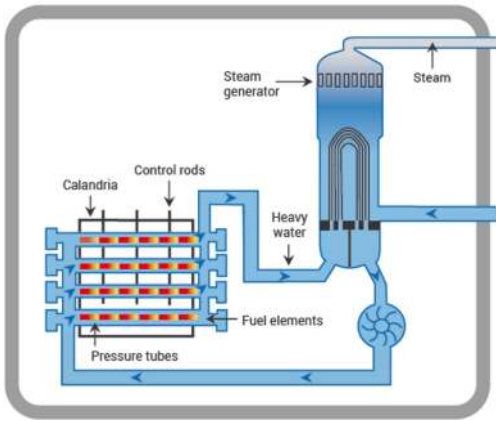
Operates with 12-15% of water in top of core as steam. This moderates reaction.

Can operate in load-following mode better than PWR

Steam passes through drier plates (steam separators) and then to turbines.

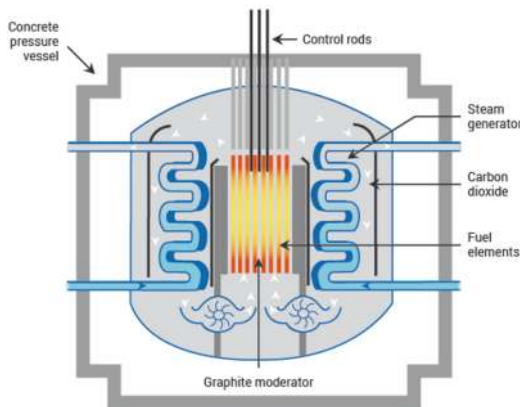
Turbine is shielded due to being radiated by primary circuit. Mostly N-16 though, that has a 7 second half-life so it can be entered once shut down.

A Pressurized Heavy Water Reactor (PHWR/Candu)



Fuel assembly of 37 half metre fuel rods, 12 bundles lying horizontally
Control rods inserted vertically. Gadolinium used as second shutdown system
Heavy water moderator provides heat as well.
Can use recycled uranium or a blend with enriched. Thorium can be used as well.

An Advanced Gas-cooled Reactor (AGR)



Graphite moderator and carbon dioxide is primary coolant.
Fueled by uranium oxide pellets enriched to 2.5 and 3.5%.
Carbon dioxide circulates through core reaching 650 C
Nitrogen added to coolant will slow down fission.
High temperature gives high thermal efficiency.
Allows for operational refueling.

Use enriched uranium

LEU → 20% or below

MEU →

HEU →

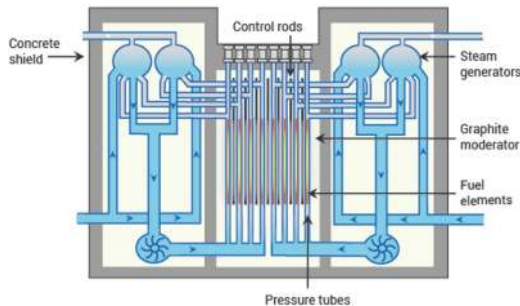
CO_2 is coolant $\text{CO}_2 \rightarrow \text{natural loop} \rightarrow \text{O}_2 + \text{C} (+\text{G})$
Graphite is moderator This means graphite slowly disappeared from reactor

Low power density: fewer reactions per unit volume.

Graphite takes radiation damage and stores it as higher energy. This can then be a fire.

Needs to be pressurized to increase density.

A Light Water Graphite-moderated Reactor (LWGR/RBMK)



Uses 7 meter vertical pressure tubes that run through a graphite moderator.
Cooled by water boiling at 290C and 6.9 Mpa.
Fuel is low-enriched uranium oxide.
Positive feedback issue can arise as excess boiling with reduce cooling and neutron absorption.

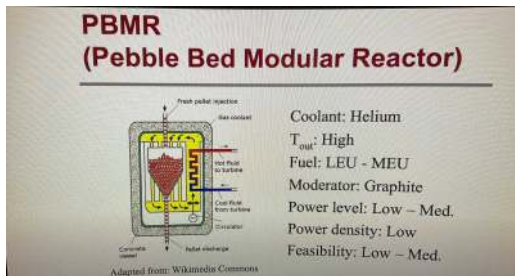
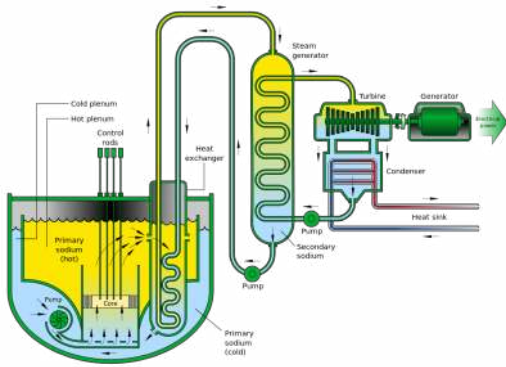
If you lose coolant, you lose moderator

Positive feedback coefficient due to graphite-tipped rods. This slowed down neutrons to even better fissioning.

Fast Neutron Reactors

Do not contain moderators as they rely on fast neutrons to cause fission.
Use plutonium as main fuel, due to fast neutrons being more efficient with Pu than U.
Produce more fissile material than they consume.

Sodium Cooled Fast Reactor



the helium transfers heat
to fluid for steam gen.
silicon carbide surrounds
pebbles.



$\sim 800^{\circ}\text{C}$
 $\rightarrow$ Helium impurities can cause
 issues.
 requires special alloys



D_2O won't slow down neutrons
 as much as H_2O , absorption
 cross-section is smaller.

Doesn't require pressurizing.
 D_2O is expensive.



$\rightarrow$ Does not need pressure
 $\rightarrow$ Needs cover gas to avoid sodium
 interaction.

$\rightarrow$ Do not slow down reactor.

$\rightarrow$ Fast fission: high energy fission

$\hookrightarrow$ U-238 is better at fast
 than thermal fission

$\rightarrow$ High boiling point

$\hookrightarrow$ $\text{Na} + \text{H}_2\text{O} \rightarrow$ dangerous reaction



x

→ Lead is very corrosive
→ Lead-Bismuth.
→ Used in soviet subway



x

Coolant and fuel are the same
Leaked salt goes subcritical.

Safety Functions

Sunday, January 16, 2022 20:40

The IAEA has 3 safety classes: safety, safety-related, non-safety.

Most common safety loops:

- Reactor trip;
- Emergency core cooling;
- Decay heat removal;
- Containment/confinement isolation;
- Containment fission product removal;
- Containment heat removal;
- Emergency ventilation;
- Emergency power supply

Safety-related loops:

- Reactor power control;
- Diverse reactor trip;
- Pressure and temperature control for normal heat removal systems;
- Fire detection;
- Radiation monitoring;
- Personnel access control;
- Display of information for planning emergency response.

Non-safety:

- Feedwater reheater control;
- Demineralizer control;
- Intake and discharge screen control

Cooling and coolants

Thursday, January 7, 2021 12:01 PM

NPPs must remove heat from the reactor core to transfer it to the steam turbines, as well as dump surplus heat.

The bigger the temp difference between the internal heat source and the external environment (reactor core vs. outside air), the more efficient the mechanical work will be. This is why most power plants are situated close to cold water sources.

* Many power plants, fossil and nuclear, have higher net output in winter than summer due to differences in cooling water temperature.

Heat Transfer steam cycle

Water is circulated continuously in a closed loop steam cycle with minimal loss. It turns to steam, goes through the turbine and then goes through a condenser which will return it to the heat source. When a conventional plant shuts down, the heat source is removed almost immediately. With NPPs however, some heat is still generated from radioactive decay. This must be actively removed with cooling systems and Emergency Core Cooling Systems (ECCS). The primary cooling is usually a main steam supply circuit that bypasses the turbine and goes directly to the condenser. Decay subsides after a day or 2.

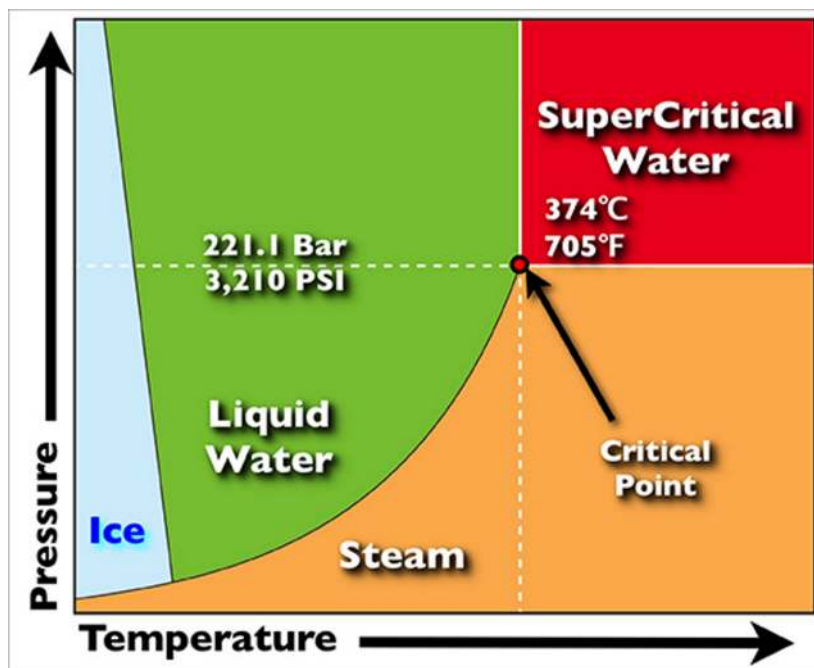
The second function for water is to cool the system and condense low-pressure steam. The surplus heat needs to be transferred to either air or water. Three ways to do this:

- Direct cooling: water from a river/lake/sea is pumped through the condensers and discharged at a slightly higher temp.
- Recirculation: If water source is not present. Pass steam through condenser and then a cooling tower.
- Dry cooling: Cooling towers in a closed circuit or forced draft air flow (like a radiator)

15% of waste heat in a fossil plant is through the stack (smoke). In NPP all waste heat is dumped into condenser. CCGTs release a large amount of heat through their exhaust and as such require less cooling.

Radioactive decay of fission products produces heat for a long time after shutdown

Water: Must be maintained at a high pressure (1000-2200 psi) to enable it to function above 100 C (avoid boiling).



Helium: must be used at high pressure (1000-2000 psi) to remove heat effectively.

Carbon dioxide: Used in early british reactors and AGRs. Denser than helium and as such higher thermal efficiency. Also less prone to leakage.

Sodium: used in fast neutron reactors, at around 550 C. Melts at 98 C and boils at 883 C. High thermal conductivity and thermal capacity. Used in secondary circuit to generate steam. Does not corrode metal. Reacts with water/steam and creates hydrogen.

Lead: transparent to neutrons which adds to efficiency. Corrosive of fuel cladding and steels. 650 C can be reached. Higher thermal conductivity than water, but lower than sodium. Lead yields polonium during reaction.

Salt: (fluoride salts) boil at 1400 C at atmospheric pressure. Low vapour pressure and high volumetric heat capacity. Low neutron absorption and good moderation capability. Does not react with air or water and is inert to most metals.

Chloride salts: candidates for secondary cooling loops

Fueling

Thursday, December 31, 2020 2:59 PM

Most reactors need to be shut down for refueling due to the vessel having to be opened up. AGR, CANDU and RBMK reactors can be refueled online because they use pressure tubes instead of vessels.

Fuel cycles are usually every 12, 18 or 24 months.

Uranium must be enriched if using light water as a moderator. This is due to the fact that light water absorbs and slows down more neutrons than other moderators (graphite/heavy water). Graphite and heavy water reactors are more efficient and do not require uranium enrichment.

During operation some U-238 is converted into plutonium-239 (Pu-239) which provides energy as well.

Most reactors use ceramic uranium oxide (UO₂ with a melting point of 2800 C) as fuel. The pellets (1 cm diam and 1.5 cm long) are arranged in a tube to form a fuel rod. Numerous rods form a fuel assembly that can be lifted in/out of the core. BWR assembly is usually 320 kg while PWR is 655 kg.

"Poisons" are used in fuel and coolant to even out performance. A poison is a neutron absorber that decays under neutron exposure. Gadolinium is the most common.

Currently a number of accident-tolerant fuels are being developed that are more resistant to melting (TRISO fuel)

I&C Design

Sunday, January 16, 2022 20:45

1. Single failure criterion: the NPP must be able to tolerate the failure of any component and the resulting degradation without serious consequences
2. Redundancy: each critical component must have a duplicate in case of failure
3. Diversity: Critical actuators must have different control algorithms/devices, technology.
4. Independence: a failure should not propagate through the system. Achieved via isolation

Non-safety / safety-related can not hamper actuation of a safety function.

Alpha, beta, gamma & neutron radiation must be monitored for:

- Solid materials: spent fuel, filters
 - Liquid: coolant, feedwater
 - Aerosols: exhaust gases
-
- Screens should be dull under normal operating conditions
 - Displays should be obvious under abnormal conditions
 - A faceplate should only show up on the display it is called from

Nuclear Reactor Trips

Saturday, February 16, 2019 5:34 PM

Whenever a direct process or calculated variable exceeds a setpoint, the reactor will be shut down:

Condition II: to prevent fuel rod damage for an anticipated transient

Condition III: to limit core damage for infrequent faults

Condition IV: to protect the reactor coolant pressure boundary for limiting fault conditions

Trip signals that active Reactor Protective System:

Manual: redundant pushbuttons

Neutron flux: protect from control rod ejection or rod drop

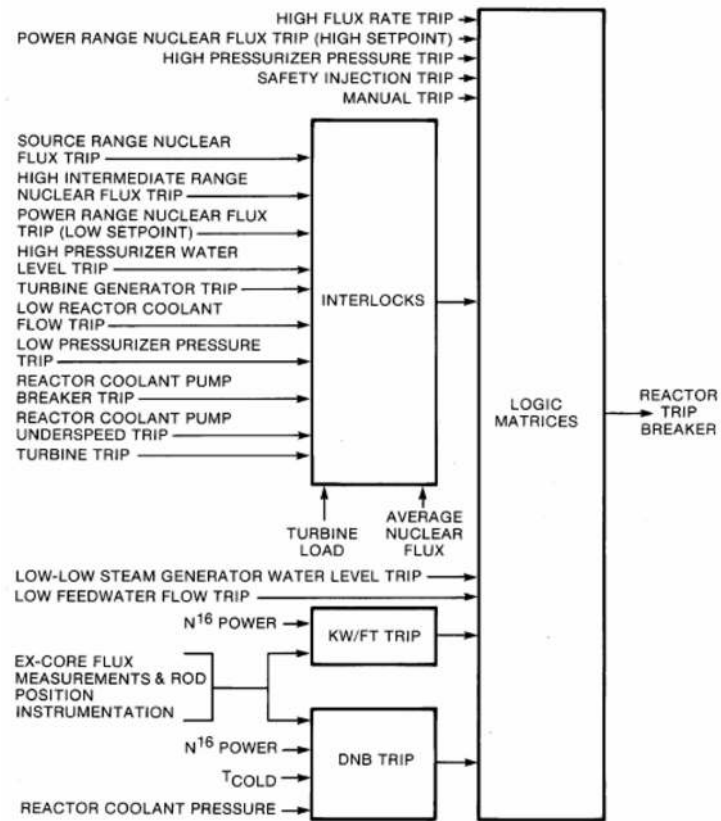
Primary coolant: protect from excessive coolant temp as a result of pump failure, low flow, boiling point close.

Pressurizer: high/low pressure, high level

Steam gen: low feedwater flow, low water level

Turbine gen: Trip from turbine can carryover to reactor trip

Safety injection signal: activation of safety injection system will trip reactor



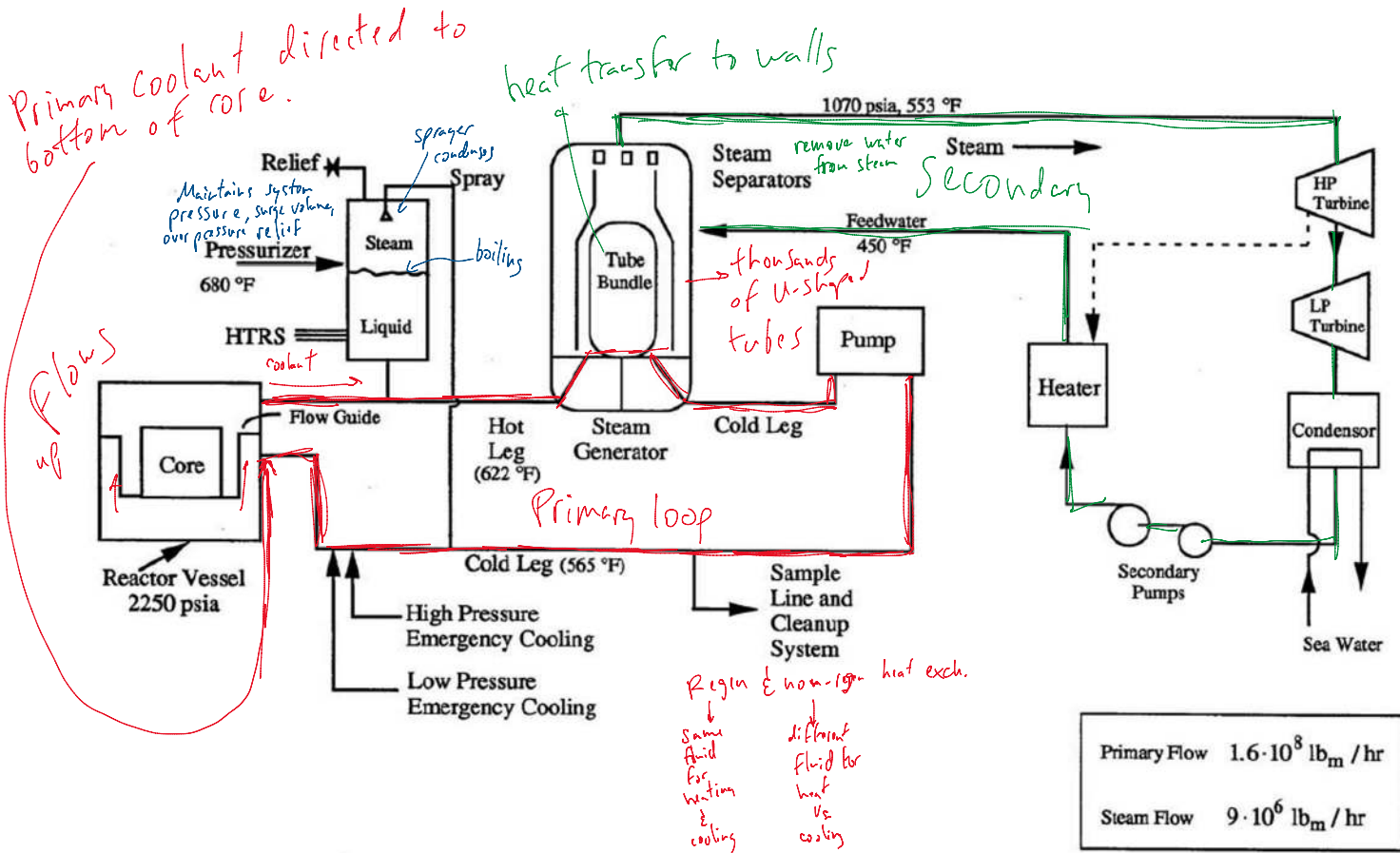
Steam Turbine

Monday, January 17, 2022 19:08

Last-stage blades (LSB) are a key element of the low pressure section of a steam turbine. For nuclear power plants, the LSB is extremely long to compensate for exhaust backpressure loss

Pressurized Water Reactors

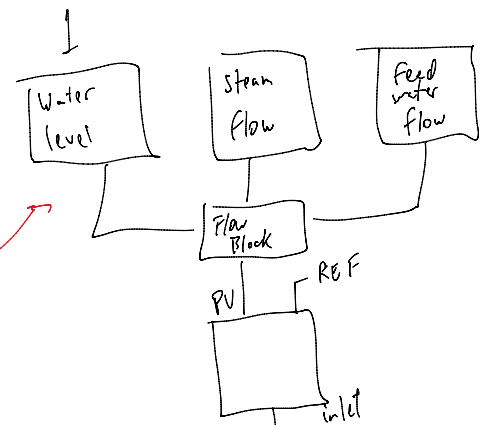
Thursday, January 27, 2022 10:49 AM



Pressurized Water Reactor (PWR)

Control Loops

- **Pressurizer Level**
 - o Manual Control.
 - o Level is constant at steady-state
 - o Transients will vary as coolant changes density
 - o Significant changes during startup (coolant heats from 70 F to 550 F, must drain water), shutdown (reverse)
- **Pressurizer Pressure**
 - o Setpoint PID with manual override.
 - o Low pressure = heaters energize
 - o High pressure = open spray valve
 - o Spray valve sprays water droplets at 565 F to 680 F steam condensing it
- **Primary Temp**
 - o Manual control or PI controllers.
 - o Keep average temp of primary coolant within 3 C (greenband)
 - o Temp changes with load.
 - o Control rods will move to counteract temp change.
- **Turbine Runback**
 - o Controller closes steam admission valve if there is over-power or over-temp
 - o Runback will activate if there is a dropped rod
- **Steam Dump Control**
 - o Allows for a sudden loss of load without necessitating reactor trip.
 - o If there is a large reduction in load, the energy in the reactor exceeds what is being removed by turbine, causing primary and steam gen temps/pressures to rise.
 - o Steam is dumped directly to the condenser.
 - o Primary coolant temp is compared to reference. If there is a sudden decrease in load, primary coolant temp rises and ref decreases. If diff $\Rightarrow$ 10%, steam will be dumped
 - o Control rods will also be driven into core to lower power/temp
- **Steam Generator Level Control**
 - o Steam generator levels fluctuate with load changes
 - o When raising power, the shell side volume increases with vapor, causing water to rise level. (Swell)
 - o If power is decreased, vapor volume decreases and water level drops. (Shrink)
 - o Three element PID used to maintain level.
- **Feed Flow Control**
 - o Speed of main feed pumps regulated by pneumatic system.
 - o Pressure drop across steam gen feedwater valves kept at specific preset value
- **Hot Wall Control**



- Three element is used to maintain level.
- **Feed Flow Control**
 - o Speed of main feed pumps regulated by pneumatic system.
 - o Pressure drop across steam gen feedwater valves kept at specific preset value
- **Hot Well Control**
 - o Secondary loop contains two reservoirs to accommodate volume changes:
 - Condenser hot well
 - Heater drain tanks



GENERAL PROCESS

Introduction

Thursday, April 1, 2021 16:06

SIS stands for Safety Instrumented System. It is controlled by IEC 61508 and 61511

The primary focus here is safety, so we divide safety into layers.

A normal control system is composed of a transmitter, controller and control valve all working in tandem to respond to dynamic conditions. It must respond to any and all value changes made to the system. This is considered the first layer of protection.

The second layer is when something happens outside of the normal process values, causing an alarm. This normally requires operator intervention to correct.

The third layer is the safety system, designed to only take action under dangerous conditions. It contains a sensor, a logic solver and Final Element (FE) to carry out the safety functions. This allows it to react in one of the following:

- Take process to a safe state if conditions are violated
- Allow process to continue if "permissive conditions" exist
- Take action to mitigate against industrial hazard

The fourth layer is active protection. It contains valves and rupture disks that provide relief to pressure/temp originated issues.

The fifth layer is passive protection, consisting of a containment structure (think of containment dome in a nuclear plant to prevent radiation).

The last layer is plant/emergency response, consisting of evacuation, firefighting, etc.

An SIS consists of safety instrumented functions (SIF). Each SIF has a safety integrity level (SIL) with a separate loop comprised of the sensor, logic solver (LS) and final control element (FE)

A logic solver is usually a controller that reads the signals from sensors and acts upon logic by providing output to the final control element. The final control element will then implement the action demanded by the LS:

Final control elements in SIL are divided into:

- On/off
- Dual purpose control and safety use
- Dual purpose in addition to on/off

Any normal control system has a Safety Integrity Level lower than 1.

Safety Integrity Level (SIL)

Probability of failure on demand (PFD): how probable system is to fail to reponse on demand. There are two ways an SIS can fail:

- nuisance/spurious trip: unplanned but safe process trip. Minimal danger but high operational cost
- Undetected condition: an unsafe condition is not detected by SIS and plant enters dangerous condition.

PFDavg will take into consideration failure rates of equipment involved. This is necessary to determine the SIL level.

Physical failure: caused by one or more hardware mechanisms degrading. Permanent and can be singled to a single component

Functional failure: due to a deterministic way to a cause.

Mean Time Between Failure (MTBF)

Mean Time to Fail (MTTF):

- Safe detected
- Safe undetected
- Dangerous detected
- Dangerous undetected

Individual devices do not have an SIL rating, but can be assessed to see if they are apt for an SIL.

Without mechanical movement, a control valve is prone to develop stickiness.

Mechanical components do not have diagnostic capabilities, so can only be used in SIL 1. A digital valve controller (DVC) increases the SIL.

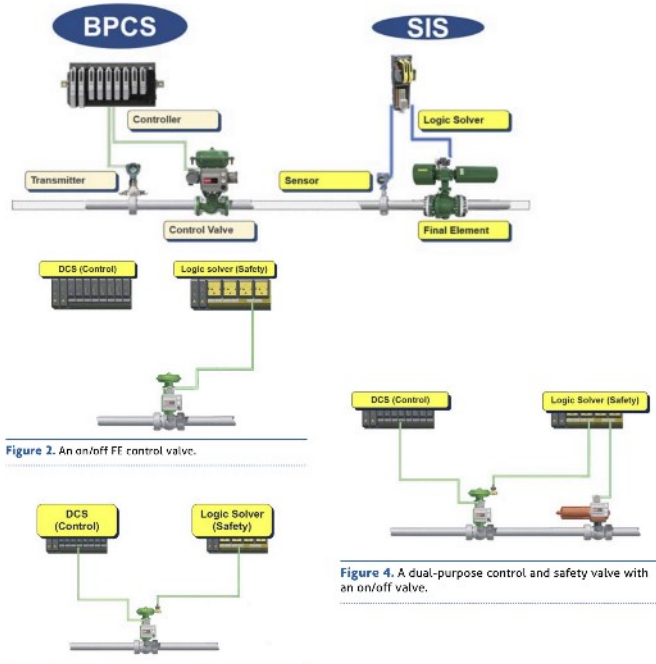
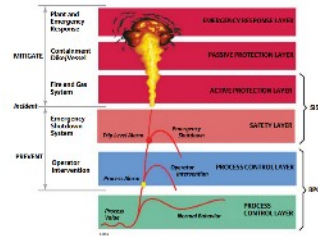


Figure 2. An on/off FF control valve.

Figure 3. A dual-purpose control and safety valve.

RRF (Risk Reduction Factor)	PFD _{avg} (Probability of Failure on Demand = 1/RRF)	SIL (Safety Integrity Level)
100000 to 10000	$\geq 10^{-5}$ to $< 10^{-4}$	4
10000 to 1000	$\geq 10^{-4}$ to $< 10^{-3}$	3
1000 to 100	$\geq 10^{-3}$ to $< 10^{-2}$	2
100 to 10	$\geq 10^{-2}$ to $< 10^{-1}$	1

$$PFD_{TOT} = PFD_{SIS} + PFD_{LS} + PFD_{FE}$$

Excitation Theory

Tuesday, August 25, 2020 9:58 AM

Excitation is the action of creating an electromagnetic field within a generator to enable mechanical to electrical energy conversion.

Regulator is in charge of controlling amount of voltage and resulting excitation field
Exciter is in source of power.

Most common methods are:

- Shunt (self-excited)
- Excitation Boost System (EBS)
- Permanent Magnet Generator (PMG)
- Auxilliary Winding

An Automatic Voltage Regulator (AVR) is used in all applications to supply DC output to the exciter stator.

- Silicone Controller Rectifier (SCR) : senses power level from stator and determines firing for exciter voltage. Not recommended for non-linear loads.
- Field Effect Transistor (FET) : senses power from stator and translates to PWM signal.

The exciter rotor AC output is rectified to a DC input for the main generator rotor.

Shunt

- Most simple setup. Stator supplies input voltage to AVR as generator rotates.
- Drawback is that AVR is affected by generator load. Increases load means voltage decrease and mor current. This pushes the AVR to collapse the excitation field.
- A short circuit will cause a loss of generator power output.

Excitation Boost System (EBS)

- Contains additional boost generator and control module
- Boost generator supplies power to controller as the generator shaft rotates.
- Controller receives signal from AVR in order to vary level of excitation current if needed.
- Not recommended for continuous power applications.
- Intended for emergency or backup applications.

Permanent Magnetic Generator

- Permanent magnet is mounted on the end of the generator shaft
- PMG supplies isolated power to AVR when shaft rotates.
- Excitation field does not collapse immediately as response to short circuit fault
- Changing load does not affect excitation field
- Voltage created on initial startup.
- Most commonly used for motors that startup / shutdown

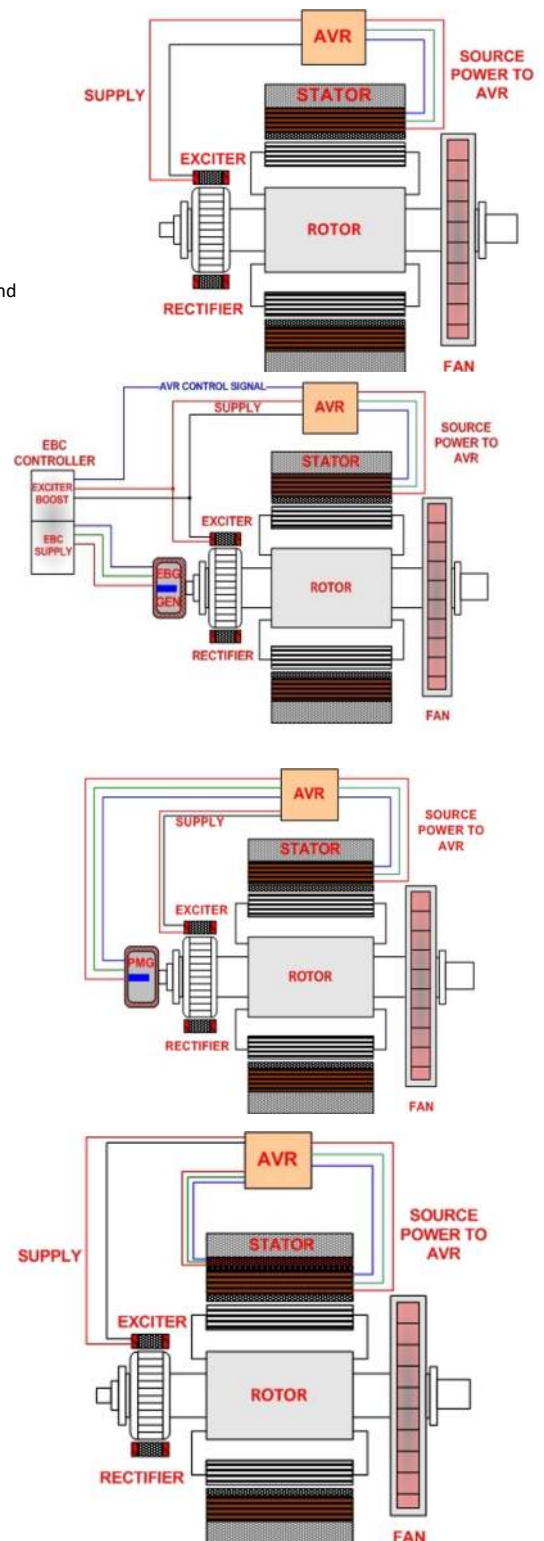
Auxiliary winding

- Creates separate excitation field. Additional single phase winding installed in stator
- As generator shaft rotates the stator mains supply voltage to AVR
- Additional windings supply more voltage in case of non-linear loads

Electromagnetic field can range from 50 amps to 9000 amps.
Modern excitation systems are static and DC current is created by rectifying AC with saturable current transformers (SCTs) and power potential transformers (PPT).

Subsystems include processor and I/O that monitor:

- Generator terminal voltage/current
- Field voltage/current
- Rotating exciter field voltage/current



Control & Automation

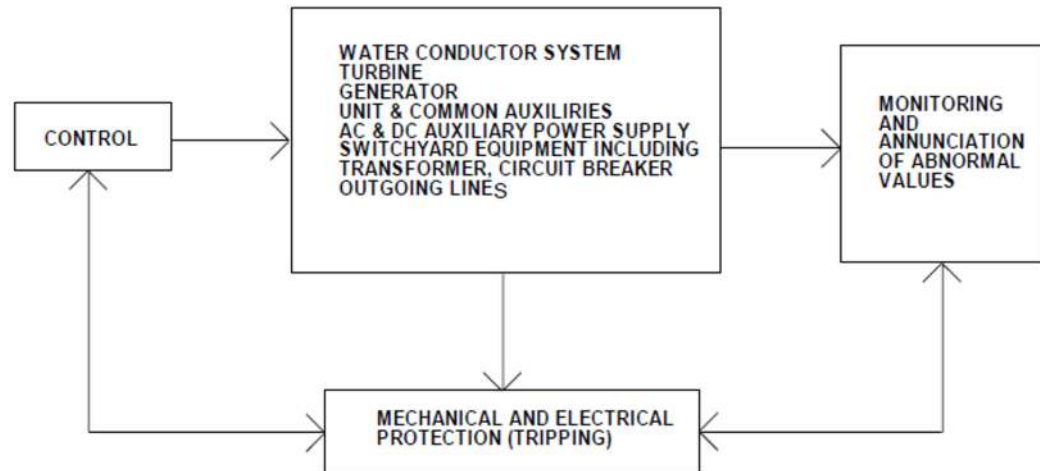
Friday, January 28, 2022 2:00 PM

Main systems:

- Start/Stop sequence
- Power factor

Main components:

- Intake gate
- Main inlet valve
- Turbine
- Governor
- Lubrication system
- Generator excitation
- Main Circuit Breakers



INTAKE

Penstock gates at intake/inlet valves:

- Emergency shutdown of unit
- Prevent leakage in turbine wicket gates / needle valve when turbine shut down
- Dewatering for maintenance

Main inlet valve

- Butterfly or spherical
- Designed for slow filling turbine
- Oil driven servo operation
- Interlocked operation so that only opened with balanced head pressure on both sides

Control Schemes

Friday, January 28, 2022 2:13 PM

Local Manual

Semi Auto

Auto

Local Manual

- Unit auxiliaries started manually by electrical push buttons.
- Display via lamp / visual inspection
- Operator at turbine governor panel starts up and controls speed.
- Used as backup mode of operation

Semi Auto (Local control from Unit Control Board)

- UCB controls governor, start/stop, unit aux, etc.
- Generator, transform bus duct relays are in central control room.

Automatic (everything from Central Control Room)

- Control functions initiated and modified from control room
- Pre-start checks
- Unit start
- Unit stop
- E-stop
- Speed level setting to adjust unit load
- Gate limit control (built into governor)
- Voltage VAR control
- Gate limit position

SYNCHRO

Must ensure voltage and frequencies of grid and generator are in sync and same phase

Manual

Lamps are connected across respective phases of incoming and running voltage buses. Manual adjustment of excitation on machine. Synchroscope indicates voltage difference / phase difference.

Auto

Continuously controls terminal voltage of incoming machine to match system

Controls speed of prime mover (governor) to match

Energizes closing coil of circuit breaker at the instant when phase difference is minimal.

Main disadvantage is during disturbance, it may take long to sync.

Self-sync

Gen breaker closed after unit has accelerated to 95 percent of rated speed. Field applied immediately.

Recommended for smaller units as it can cause disturbances

AUTOMATIC LOAD CONTROL

Mechanical Governors

Automatic turbine regulation achieved by governor speed droop settings.

Time delay between actual load change and governor response.

Actual speed can deviate from rated speed.

Speed droop varied between 0 and 6%

Analogue Electronic Governors

Respond to change in rotational speed AND acceleration during load variations.

An abrupt variation of load causes a small temporary change.

Feedback of servomotor provides integral function for limiting temp speed droop.

Frequency control provided by speed level setting every half-hour

Machine Running Supervision

Operating personnel must ensure:

- Load-frequency control
 - Governor is set to grid freq by setting speed droop to zero
 - Frequency will stay the same regardless of load changes
 - This is impossible in an interconnected system, as several governors can't match up
- Reactive Power and Voltage Control:
 - Automatic voltage regulator holds terminal voltage to a scheduled value
 - Power factor control
- Supervision of alarm and protective equipment

VIBRATION MONITORING

Belts

Thursday, March 25, 2021 11:35

The V-belt is the most common in the industry. Vbelts operate at speeds up to 6600 feet per minute. Cog belts and ribbed vbelts are faster

Measurement point orientations for Belt Drives

The most optimal vibration orientation for detecting belt defects is in-line (parallel).

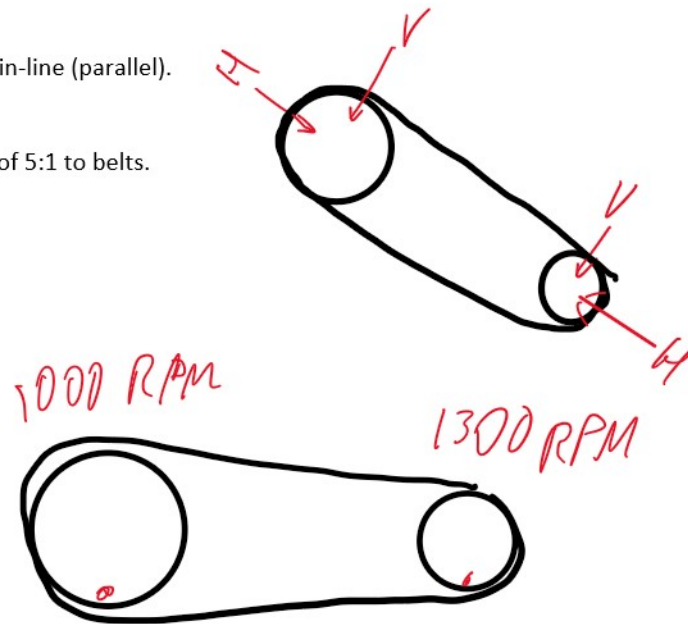
Belts have a design life of approx 25,000 hours. Sheaves have a life of 5:1 to belts.

Belt vibration causes:

- Belt defects (cracks, broken pieces, hard spots)
- Shaft misalignment
- Sheave groove misalignment
- Belt resonance
- Belt too tight or too loose
- Bent shaft

Vibration freq

- Belt Operating freq
- Drive sheave operating freq
- Driven sheave operating freq



Several belt drive problems do not cause vibration at calculated freqs.

Belt freq (CPM) = $3.1416 \times \frac{\text{Sheave diam} \times \text{Shaft rpm}}{\text{Belt length}}$

Belt length

Belt problems in vibration data

Defects will produce freqs at belt freq and harmonics.

Installation and operating problems will produce vibration at operating freq of the sheave associated.

A defect in the belt will impact each sheave once per belt rotation. In a 2-sheave system, this means an impact freq of 2x the belt freq. There will normally be harmonics in the spectral data.

Installation and operational problems

Sheave misalignment: shafts should be parallel and groove aligned for a straight belt path. If misaligned, there will be excessive power consumption, accelerated component wear. 1 x RPM of driver and driven in axial direction.

Imbalanced sheave causes vibration at 1 x RPM of faulty sheave in radial direction.

Offset: shaft centerlines are parallel, but sheaves are offset from the position of either the driver/driven.

Vertical angular: Shaft is not parallel to vertical plane.

Horizontal angular: shaft not parallel to horizontal plane.

Belt Strand vibration is caused by loose tension or high tension. Loose will vibrate randomly at low freq. High tension will appear at high freq.

Intro to vibration

Wednesday, March 24, 2021 16:14

Vibration is the movement of a body about a reference. The reference is the position of the mass at rest without forces acting upon it.

Forcing functions are either periodic or random.

Periodic is movement occurring in a repetitive manner

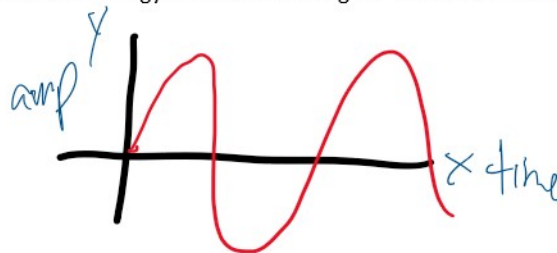
Random is transient/momentary and non-periodic events.

Some examples of forcing functions are:

- Imbalance of rotating comps
- Misalignment
- Gear wear
- Belt defects
- Electrical problems

Vibration is measured using transducers, placed either on the shaft or the casing. Shaft readings are the preferred method for sleeve bearings, while casing is preferred for anti-friction bearings.

A transducer will convert mechanical energy to an electrical signal. The waveform is a display of amplitude vs. time.



The amplitude of a vibration signal will help determine the severity.

This measurement will represent one of the following:

Displacement

Distance body moves in relation to reference point (Pk - Pk)

Expressed in mils

Velocity

Rate at which displacement occurs (Pk)

Expressed in inches per second (in/sec)

Acceleration

Change of velocity per time.

Measured as amount of energy in area under curve (Root Mean Square. RMS)

Expressed in units of G's (1g = 386 in/sec²)

Frequency

How often an event occurs per unit of time. The rate at which a machine component vibrates is known as vibrating frequency. A period in the waveform is the amount of time it takes to complete one cycle of vibration.

$$F = \frac{1}{T}$$

Frequency Domain Once the frequency is calculated, it can be plotted in the frequency spectrum via a Fast Fourier Transform (FFT). The spectrum represents the amplitude and frequency of periodic cycles in the waveform.

Frequency Units

- Cycles per minute (CPM) how many times a cycle occurs in one minute
- Cycles per second (CPS or Hz) how many times a cycle occurs per second
- Orders: how many times a cycle occurs per shaft revolution.

$$Hz = \frac{CPM}{60} \quad \text{order} = \frac{\text{freq}}{75}$$

Fast Fourier Transform

Conversion from time to frequency domain. This allows for conversion from waveform to spectrum. The spectrum is the primary choice for analyzing machine vibration.

Fast Fourier Transform

Conversion from time to frequency domain. This allows for conversion from waveform to spectrum. The spectrum is the primary choice for analyzing machine vibration.

Amplitude relations

- Root Mean Square (RMS): $0.707 \times$ true peak value
- Peak to Peak (Pk-Pk): 2 times true peak
- Peak (Pk): 1.414 times RMS

Acceleration $\rightarrow$ RMS

Velocity $\rightarrow$ Peak

Displacement $\rightarrow$ Pk - Pk

Frequency = Hz

Velocity is the unit of choice for spectral analysis due to its flat behavior across freqs. The spectrum therefore will be stored and displayed in units of velocity.

Displacement is the preferred amplitude when looking for faults and machine characteristics in lower range of spectrum

Since amplitude of force is a good indication of severity, acceleration is the best for time waveform.

Misalignment

Wednesday, March 24, 2021

11:19

Condition in which two connected shafts are not parallel.

Belt drive equipment shafts should be parallel, while direct coupled equipment share a common axis.

Types:

- Offset:
- Angular:

Causes:

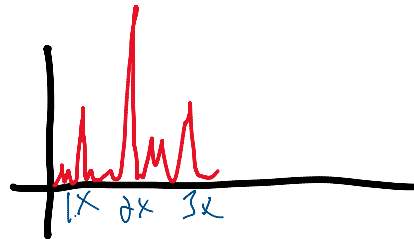
- Improper base prep
- Soft foot
- Pipe strain
- Thermal activity

Waveform

- Periodic sinusoidal
- One or two dominant events with up to eight visible events

Spectrum

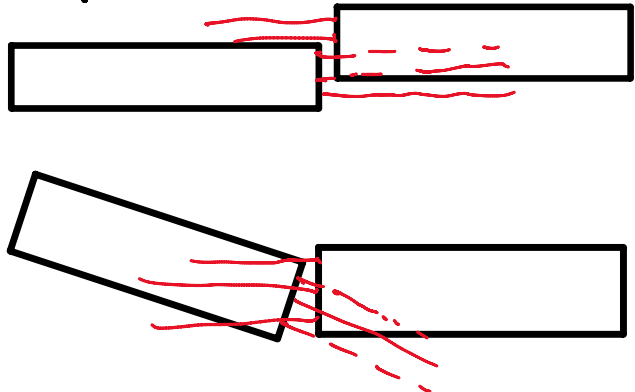
- Increase amp of 1x, 2x peaks
- Harmonics up to 6x or 8x
- Locked or damaged coupling could be 3x
- If 2x peak is 50% greater than 1x, misalignment is present



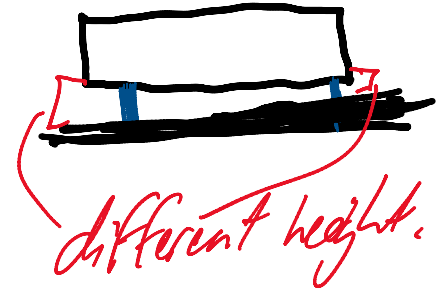
Offset: parallel offset

Angular: lines intersect

Combination: more common to see combo of both above



Soft Foot: supporting members of asset do not equally support it.



Resonance

Wednesday, March 24, 2021 06:14

Every piece of equipment has a natural frequency. If an external force is applied (bump or impact), the system will respond by vibrating at its natural frequency until force has dissipated.

The properties that govern natural frequency are mass, stiffness and damping.

Mass and stiffness determine natural frequency.

Damping absorbs energy and controls amplitude.

The entire structure of combined components, this will have a system natural frequency.

A bump test can be used to test for a natural frequency. Striking the component with a hammer and using an accelerometer to measure vibration.

Resonance is the amplification of the amplitude by natural frequency.

Resonance is a condition where the periodic frequency of a forcing function matches a natural frequency of a component.

A vibrating frequency within 20% of natural frequency will cause resonance.

A forcing function (eg imbalance) can excite the natural frequency.

*Turbines must be run faster through their natural frequency to avoid excessive vibration.

Mass can be added to increase natural frequency. Stiffness can be added as well.

Rolling Element Bearings

Thursday, March 25, 2021 08:43

A rolling element (anti-friction) bearing defect is an imperfection on one or more contact surfaces of the bearing. This is usually due to improper lubrication or loading. Some causes:

- 43% improper lubrication
- 27% improper mounting
- 21% Other sources (improper application, manufacturing defects)
- 9% Normal life expectancy
- Imbalance and misalignment account for 90% of premature failure
- 10% of all bearings are defective before installation (storage)

Waveform

- Impacting
- Excessive g swing

Spectrum

- Non-sync harmonic family
- Harmonics noticeable at higher freq in early stages
- Excessive broad band energy
- Sidebands often seen around harmonics

The best indicator of defect is the presence of high freq, non-sync peaks in the spectrum. The best indicator of severity is the waveform.

The worse the defect gets, the more non sync peaks will appear in the spectrum closer to the fundamental bearing defect freq.

Once the bearing fails, the harmonics may be replaced by broadband energy.

BPFI (Ball pass Frequency Inner) is the highest of the four calculated bearing defect freqs

FTF (Fundamental Train Freq) is sub sync and the lowest.

The waveform Pk to Pk is the most helpful.

BPFO - Ball Pass Frequency Outer Race

Number of rollers that come in contact with single location on outer race in one rotation

BPFI - Ball Pass Frequency Inner Race

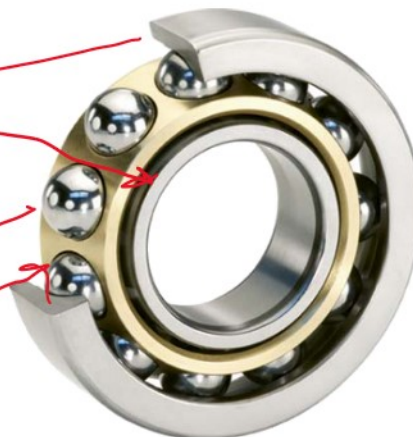
Number of rollers that come in contact with a single location on inner race in one rotation

BSF - Ball Spin Frequency

Number of rotations made by roller during one rotation

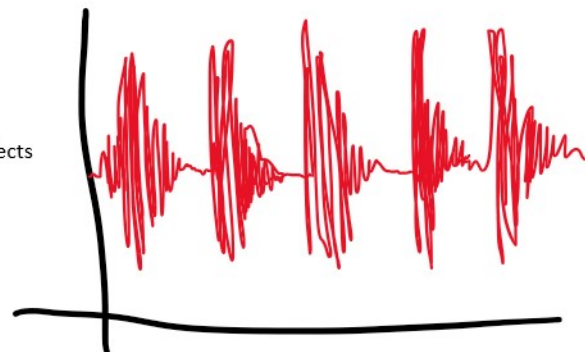
FTF - Fundamental Train Frequency

Number of cage rotations



Bearing components usually fail in the following order:

- Race defects -> ball/roller defects -> cage defects
- Inner race defects and failures occur at much lower amplitudes than outer race defects
- Early faults generate defect freqs and harmonics for one of the races.
- A defect will show as a school of angel fish swimming in the right direction.



Data Analysis

Tuesday, March 23, 2021 11:31

Crest Factor: tells you if individual peak values are above average peak values. A large crest factor can mean an anomaly in the data.

The waveform data is the source used to generate the Spectral and Trend data.

Length of waveform is important. It should be long enough to include 10 - 15 shaft rotations.

Time Waveform

Random vs Periodic

Periodic appear as repetitive cycles, may occur at any frequency. Easiest to be recognized

Random vibrations will appear as events that do not have repetitive pattern. Appear once or twice in stored waveform

Sinusoidal: little to no impacting. One noticeable event per shaft rotation

Impacting: sharp spikes well above average RMS value. High crest factor. Indicates mechanical defects.

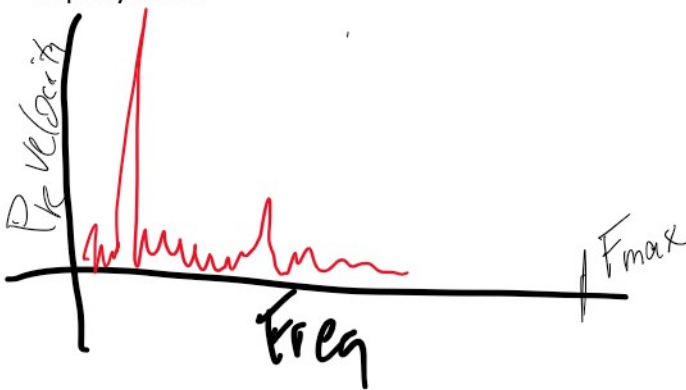
Asymmetrical: lack of symmetry between positive and negative. Bearing with higher degree of freedom in one direction vs the other.

Truncation: asymmetrical short cut-off peaks on one side and high peaks on the other. Rubbing or loose condition

Transient/random: once or twice in non-repetitive manner.

Modulation: increase/decrease in amplitude/frequency across waveform.

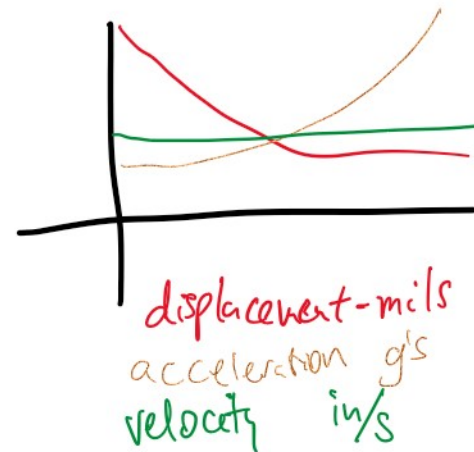
Frequency Domain



When viewing the spectrum display, observe the Amplitude units, plot scale and overall value.

The most common spectral amplitude unit is velocity

Due to velocity being consistent across the spectrum



Frequency Patterns: machinery fault conditions can cause different patterns. Pattern recognition is used to identify defects (imbalance, misalignment, looseness)

Calculated freqs: calculated from geometry:

Normal: vibration peaks that appear during normal running (such as gear mesh)

Defect (Fault): appear on when there is a problem

Fundamental: pre-calculated freq normally based on turning speed

Dominant Peak: high amplitude peak

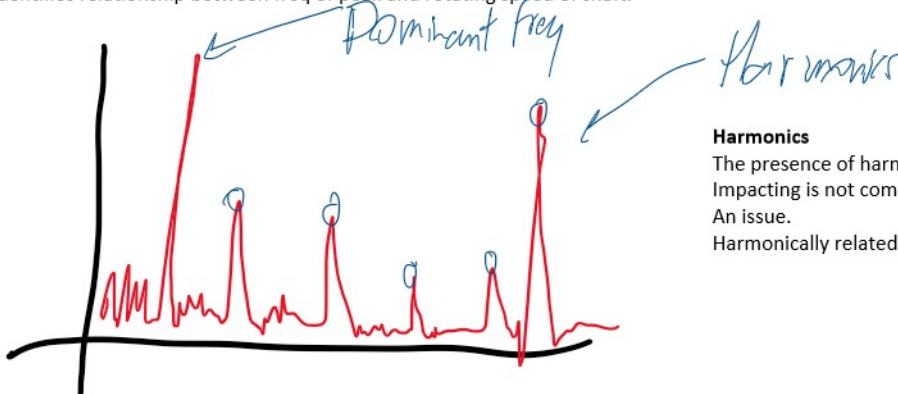
Harmonics: integer multiples of any frequency, caused by harmonic distortion

Sidebands: equally spaced peaks above and beyond a calculated frequency. Caused by modulation of carrier freq

Carrier freq: Vibration peak that changes in amplitude or freq. Center between sideband peaks.

Broadband energy: are in spectrum where peaks are so close they appear as single mound. Also raised noise floor.

Order: identifies relationship between freq of peak and rotating speed of shaft.



Harmonics

The presence of harmonics is an indicator of impacting. Since Impacting is not common for most industrial equip, it is likely an issue.

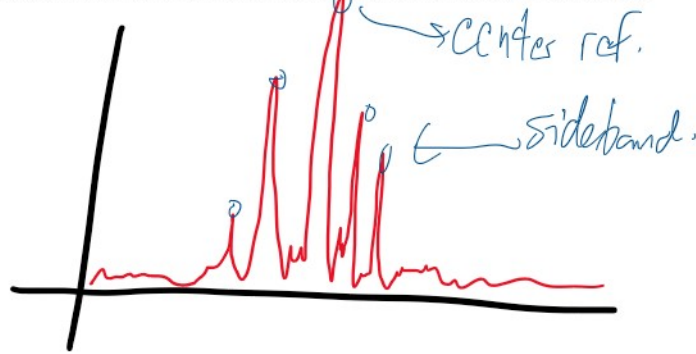
Harmonically related peaks are caused by the same defect.

Sidebands

The center frequency in a waveform is known as the carrier frequency. The carrier freq is usually modulating in amplitude. When applying FFT, the carrier freq will appear as a single peak with one or more sideband peaks. If sidebands are higher than center peak, there is a big issue.

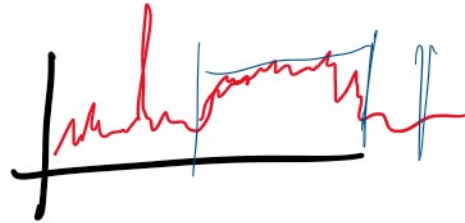


more sideband peaks. If sidebands are higher than center peak, there is a big issue.



Broadband Energy

Mound of peaks caused by random non-periodic impacting in waveform
Appears in localized area.



The most important data for analysis is turning speed.

Energy is divided into three categories: synchronous, non-synchronous and subsynchronous

Synch: spectral peaks at $1 \times TS$

- Imbalance (ALWAYS $1 \times TS$)
- Pitch line run out
- Bent shaft
- Misalignment
- Looseness
- Blade freq. depends on number of blades

Non-synch: spectral peaks above TS, not whole number multiples

- Vibration from another component or machine
- Higher multiples of belt freq
- Rolling element bearing defects
- System resonances
- Electrical faults

Subsynch: lower than TS

- Vibration from another component in machine rotates at slower speed
- Drive belt
- Hydraulic instability
- Shaft rub
- Cage freq of anti-friction bearings.

The overall value can be useful in determining the condition of the machine. Since it represents an average, it gives little indication of whether it is a mechanical problem or where the source is.

The frequency of the vibration indicates the cause/nature of the problem, while amplitude indicates the severity.

Analysis process

1. Locate the peak that is equal to the turning speed. If there is not enough vibration to cause a peak, a strobe light can be used to determine $1 \times TS$.
2. With $1 \times TS$ marked, set RPM. Once RPM is defined, analysis can be continued.
3. Determine subsynch, sync and nonsynch components
4. Identify harmonic relationships within spectrum. Harmonics are an indication of impacting, while the spacing between them represents the source of the forcing function. Sidebands will indicate modulation, with the spacing between center and sidebands lead to the source of the modulation.
5. Determine the severity. Load, speed and design have an effect on this. Comparing to a baseline or other similar machine is useful.

Data Collection

Tuesday, March 23, 2021 06:05

A vibration transducer is also known as a probe or a sensor

There are different types of transducers for each measurement type, the most common sensitivities are:

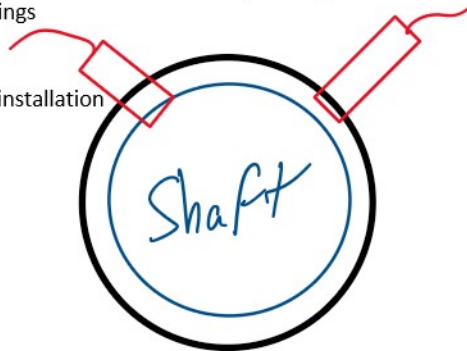
- Displacement: 200 mv/mil
- Velocity: 500 mv/ips
- Accelerometers: 100 mv/g

Displacement Probe

Measures actual shaft movement relative to tip of the probe

Ideal for journal bearings

Requires permanent installation



Velocity Probe

Most common is the seismic transducer.

Made up of a coil suspended on a spring between magnets. The magnet moves within the coil generating an analog signal

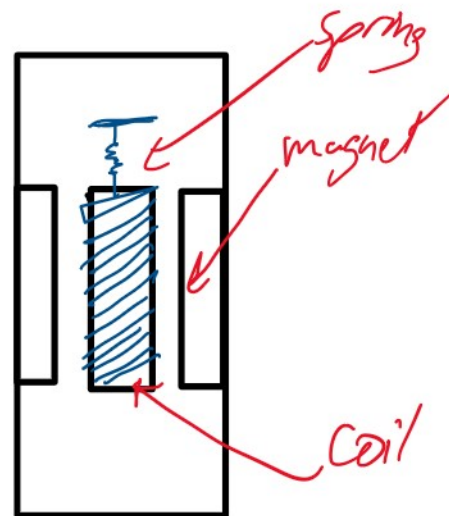
Gravity will affect the output and must be considered when applying.

Excellent signal to noise ratio, produces own output signal (no power supply)

Very large and rugged.

Excessive external temps affect response

Must be oriented horizontally to obtain best results



Accelerometers

Use piezoelectric properties to measure acceleration.

Standard is 100mV/g

Best for use when high forces are involved with low displacement/velocity (gears)

Measures force as a voltage, which is converted to g's

Most common transducer used.

Possess a broad frequency range from 1 Hz to 30 kHz and higher

Very small and lightweight

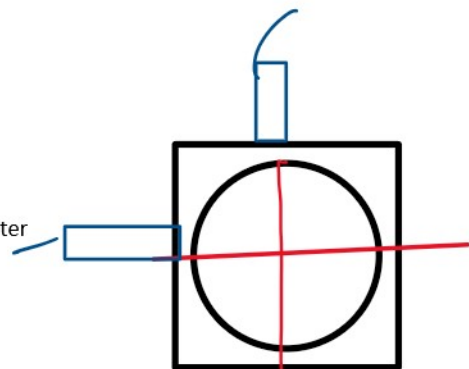
Max temp is 250

Requires external power supply provided by data collector

Stud mount provides best linear response

Transducer Location

Transmission path is the path between sensor and sensed. The closer the better



Fundamentals of Vibration

Tuesday, February 2, 2021 3:25 PM

Preventive Maintenance:

- Routinely Scheduled
- Testing
- Measurements
- Risk factor: failures between schedules
- Risk factor: human error during activities

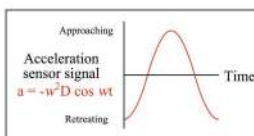
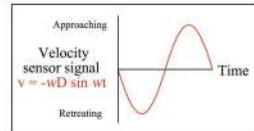
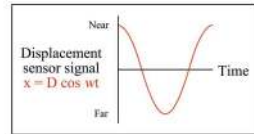
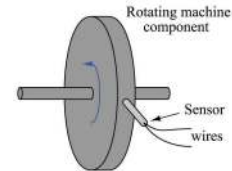
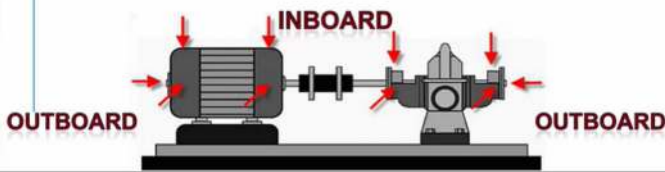
Labeling Measurement Points

Motor

1. MOH – Motor Outboard Horizontal
2. MOP – Motor Outboard Peakvue
3. MOV – Motor Outboard Vertical
4. *MOA – Motor Outboard Axial
5. MIH – Motor Inboard Horizontal
6. MIP – Motor Inboard Peakvue
7. MIV – Motor Inboard Vertical
8. *MIA – Motor Inboard Axial

Pump

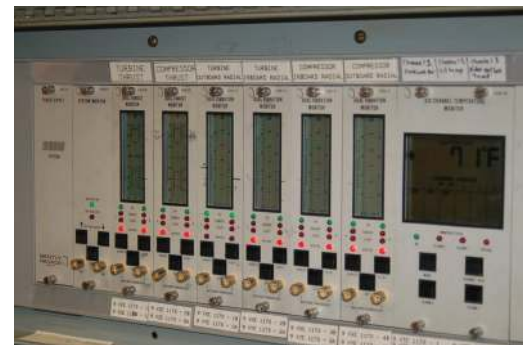
1. POH – Pump Outboard Horizontal
2. POP – Pump Outboard Peakvue
3. POV – Pump Outboard Vertical
4. *POA – Pump Outboard Axial
5. PIH – Pump Inboard Horizontal
6. PIP – Pump Inboard Peakvue
7. PIV – Pump Inboard Vertical
8. *PIA – Pump Inboard Axial



Helpful Hints

Use your **natural** data collectors – take particular note of:

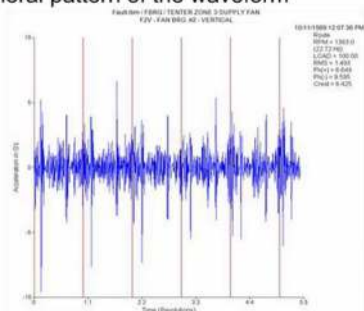
- Overall Appearance
- Condition of foundation – Grout, mounting bolts
- Structural integrity
- Welds
- Existing Leaks – liquid, steam, powder, lubrication
- Instrumentation – operating parameters such as pressure, temperature, flow, etc.



How do we identify the cause of a fault from a waveform?

- The waveform is an excellent way to determine the severity of the present vibration
 - By looking at the general pattern of the waveform

- Erratic?
- Sinusoidal?
- Non-sinusoidal?



Analysis Parameter Sets

AP sets determine how the analyzer **acquires** and **stores data**. Analysis Parameter sets define the following information for the analyzer:

Spectral bands	Special Time Waveform
Maximum frequency (Fmax)	Windowing
Resolution	Low Frequency Cutoff
Averaging methods	# Averages
Signal processing	Filters

Synchronous - Causes

Lower Multiplies of Turning Speed (N orders <8)

- Unbalance
- Pitch Line Run-Out
- Misalignment
- Bent Shaft
- Looseness
- Vane Pass

Synchronous - Causes

Higher Multiplies of Turning Speed (N orders > 8)

- Gear Mesh
- Blade Pass
- Several Motor Frequencies

Sub-synchronous - Causes

Frequencies that occur below 1XTS (Shaft speed)

- Other components in machine that are rotating at a lower speed
- Another machine in the area
- Machines with belts have a primary belt frequency and often a 2 X Belt Frequency
- Hydraulic Instability such as oil whirl and oil whip
- Motor Rub, shaft rub, compressor wheel rub
- Cage Frequency (FTF) of anti-friction bearings.

Non-Synchronous - Causes

Frequencies that are non-integer multiples of Turning Speed (Must be greater than turning speed)

- Component on an adjacent machine
- Multiples of belt frequencies
- Rolling element bearing defects
- Electrical
- Lube pumps
- Sliding surfaces

Gear defects

Thursday, March 25, 2021 09:17

Action of two gears rotating so they share a common tangent during rotation.
Gear boxes are used to reverse shaft rotation, speed up/slow down, change angle.

Misaligned gears, chipped/broken teeth, stress fractures and worn gears are all common:

- Improper lubrication
- Bearing failure
- Water intrusion
- Overheating

Spectrum

- Peak at Gear mesh freq (GMF)
- Harmonics at gear mesh freq with turning speed sidebands

GMF = number of teeth x turning speed

Gear mesh is expressed as CPM, not RPM!

Waveform

- Impacting is normal
- Once per revolution pulses with cracked/broken tooth

In most cases it is best to replace the entire gear set, not just the faulty gear. Bearings should also be replaced.

Gear Signatures

GMF: appears regardless of condition, amplitude changes with load

Sidebands: indicate which gear is bad by spacing between peaks

Gear Natural Freq: excited by defects

Sidebands around gear mesh freq will be spaced equal to the turning speed of the shaft that the bad gear is on.

Gear mesh is sync energy

An increase in amplitude of gear mesh sidebands indicate problems

Gear misalignment shows as 2x GMF

Imbalance

Wednesday, March 24, 2021 08:30

Imbalance is when the center of rotation differs from the center of mass.

This results in displacement of the support components, eg bearings

Causes:

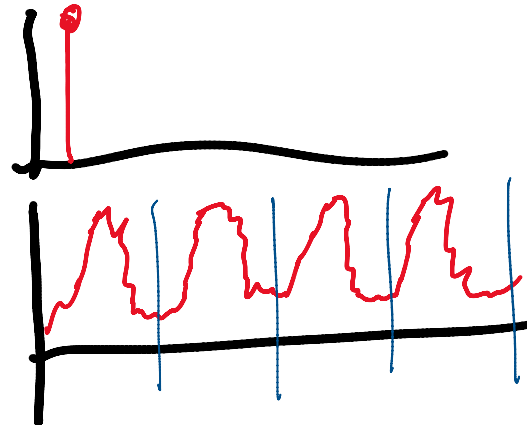
- Material build up
- Wear
- Broken/missing parts
- Improper assembly
- Thermal distortions

Spectral Characteristics

- High amplitude at $1 \times TS$
- Little to no harmonics of $1 \times TS$

Waveform characteristics

- Periodic sinusoidal
- One dominant event per rotation
- Little to no impacting



Directional in Nature, Usually horizontal.

Amplitude usually higher in direction where machine has freedom of movement

Centrifugal force is affected by speed, and imbalance will increase with speed.

Imbalance is synchronous energy as it occurs at $1 \times TS$. It is not the only forcing function that results at $1 \times TS$ though

Journal Bearings

Thursday, March 25, 2021 08:26

Excessive clearance, improper load and improper lubrication can cause high vibration levels

Excessive clearance allows small excitation force.

Dominant frequency can occur at 1x, 2x, 3x.

Data should be collected at radial and axial.

Radial -> info on plain bearings

Vertical -> excessive clearance

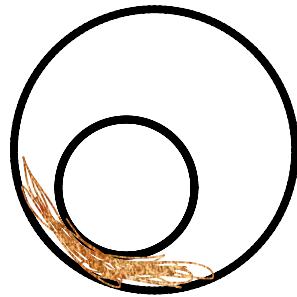
Axial -> thrust bearings

The larger the clearance, the larger the amplitude of the harmonics

Oil whirl = lubrication instability where shaft moves in oil around inner clearance of bearing.

Oil whirl typically shows up in vibration spectrum between .42 and .48 orders

Sleeve bearing designs minimize the effect of oil whirl, by forcing the oil and shaft to move around the bearing at different rates.



Looseness

Wednesday, March 24, 2021 11:58

Mechanical looseness is the condition in which a rotation element's fasteners are no longer held fast in place

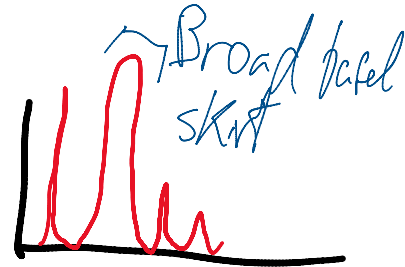
- Base mount
- Split casings
- Bearing caps/supports
- Impellers
- Fans
- Bearings
- Couplings

Waveform

- Non-periodic, non-sinusoidal

Spectrum

- Turning speed harmonics (in the early stages, once it loosens more this will not be present)
- Occasional fractional harmonics
- Broad based skirts on spectral peaks
- Possible elevated noise floor (more in later stages)



NETWORK

BOOTP

Monday, June 06, 2016 3:47 PM

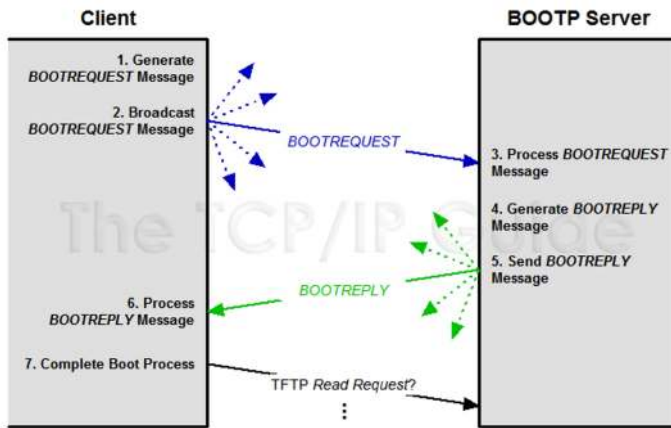
http://www.tcpipguide.com/free/t_BOOTPDetailedOperation.htm
<http://www.freebsd.org/cgi/man.cgi?query=bootpd&sektion=8>

Used to configure devices that have no internal storage.
Client/server nature, single exchange of message between client and server.
Client is device to be configured, server is network device used to respond to requests.
Server contains table with MAC - IP

Uses UDP transport layer over port 67. Server listens on this port for client broadcasts.

Client knows it's address, reply can be sent directly

Client doesn't know address: server will assign it and add to ARP table.



Message format:

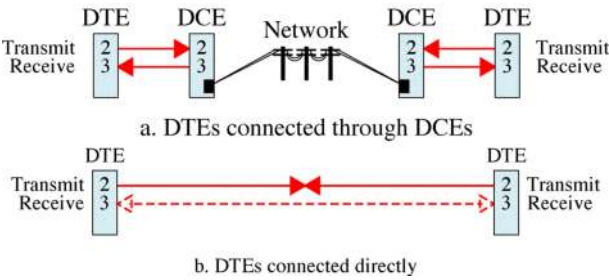
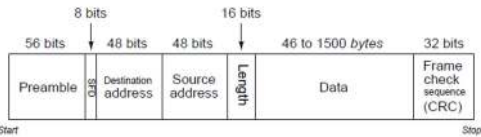
http://www.tcpipguide.com/free/t_BOOTPMessageFormat.htm#Table_186

Vendor specific info field used to pass parameters

Ethernet

Thursday, April 14, 2016 2:38 PM

- Standards under 802.3
- 802.3a-1985 10BASE2 "thin" Ethernet
- 802.3d-1987 FOIRL fiber-optic link
- 802.3i-1990 10BASE-T twisted-pair cable Ethernet
- 802.3u-1995 100BASE-T "Fast" Ethernet and Auto-Negotiation
- 802.3x-1997 Full-Duplex standard
- 802.3ab-1999 1000BASE-T "Gigabit" Ethernet over twisted-pair cable

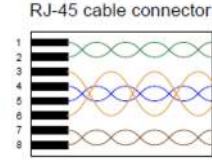


DTE: Data terminal equipment (serial port or terminal connection)
DCE: Data communication equipment (modem, data converter)

MAC address
48-bit binary, divided into 6 bytes, written as hexa 00:00:00:00:00:00

Repeaters (hubs) (Layer 1)
Coaxial cables connect devices passively. A hub is an active device that re-broadcasts a signal. A repeating hub can broadcast a signal to all ports. s.

Cabling
UTP wiring and RJ45 connectors.



10 Mbps

Pin number	Assignment	Abbreviation
1	Transmit Data (+)	TD+
2	Transmit Data (-)	TD-
3	Receive Data (+)	RD+
4	(not used)	
5	(not used)	
6	Receive Data (-)	RD-
7	(not used)	
8	(not used)	

100 Mbps

Pin number	Assignment	Abbreviation
1	Pair "A" (+)	BLDA+
2	Pair "A" (-)	BLDA-
3	Pair "B" (+)	BLDB+
4	Pair "C" (+)	BLDC+
5	Pair "C" (-)	BLDC-
6	Pair "B" (-)	BLDB-
7	Pair "D" (+)	BLDD+
8	Pair "D" (-)	BLDD-

Switch (Layer 2)
Hub that can route signal to desired ports

DHCP

Thursday, November 05, 2015 10:54 AM

Provides IP host with it's IP address, subnet mask and default gateway.
Based on BOOTP, when bootp worked for static IPs, changing IP addresses became an issue.
Client/server based protocol

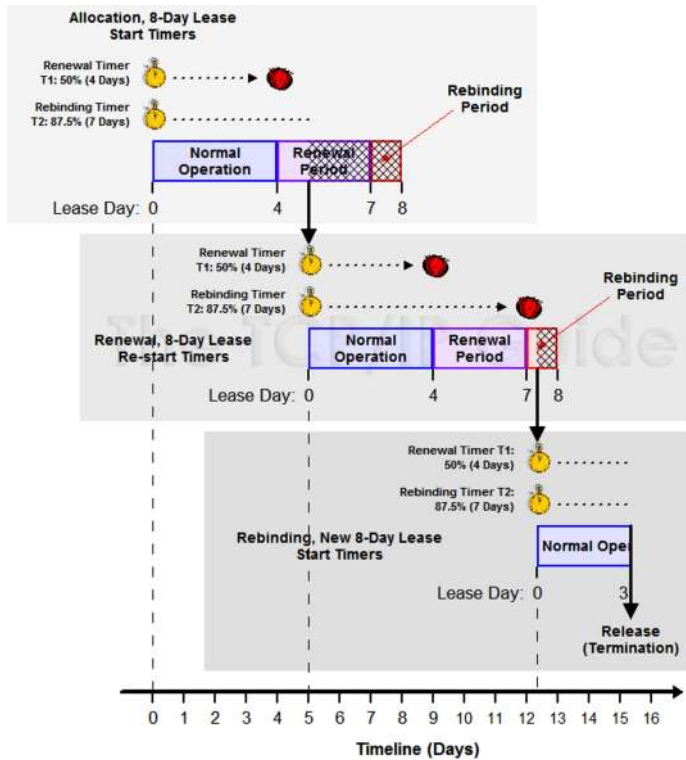
Address allocation

Manual: single device, administrator pre-allocates

Auto: address automatically selected from pool and assigned

Dynamic: address automatically assigned for limited period of time, until device no longer is using it. address *leasing*

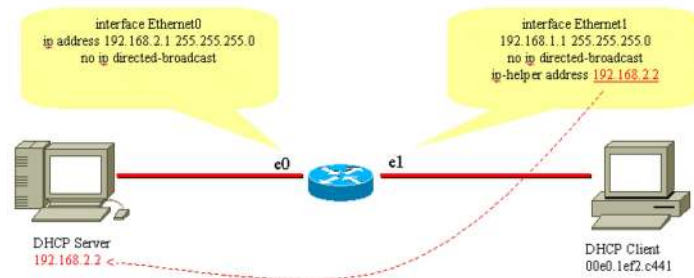
LEASING



Each time client powers up, it must confirm it contains lease of address.

DHCP server contains a pool of addresses considering the following:

- number of clients
- Stability and frequency of use per client

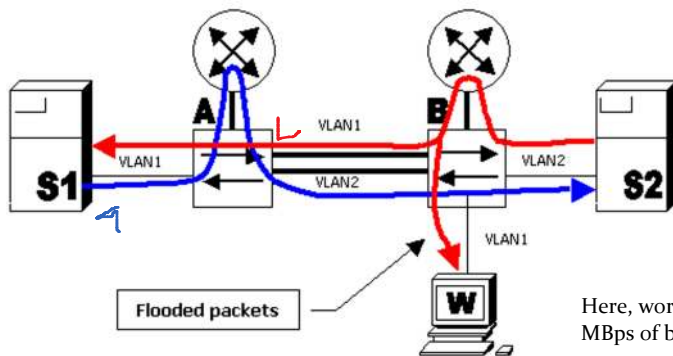


Packet flooding

Thursday, November 05, 2015 10:18 AM

If there is no destination MAC address in the incoming VLAN (L2 forwarding table), the frame will be sent to ALL forwarding ports in the VLAN.

Asymmetric routing



Packets follow different paths depending on their direction

To detect this, look at capture of packets. If unicast packets are repeatedly seen on port (and don't involve connected workstation), flooding is probably occurring.

Here, workstation W receives all comms between S1 and S2. If the backup is taking up 50 MBps of bandwidth, and the comp is limited to 10MBps, it will disconnect.

Another cause can be spanning tree protocol Topology changes. The forwarding table ages and if the address is not relearned, flooding will occur.

Forwarding Table Overflow
Rare, as most modern switches have large enough tables.

ASA Routers

Tuesday, May 8, 2018 7:22 PM

Cisco ASA 5515

ASA routers default to block traffic and secure.
Defaults:

Physical interface: Disabled
Speed/Duplex: Auto-negotiate
VLAN: SKIP

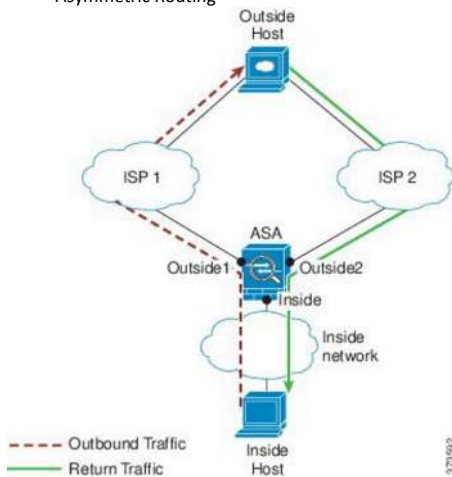
Security Level: 0 lowest, 100 highest
Network access: permit higher to lower traffic

Firewall modes: Transparent and Routed
Transparent mode has no security level
Routed mode has security level

Context mode:

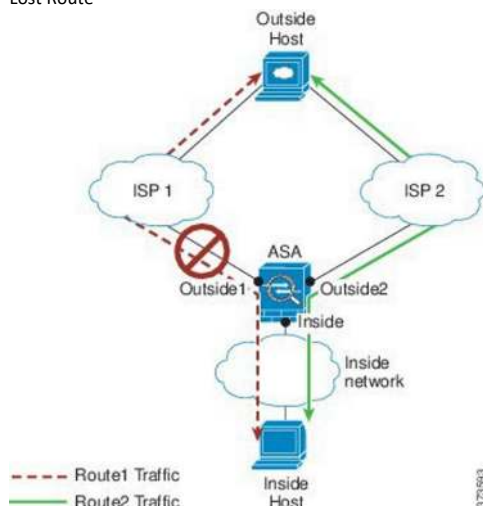
Traffic Zone:

- Asymmetric Routing



Connection made with ISP1 through one interface and returns through ISP 2. Normally would get dropped. Zoned solution makes it so both interfaces act in same zone and accept connection.

- Lost Route

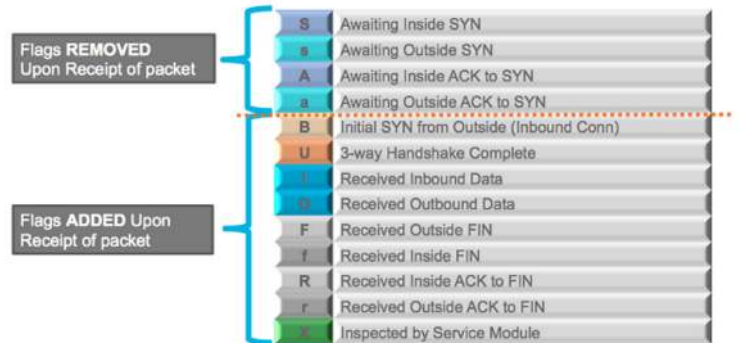
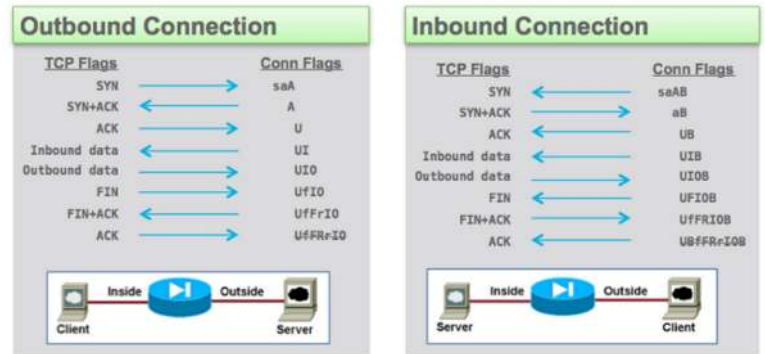


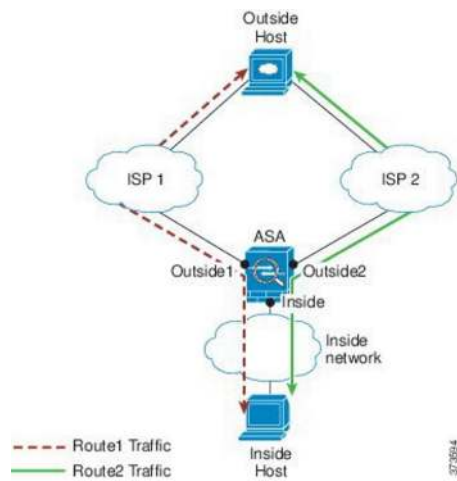
Lost route due to broken cable to ISP1. UDP would re-establish automatically through ISP 2, but TCP needs a reconnect. The zone will detect lost route and automatically connect through ISP 2.

- Load Balancing

TCP Connection Flags:
Connection

Show conn protocol tcp shows all active TCP connections and states





Load balancing across interfaces is only possible with zoning.

Firewall

ASA by default allows traffic from inside network (higher sec lvl) to outside network (lower sec lvl)
Access rules are then created to allow outside to inside traffic and limit inside to outside traffic.

Objects: reusable components in config. Similar to macros

- Network object: **object network nameOfObject hostIP subnetIP range hostname**
- Service object: **object service nameOfObject nameORnumberOfProtocol(IP specifies all protocols) service(TCP/UDP) sourcePort DestPort operator (lt, gt, eq, neq, range)**
- User Groups: specify users to which rules apply

Lt = less than
Gt = greater than
Eq = equal
Neq = not equal
Range = range of ports

ACLs:

Implicit Permits:

Routed Mode: ipv4 & ipv6 from higher sec lvl to lower sec lvl

Transparent Mode: ipv4/ipv6 from higher to lower and bi-directional ARP / BPDU

Implicit deny: All ACLs have an implicit deny at the end. If a type of traffic is not permitted it will be dropped.

For TCP and UDP connections for both routed and transparent mode, you do not need an access rule to allow returning traffic because the ASA allows all returning traffic for established, bidirectional connections. For connectionless protocols such as ICMP, however, the ASA establishes unidirectional sessions, so you either need access rules to allow ICMP in both directions (by applying ACLs to the source and destination interfaces), or

- Inbound & Outbound
- Extended: main type, used to permit/deny traffic through device.

Access Control Entry Order: ACLs follow logical top to bottom order. Allows on top of denies take precedence.

IRIG (Time Sync)

Friday, February 11, 2022 1:39 PM

Inter-range instrumentation group timecodes created by the U.S. military's Inter-Range Instrumentation Group

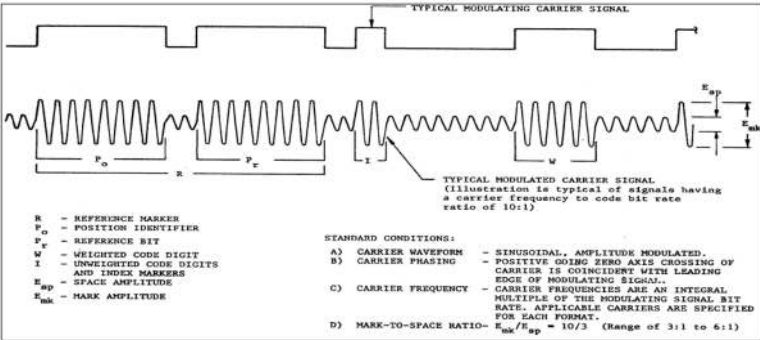
TABLE 3-1. BIT RATES AND INDEX COUNT INTERVALS OF THE TIME CODE FORMATS		
Format	Bit Rate ¹	Index Count Interval
A	1 kpps	1 millisecond
B	100 pps	10 milliseconds
D	1 ppm	1 minute
E	10 pps	0.1 second
G	10 kpps	0.1 millisecond
H	1 pps	1 second

¹ See the abbreviations and acronyms page at the beginning of this document for bit rate definitions

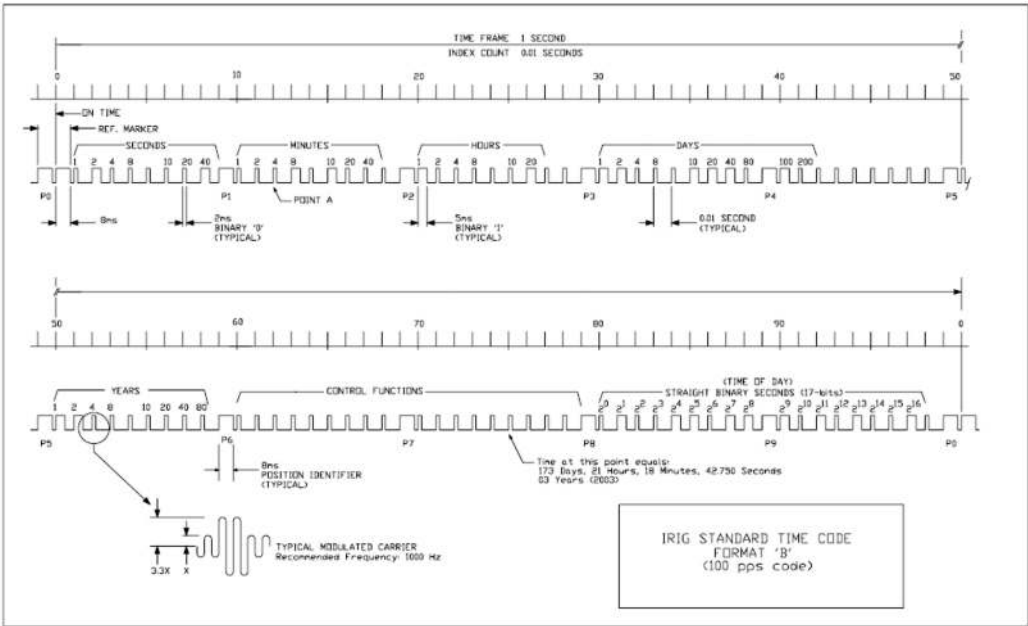
TABLE 3-5. TYPICAL MODULATED CARRIER SIGNAL FORMATS FOR A, B, E, D, G, AND H								
FORMATS					MARK INTERVAL NUMBER OF CYCLES			
FORMAT	SIGNAL NO.	TIME FRAME RATE	CARRIER FREQUENCY F	SIGNAL BIT RATE ER	RATIO F/ER	CODE "0" & INDEX	CODE "1"	POSITION IDENTIFIER & REF.
A	A130, 132, 133, 134	10 per sec.	10 kHz	1 kpps	10:1	2	5	8
B	B120, 122, 123, 127	1 per sec.	1 kHz	100 pps	10:1	2	5	8
D	D111, 112, 121, 122	1 per hr.	100Hz	1 ppm	6000:1 60000:1	1200 12000	3000 30000	4800 48000
E	E111, 112, 121, 122, 125	6 per min.	100Hz	10 pps	10:1 100:1	2 20	5 50	8 80
G	G141, 142, 126	100 per sec.	100 kHz	10 kpps	10:1	2	5	8
H	H111, 112, 121, 122	1 per min.	100 Hz	1 pps	100:1 1000:1	20 200	50 500	80 800

Pulse Rise Time: dc level shift bit, between 10 - 90%
Jitter: pulse to pulse variation at 50% amplitude
Bit Rates and Index Count: Bit rates are repetition of "on" bits. Index count is time interval between leading edge of two consecutive bits.

- Time Code Words
- Binary Coded Decimal (BCD): time of year and year
 - o 0h,0m, 0s, 0ps at 24:00 hours
 - o Counts year cycles up to 2099
 - Straight Binary Seconds (SBS) time of day (seconds of day)
- Index Markers:** Occur at each index count position



- IRIG - Time Format B:
- 74-bit with:
 - o 30 bits of BCD
 - o 17 of SBS
 - o 18 bits for control functions



- Leap Year
- Uses Gregorian calendar, the average year is 365.2425 days in length.
 - Years divisible by 4 but not 400

DATALINKS

Serial Comms

Monday, April 11, 2016 1:47 PM

Non return to zero encoding **Manchester encoding:** square signal, counting up and down transitions to represent states
-RS232 -Fieldbus
1 = -3 V -Profibus
0 = +3 V -10 Mbps ethernet

Frequency Shift Keying: sine waves at different frequencies indicate states. two cycles at 2200 represent 0, one cycle at 1200 is 1.

Master-Slave
Master is the only one with permission to transmit data. Slaves must respond to query from master. A slave device with data must wait until polled. Death of master means death of network
-MODBUS
-HART

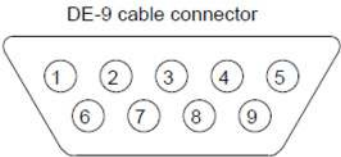
TDMA
Time division multiple access. Each device is assigned a time slot to be master. Any device on network can fail without affecting the whole. However, they still depend on a kind of master to assign time slots.
-WirelessHart
-ISA100.11a radio instrumentation
-GSM cell phone network

If bit rate differs from sending device to receiving device, a totally different message could be sent. Baud rate is also used

Parity checking is used to make sure the entire message is sent to the receiving side, and to advise if it is corrupted.

Token Passing
Token acts as temporary authorization to transmit. Any device with token is a master device. After sending out its data, it passes on token
-IEEE 802.5

CSMA
Carrier Sense Multiple Access. No dedicated master/slave, no predetermined order or schedule. Any device may talk at any time when network is free. Simultaneous communications are handled with collision detection, which causes both deviecs to cease tranmission and re-establish after a time delay.
Bit-wise arbitration is also employed to give certain ranks.
Deterministic nature, as each device broadcasts the approx time it will take relaying message.



Pin number	Assignment	Abbreviation
1	Carrier Detect	CD
2	Received Data	RD
3	Transmitted Data	TD
4	Data Terminal Ready	DTR
5	Signal Ground	Gnd
6	Data Set Ready	DSR
7	Request To Send	RTS
8	Clear To Send	CTS
9	Ring Indicator	RI

DNP3

Wednesday, April 13, 2016 5:17 PM

https://www.trihedral.com/help/Content/D_Tags/Dev_DNP3Addressing.htm

Master polls *outstations* for data. Both Master and outstation have complete databases. Array elements known as *indexes*.

Objects are the encoded bits of each index in a message. Data types are grouped in object groups:

Variations: further classification by action or specific data type (Binary Input change with time, 16 Bit Analog Inputs)

Variation 0: request for any variation, wildcard.

Class: Priority level:

Class 0: static data , will be sent every time value changes.

Class 1, 2, 3: Priority for event objects. High (1) to low (3)

Polling:

Integrity: collects all static data with present values

Class/Event: only collects flagged values (important changes)

Unsolicited response: set at slave, can cause collision as it takes priority over event/integrity polls.

Event Buffer:

Queries and received events are stored in slave memory. Relay device polling is async, and goes to the buffer, to be emptied by the master.

Sequence of event: Stores all received data in buffer, if buffer full, new data is discarded. Used by counter and binary.

Most recent event: Each point has a space and when a new value comes in it is overwritten. Used by analog.
First come first serve.

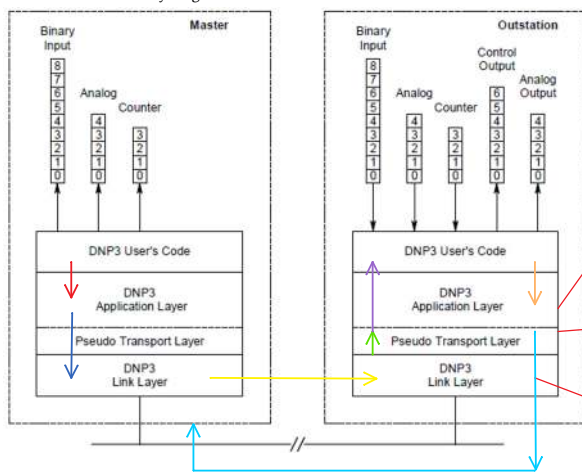
Every communication begins with a Link Reset from the Primary (Master)
Lower priority data can be sent without an ACK.

Message Format

Object:Index:Variation

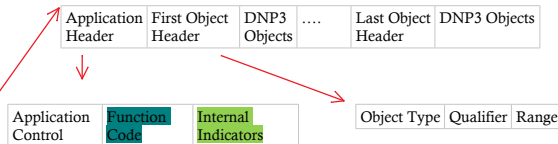
1	Binary Input
2	Binary Input Change
10	Binary Output
20	Binary Counter
30	Analog Inputs
40	Analog Output Status
41	Analog Output Block
50	Time and Date
60	Class data
70	File ID
80	Internal Indicators
100	Floating Point

Packet contents with Layering



DNP3 message path

Masters user layer requests application layer what function to perform. Then, goes down to link layer - send to outstation link layer, which checks for errors. Goes up to transport/application layer where message is assembled. Application layer indicates which groups/variations were requested and passes it on to users layer.
Outstations users layer fetches data and presents it to application layer. Application layer uses this info to format user layer data into objects. Data passed down and back to Master.



Breaks down long application layer messages into smaller packets for link layer to
Also reassembles received broken down packets.
Requires only one octet, is part of application layer

DIR	PRM	FCB	FCV	Function Code
1-from Master 0 -from Slave	0-pri to sec 1-sec to pri	0	DFC	

Data types and variations

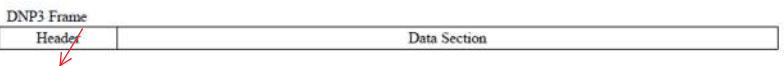
Objects are the encoded bits of each index in a message.

Internal indicators (Application layer)

Index	Error/Condition
IIN1.0	Broadcast received successfully
IIN1.1	Class 1 event data available
IIN1.2	Class 2 event data available
IIN1.3	Class 3 event data available
IIN1.4	Time sync required from Master
IIN1.5	Local mode
IIN1.6	Device trouble "Internal fail"
IIN1.7	Device restart
IIN2.0	Function unknown
IIN2.1	Object unknown
IIN2.2	Parameter error
IIN2.3	Buffer overflow
IIN2.4	Requested operation is already running
IIN2.5	Config corrupted
IIN2.6	Always 0
IIN2.7	Always 0

Function Code (Hex)

0 Confirm	10 Initialize application
1 Read	11 Start application
2 Write	12 Stop application
3 Select	13 Save configuration
4 Operate	14 Enable unsolicited
5 Dir operate	15 Disable unsolicited
6 Dir operate - No resp	16 Assign class
7 Freeze	17 Delay measurement
8 Freeze - No resp	18 Record current time
9 Freeze clear	19 Open file
A Freeze clear - No resp	1A Close file
B Freeze at time	1B Delete file
C Freeze at time - No resp	1C Get file information
D Cold restart	1D Authenticate file
E Warm restart	1E Abort file
F Initialize data	



Header	Sync	Length	Link Control	Destination Address	Source Address	CRC
--------	------	--------	--------------	---------------------	----------------	-----

Where frame begins 05 64 default # of octets remaining send/receive link layers 65520 addresses available

MODBUS

Tuesday, December 29, 2015 9:12 AM

Only master can initiate transactions
Slaves respond by supplying data to master..
MODBUS reads values by issuing request to read a range. All data is present value. No concept of events/time.

Address 0 is used as broadcast address to all slaves.
When responding, slaves put address in first frame.

The **function field** is used by the master to indicate what action to take, the slave uses it for error responses. Error free responds same function code. For the error response, most significant bit is set to 1.

The **Data field** contains extra data needed for the slave to process a request. This could be discrete and register address, quantity of items, etc. If there is an error, it will contain the code. It may also be empty.

Error checking can be LRC (ASCII mode) or CRC (RTU mode)

Data value ranges. As many as 65536 per range

Memory Block	Data type	Master	Slave
Coils	Bool	R/W	R/W
Discrete inputs	Bool	R	R/W
Holding register	UnsignedW	R/W	R/W
Input Register	UnsignedW	R	R/W

RTU

Each byte is two Hx characters
1 start bit, 8 data bits, CRC
Start with 3.5 second silence
Ends the same.
If 1.5 seconds pass, frame will be discarded.

Start	Address	Function	Data	CRC	End
3.5 char time	8 bit	8 bit	N*8 bit	16 bit	3.5 char time

ASCII

Each byte is sent as 2 ASCII (Hx) characters.
1 start bit, 7 data bits, LRC
Start with colon, (5A hex)
End with CRLF (oD, oA)

Start	Address	Function	Data	LRC	End
:	2 chars	2 chars	N chars	2 chars	CRLF

01 Read Coil Status (0X)

reads status of discrete outputs (ON/OFF) in slave.
No broadcast.
Coils start at coil 0

Query:
Specifies starting coil and number of coils to read
Response:
One coil per bit.
Each Data byte tells the status of 8 coils. If less than 8, it will pad with zeros.

02 Read Input Status (1X)

reads status of discrete inputs in slave
No broadcast.

Query:
Specifies starting input and number of inputs to read
Response:
One input per bit.
Each Data byte tells the status of 8 inputs. If less than 8, it will pad with zeros.

03 Read Holding Registers (4X)

Reads binary contents of holding registers
No broadcast.

Query:
Specifies starting register and number of registers to read
Response:
Two bytes per register.
First byte contains high order bits, second contains low order

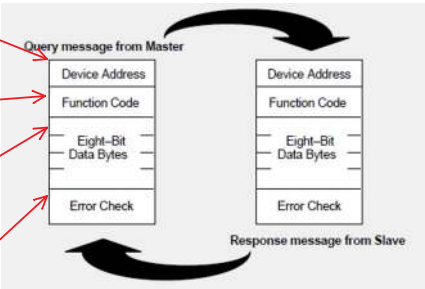
04 Read Input Registers (3X)

Reads binary contents of input registers
No broadcast.

Query:
Specifies starting register and number of registers to read
Response:
Two bytes per register.
First byte contains high order bits, second contains low order

MODBUS does not indicate how registers are sent, they could be high-byte first, low-byte first, signed, unsigned. That's why swap words/bytes is important.
Advantages: simplicity for small devices.

1 Master with up to 247 slaves. Master writes info on slaves
Typical transmission speed is 9600 baud (bits/s)



Parity Checking: applied to each character -> checks if number of 1 bits is even or odd
Frame checking: applied to entire message
Master generates and slave checks.
LRC is 1 byte
CRC is 2 bytes

16 bit un-signed = 0 to 65536
16 bit signed = -32767 to 32767
32 bit unsigned = 0 to 4294967295
32 bit signed = -2147483648 to 2147489648

Different byte, word ordering
AE41 5652 *High byte, high word first*
5652 AE41 *High byte, Low word first*
41AE 5256 *Low byte, high word first*
5256 41AE *Low byte, low word first*

TCP/IP

Wrapped in TCP/IP
Comms over port 502/503
Checksum not included

Modbus over TCP/IP

Checksum included in message

Transaction ID	Protocol ID	Message Length (bytes)	Unit ID	Function	Data
2 bytes	2 bytes	2 bytes	1 byte	1 byte	N chars

MSB: 0 is the normal response, 1 is for exception code

Modbus Exception Codes

Code	Exception	Description
01	Illegal Function	The function code received in the query is not allowed or invalid.
02	Illegal Data Address	The data address received in the query is not an allowable address for the slave or is invalid.
03	Illegal Data Value	A value contained in the query data field is not an allowable value for the slave or is invalid.
04	Slave/Server Device Failure	The server failed during execution. An unrecoverable error occurred while the slave/server was attempting to perform the requested action.
05	Acknowledge	The slave/server has accepted the request and is processing it, but a long duration of time is required to do so. This response is returned to prevent a timeout error from occurring in the master.
06	Slave/Server Device Busy	The slave is engaged in processing a long-duration program command. The master should retransmit the message later when the slave is free.
07	Negative Acknowledge	The slave cannot perform the program function received in the query. This code is returned for an unsuccessful programming request using function code 13 or 14 (codes not supported by this model). The master should request diagnostic information from the slave.
08	Memory Parity Error	The slave attempted to read extended memory, but detected a parity error in memory. The master can retry the request, but service may be required at the slave device.
0A	Gateway Problem	Gateway path(s) not available.
0B	Gateway Problem	The target device failed to respond (the gateway generates this exception).
FF	Extended Exception Response	The exception response PDU contains extended exception information. A subsequent 2 byte length field indicates the size in bytes of this function-code specific exception information.

15 (0F) Force Multiple Coils

Forces each coil to either ON/OFF
Sends starting address and number of coils
Returns slave address, function code, starting address and quantity of coils

16 (10) Preset Multiple Registers

16 bit values, data packed as 2 bytes
Returns slave address, function code, starting address and quantity of coils

17 (11) Report Slave ID

Returns description of type of controller, current status of slave
Run indicator

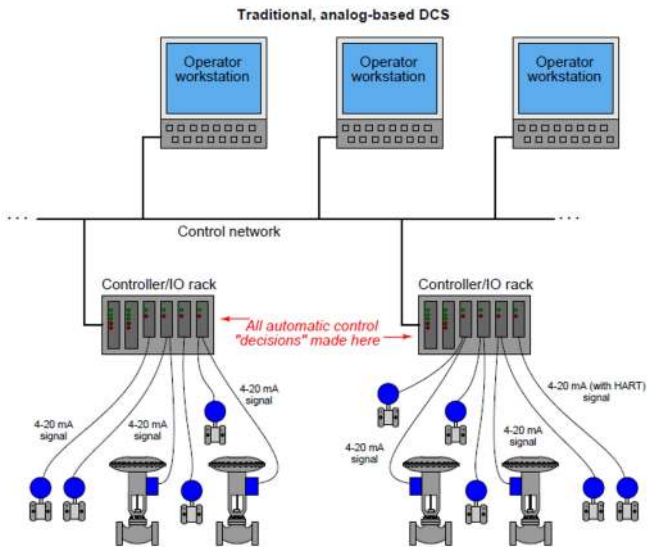
20 (15) Read General Reference

Returns contents of registers in extended memory File (6xxxxx)

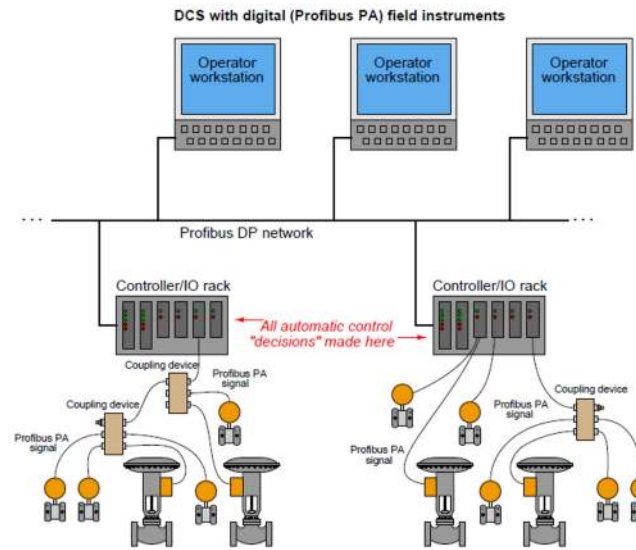
24 (18) Read FIFO Queue

reads FIFO queue of 4xxxx registers
Up to 32 registers can be read

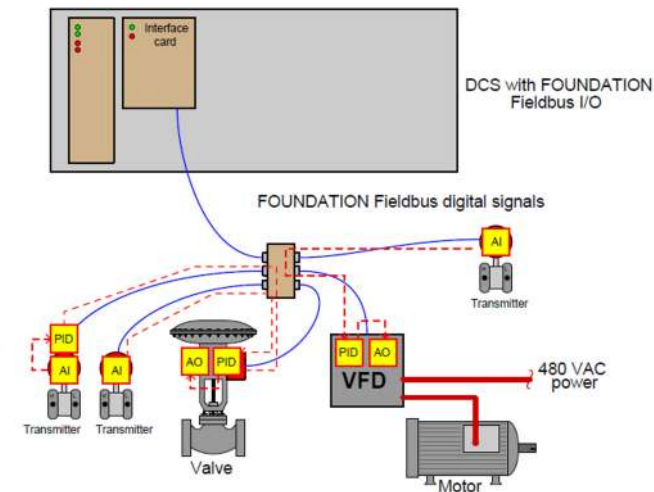
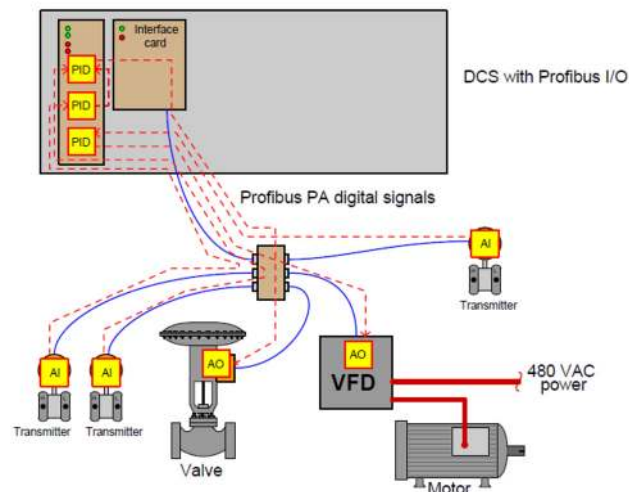
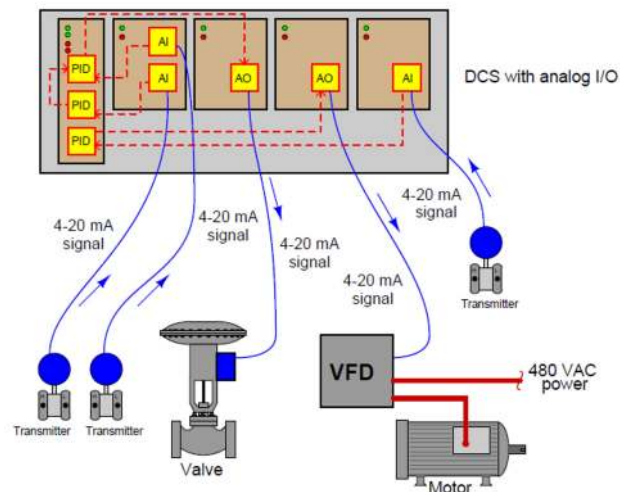
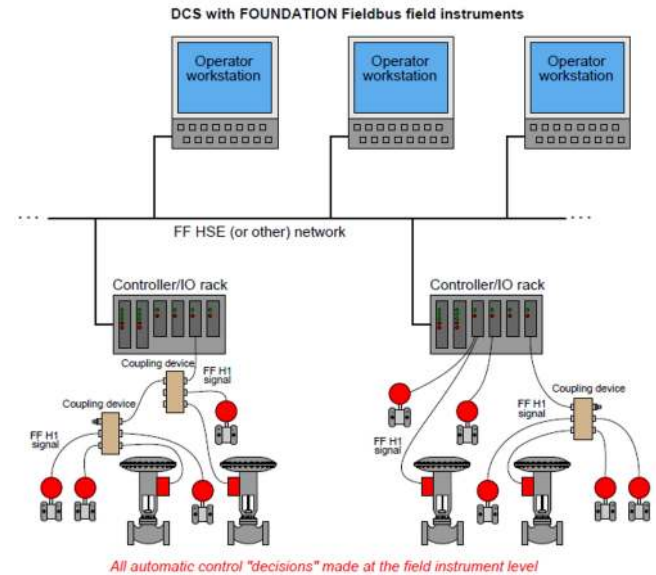
TRADITIONAL DCS



PROFIBUS PA



FOUNDATION FIELDBUS



Everything is Digital, all ADC/DAC is done in the device.
Far higher than HART speeds
Multiple instruments on common networks leading to DCS
All automatic control is done on DCS side

Everything is done on devices, DCS is only there for monitoring, historizing data, configuration loading.

Fieldbus

Friday, April 15, 2016 11:52 AM

Two network levels:

H1: low speed field, for devices.

H2: high speed plant, for workstations (replaced by HSE)

Physical layer

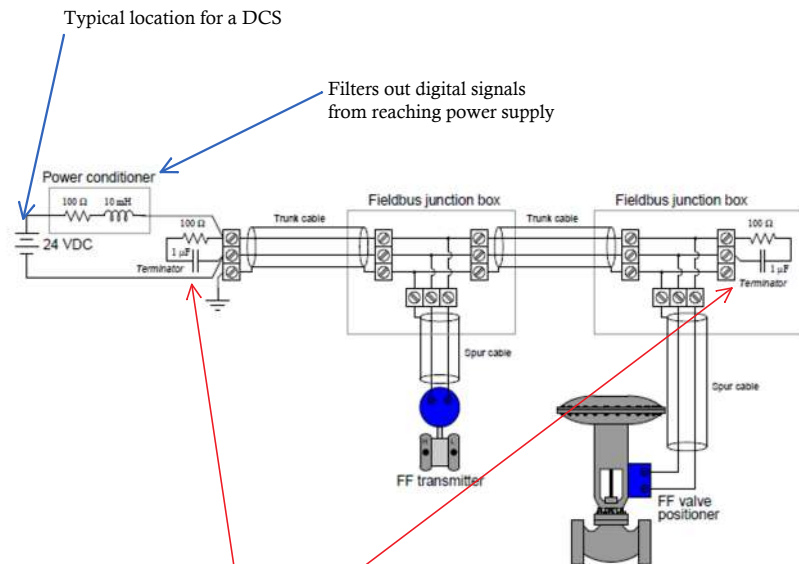
Two-wire ungrounded cable
100 ohm impedance
DC power over same data wires
31.25 kbps data rate
Differential voltage signaling
Manchester encoding

Segment:

DC power supply
Power conditioner
Two terminator resistors
shield twisted pair cable
two or more FF instruments

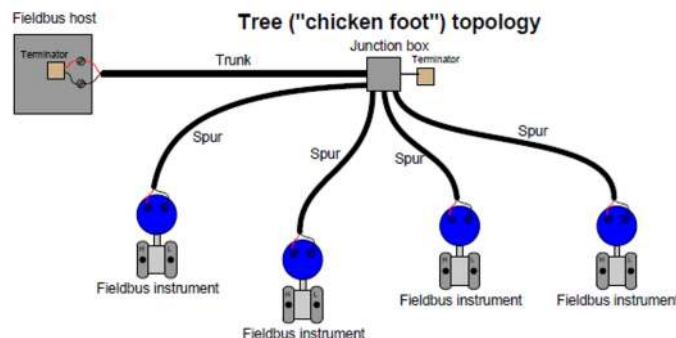
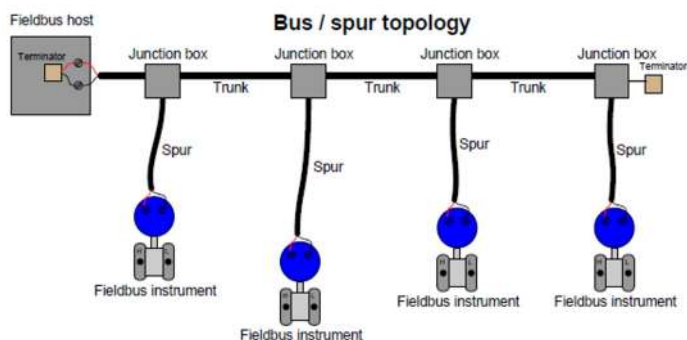
Bricks

Coupling devices are used for connecting multiple devices, contain plugs and sockets.
May be weak to RF interference.



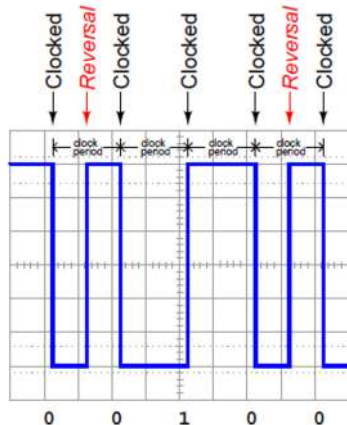
Terminating resistors,
No matter how many devices are connected, only two used.

Avoids issues with disconnected devices in daisy chaining



Electrical
Devices are polarity-insensitive.
Each device draws at least 10 mA of current.
Operating voltage range: 9 -32 VDC

Manchester encoding



Cable types

Cable Type	Type A	Type B	Type C	Type D
Wire size	AWG 18	AWG 22	AWG 26	AWG 16
Char. Impedance	100 Ω ± 20%	100 Ω ± 30%	—	—
Shielding	1 for each pair	1 for entire cable	none	none
Twisted pairs	Yes	Yes	Yes	No
Max. length	1900 m	1200 m	400 m	200 m

DATA LINK layer

Broadcast network. All transmission is sensed by all devices.
Master/slave for cyclic comms
Delegated token for acyclic
Dedicated scheduler for segment comms
8 bit address field
Max 32 devices on segment

Each segment has a Link Active Scheduler (LAS) to determine what data goes through

Addressing

Address range (decimal)	Address range (hexadecimal)	Allocation
0 through 15	00 through 0F	Reserved
16 through 247	10 through F7	Permanent devices
248 through 251	F8 through FB	New or decommissioned devices
252 through 255	FC through FF	Temporary ("visitor") devices

Unused addresses should be indicated to LAS to skip.

Delegated token for access

Dedicated scheduler for segment comms

8 bit address field

Max 32 devices on segment

Each segment has a Link Active Scheduler (LAS) to determine what data goes through when. Periods of time between transmissions are used for processing control.

Scheduled unscheduled communications

Virtual communication Relationships

Publisher/subscriber (scheduled): BNU

Client/server (unscheduled) QUB

Source/Sink (unscheduled) QUU

16 through 247	10 through F7	Permanent devices
248 through 251	F8 through FB	New or decommissioned devices
252 through 255	FC through FF	Temporary ("visitor") devices

Unused addresses should be indicated to LAS to skip.

Device identifier (FF MAC)

First 6 bytes	Middle 4 bytes	Last 22 bytes
Manufacturer code	Device type code	Serial number

WINDOWS

COM

Wednesday, January 18, 2017 5:08 PM

COM follows an OOP programming model, objects must be instantiated, can be pointed to and de-allocated

Uses Local procedure Calls (LPC)

- 1- COM objects can run across processes and computers
- 2- COM methods can be called across a network
- 3- COM object names must be unique.
- 4- Each COM object has two interfaces: IUnknown and actual method

Interface: collection of functions used to communicate client with server. Must be used to access object.

- Strongly typed, named I*. Never change after publish
- Must contain:
 - o QueryInterface(): gets an interface from a COM object.
 - o AddRef(): called to increase reference count
 - o Release(): called to decrease ref count
- Use VTABLE to point to methods.

IUnknown: Parent of all COM interfaces, contains create, reference and release functions

GUID: hex string used to identify COM objects and interfaces

COM Server

- DLL: in process, server loaded during runtime, part of same application. Best performance
- EXE: standalone, private process space, handled by SCM, local or remote (DCOM)
- Service based.

COM Client

- Any program that uses COM to call methods on a COM server
- Client Methods
- CoInitialize(o) -> always initializes the COM library
 - CoCreateInstance(): locates (registry) and connects to server. Creates object. Returns point to object
 - Uses SCM for object location and starting
 - Release(): releases and disconnects server

Proxy/Stub

The client program communicates with a proxy as if it was directly communicating with the server.

The proxy communicates with the stub on the server side. The communication appears to be interprocess, although it is over the network.

IDispatch: interface to use COM through non-pointer languages. Locates and calls COM methods without knowing definition. Use Invoke() method to call functions.

COM Registry

For a COM server to operate, it must be registered within Windows.

Servers can self-register, or use REG scripts

Kept under HKEY_CLASSES_ROOT:

AppID: each AppID is a COM server

CLSID: ID of a COM class

Interfaces: info about proxy/stub

ProgID: programID

Typelib: type library info

The only required key is the CLSID.

Callback interface

Most interfaces are driven by the client, sometimes the server must provide responses.

A callback interface can be used to create an async link to the client

Client starts Server

Server started, dll loaded, executable run by SCM

Client request COM objects and interfaces

Server creates COM Object

Server creates COM interface

Server increments reference count for active interfaces

Server returns interface to client

Client originates all method calls to server

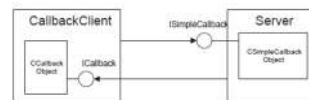
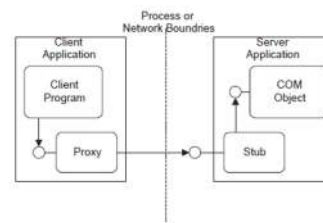
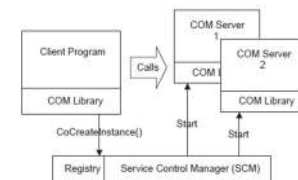
Server executes method on object

Client releases server interfaces, server can shut down

Server decrements interface reference count

If reference count = 0, COM object can be deleted

If there are no active connections, shut down server



DCOM

Thursday, January 19, 2017 12:54 PM

Client calls `CoCreateInstanceEx()` first, passing the description of the server and requests a CLSID and interface. This request is handled by the SCM, which creates the objects on the local and remote computers. When remote COM object is created, all calls are handled between the proxy and stub over RPC.

RPC runs over TCP/IP, UDP, NETBIOS, or others. The standard is UDP.

A client must be registered to enable remote access to COM objects.

Access permissions handle the security for DCOM, under the App ID "access permission"

Authentication and impersonation both apply here.

Disconnection

For client, if a COM call is made after disconnection, it will get an error. It may crash. The client won't see the disconnect until it makes a call.

The server will never know about the disconnect, and will wait for requests until prompted.

Delta ping is used by COM to solve this.

Access Denied errors

Server system rejects a connectino:

- Check security, file protection, network access. Launch permissions.
- Make sure server identity is not set to "Launching user"
- Server may be registered but exe is missing
- Try re-installing, re-registering server
- File and Print Sharing
- Check parameters of `CoInitializeSecurity`

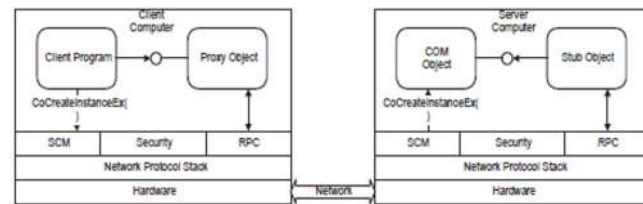
RPC Server Unavailable

- Test connection with Ping and TRACERT
- Invalid server name, name resolution issue, use IP address
- Server may be offline, or the RPCSS service is stopped
- Check server dcom settings

Troubleshooting

Make sure both client and server work locally. Install both on the server machine.

Make sure you can browse the remote computer. Ping and Tracert



Impersonation

Wednesday, January 18, 2017 2:01 PM

Used to run threads out of usual security context, to meet requirements of client/server applications.

Every process has a *primary token* that describes its security context. The token also contains the session ID, integrity level and UAC virtualization state. This token is created by the LSASS during logon. It is then determined if the user is part of a powerful group/privilege.

Child processes inherit their creator's token.

The server tells the SRM it will be adopting the security profile of a client. The SRM still runs security validations, but now in the context of the impersonated user.

Windows API uses *ImpersonateNamedPipeClient*

A thread uses *ImpersonateSelf*

DDE uses *DdeImpersonateClient*

RPC uses *RpcImpersonateClient*

SSPI uses *ImpersonateSecurityContext*

COM uses *CoImpersonateClient*

Once the server thread finishes the task, the primary security context takes over again.

The client can limit the level of impersonation with security quality of service (SQOS).

A server must obtain a users account name and password when remotely impersonating a client.

Impersonation levels:

- Anonymous: server can't impersonate or identify client
- Identify: server gets identity (SID) of client, but can't impersonate
- Impersonate: server identifies and impersonate
- Delegate: server impersonates on local and remote systems

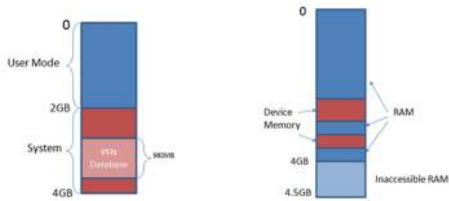
Memory

Thursday, May 3, 2018 12:59 PM

Physical Memory

Actual installed memory.

Memory manager keeps track of pages in the PFN to map into virtual memory space. There are ways to increase memory capacity, but they decrease performance.



Limits:
X86 -> 4GB
Actually lower effective limit
32 bit systems don't know how to handle

Available memory: memory not tied to a specific process, kernel or driver

Virtual memory

Separates programs view of memory as physical and allows more processes to execute concurrently. No direct connection to physical memory.

Process Address Space: Each process has its own virt mem. Limits are defined for user mode and kernel mode space.



Applications use Heap APIs, .NET garbage collector and C libraries (VirtualAlloc function) for virtmem allocation.

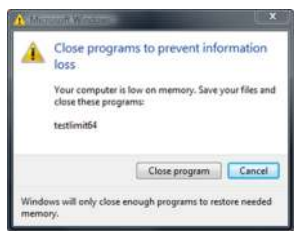
Certain applications perform better with larger address space, such as SQL Server and Active Directory. These applications contain a flag that indicates the ability to do so.

X64 bit windows uses a different model for virtual memory allocation:



Committed Memory: a process can commit virtual memory to maintain data in process stores. The *commit limit* is the sum of physical memory and page files. The sum of all committed memory is known as the *commit charge*.

When system virtual memory runs low, applications may fail and strange error messages may appear. A standard windows error prompt may appear as well:



Process Committed memory

Certain memory allocations do not count towards commit limit:

- Reserved virtual memory
- File mapping view
- Private
- Pagefile backed

Private virtual memory: cannot be shared among processes. Easy to attribute

Page-file backed memory: Can be shared

Paging File:

the only way to reasonably size the paging file is to know the maximum total commit charge for the programs you like to have running at the same time. If the commit limit is smaller than that number, your programs won't be able to allocate the virtual memory they want and will fail to run properly.

Set the paging file minimum to be that value minus the amount of RAM in your system (if the value is negative, pick a minimum size to permit the kind of crash dump you are configured for). If you want to have some breathing room for potentially large commit demands, set the maximum to double that number.

Some say no paging file is better, but having a paging file allows for writing to a modified list and making memory available to other processes

Device manager can be used to verify which devices occupy more memory space.

Performance Monitor: %Available Bytes%

Pools
Paged & nonpaged

A pool manager works in kernel mode, and uses virtual memory space for allocation. The pool manager allocates space to a requesting application and stores a pointer.

Non-paged: used by kernel and device drivers to store data that can be accessed when the system can't handle page faults. This includes interrupt service routines and deferred procedure calls (DPCs). Page faults are blocked during these for data structure protection. Assigned to physical memory. Common structures stored in non-paged pool: sync objects (mutex's, semaphores, events) file references, I/O request packets.

Paged: OS can write to paging file for repurposing. Whenever a driver references paged pooled memory, a *paged fault* occurs, and the memory manager reads data back into physical memory. The registry is the largest consumer of paged pooled memory.

When pool runs out, applications start failing, I/O operations drop and data can be lost. Drivers should be able to exit without crashing the entire OS, but third party drivers may not.



Pool leakage: when an application uses memory but does not release it when unused. The end result is a hanging computer...

As a *general* rule of thumb, any process with a handle count higher than 10k should be investigated for a possible leak

From <<https://blogs.technet.microsoft.com/askpfeplat/2014/03/09/another-troubleshooting-adventure-more-real-life-memory-pool-leaks/>>

DPC: higher priority tasks (higher IRQL) can steal processor time from lower priority tasks.

IRQL: Interrupt request level. Hardware-independent means of controlling priority of interrupts.

Registry

Friday, January 20, 2017 11:32 AM

Contains configuration and identification for applications, dlls, exe, etc

Registry startup

- Initial boot reads config data of boot drivers to load into memory.
- Boot Configuration Database is stored in registry hive
- Kernel initialization, specifies what device drivers to load
- Explorer and other components read user preferences from registry
- Applications read system settings.
- Registry provides async callbacks to notify changes. Some applications however, constantly monitor.

HKLM\BCD00000000 is the Boot config database, replaces the boot.ini file.

Value Type	Description
REG_NONE	No value type
REG_SZ	Fixed-length Unicode string
REG_EXPAND_SZ	Variable-length Unicode string that can have embedded environment variables
REG_BINARY	Arbitrary-length binary data
REG_DWORD	32-bit number
REG_DWORD_BIG_ENDIAN	32-bit number, with high byte first
REG_LINK	Unicode symbolic link
REG_MULTI_SZ	Array of Unicode NULL-terminated strings
REG_RESOURCE_LIST	Hardware resource description
REG_FULL_RESOURCE_DESCRIPTOR	Hardware resource description
REG_RESOURCE_REQUIREMENTS_LIST	Resource requirements
REG_QWORD	64-bit number

Root Key	Description
HKEY_CURRENT_USER	Stores data associated with the currently logged-on user
HKEY_USERS	Stores information about all the accounts on the machine
HKEY_CLASSES_ROOT	Stores file association and Component Object Model (COM) object registration information
HKEY_LOCAL_MACHINE	Stores system-related information
HKEY_PERFORMANCE_DATA	Stores performance information
HKEY_CURRENT_CONFIG	Stores some information about the current hardware profile

H stands for handle

Root Key	Abbreviation	Description	Link
HKEY_CURRENT_USER	HKCU	Points to the user profile of the currently logged-on user	Subkey under HKEY_USERS corresponding to currently logged-on user
HKEY_USERS	HKU	Contains subkeys for all loaded user profiles	Not a link
HKEY_CLASSES_ROOT	HKCR	Contains file association and COM registration information	Not a direct link; rather, a merged view of HKLM\SOFTWARE\Classes and HKEY_USERS\<SID>\SOFTWARE\Classes
HKEY_LOCAL_MACHINE	HKLM	Global settings for the machine.	Not a link
HKEY_CURRENT_CONFIG	HKCC	Current hardware profile	HKLM\SYSTEM\CurrentControlSet\Hardware Profiles\Current
HKEY_PERFORMANCE_DATA	HKPD	Performance counters	Not a link

RPC

Wednesday, September 07, 2016 6:00 PM

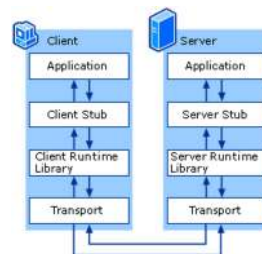
Computer calls for procedure at remote address space. Acts as if local however.
Request-response, initiated by client.
Stub called locally, parameters pushed onto stack
Client stub messaged to server
Server stub receives message and unpacks stub.
Made up of remote program number, version and remote procedure number

A DLL is reserved for each remote procedure. These DLLs marshall parameters and functions over the network.
Active Directory server name publishing is done over RPC

Async RPC: allows RPC to call a function before receiving response to the previous.

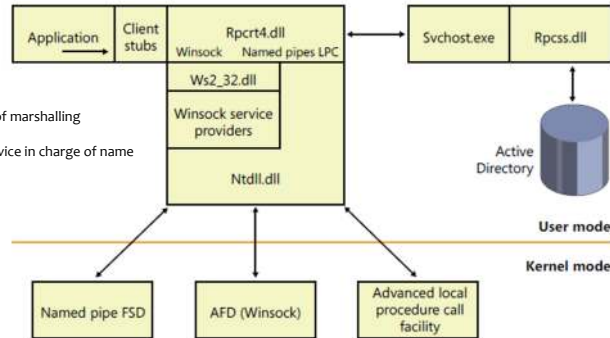
RPC uses SSPs to indicate authentication levels, and can use impersonation.

RPC run-time DLL (%SystemRoot%\System32\Rpcrt4.dll): in charge of marshallng functions over network and local
RPC subsystem (RPCSS—%SystemRoot%\System32\Rpcss.dll): a service in charge of name lookup, registration, dynamic endpoint mapping
kernel-RPC driver (%SystemRoot%\System32\Drivers\Mrpc.sys)



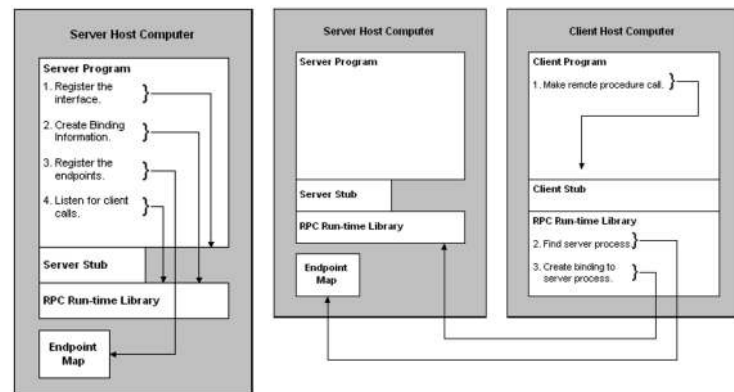
Portmap: maps RPC services to ports they are listening on. Client contacts portmap on service to determine proper ports.

Service Name	UDP	TCP
HTTP	80, 443, 593	80, 443, 593
Named Pipes	445	445
RPC Endpoint Mapper	135	135
RPC Server Programs	<Dynamically assigned>	<Dynamically assigned>



Portmap: server that converts RPC program number into DARP port numbers. When RPC starts, it tells portmap which port number it is listening to . When a client wants to make an RPC call, it contacts portmap for the port number. Use port 111

Rpcbind: utility that converts RPC program numbers into universal address. Checks with NAT.



Thursday, September 01, 2016 1:09 PM

Local security reference monitor (SRM) (\$SystemRoot\System32\lsisrv.exe): access checks and privileges.
Local Security Authority subsystem (LSASS) (\$SystemRoot\System32\lsass.exe): security policy, user auth, Message logging.
LSASS policy DB: Stored under HLM\SECURITY.
Local Security Authority (LSA) (\$SystemRoot\System32\lsass.exe): Manage user names and groups.
Interactive Local Computer (Winlogon): responsible for responding to SAS and creating first process at log on.
Logon user interface (LogonUI): presents user interface.
Credential providers (CPI): COM objects that run and obtain namepss, etc.
Network Logon (netlogon): Secure channel to domain controller.
Kernel Security Device Driver (KSecDD): advanced local procedure call to connect with LSASS.
AppLocker: locks applications to certain users. Uses SVCHOST process.

Object Security and protection:
Kernel objects are trusted and not checked.
User objects are checked and protected.
Drivers usually are not checked.

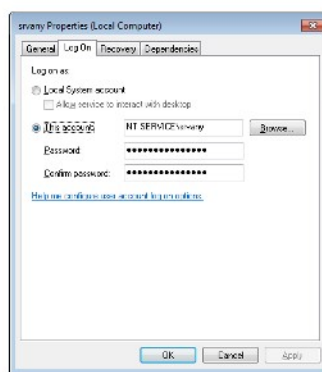
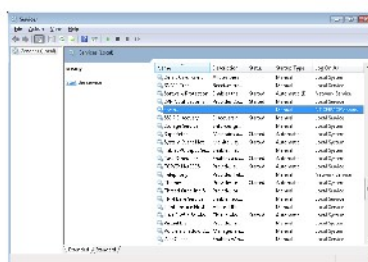
Tokens
Used to validate access (via ACLs) to accounts. SID's represent accounts.

When a thread opens an object, it must specify the actions it wants to perform beforehand. The object manager calls the SRM to check and delegate access. If an object does not use default security, it must manage its security settings. The I/O manager defines the file object type.

A process inherits the integrity level of its parent
Processes have an inherent integrity level.
If you disable UAC, objects will still be able to maintain integrity settings.

Identifies security context of a process/thread. Includes session ID, integrity level, UAC state. LSSASS creates a user token at logon. It checks multiple account levels and privileges. The group SID in token determines the group a user is in. The privilege array contains a list of rights associated to the token.

RESTRICTED TOKENS
Copy of a token, with modifications
FILTERED ADMIN TOKEN
Integrity medium. Deny only SID.

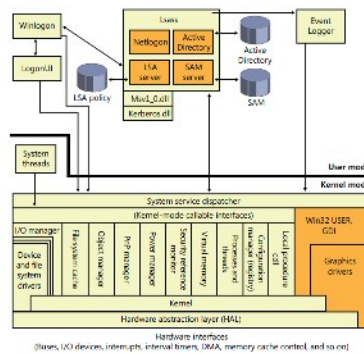


- Take over the job, writes down the time security descriptor
- Restore files
- Load/unload device drivers
- Create token
- Act as part of operating system

User Right	Role
Deny logon locally, Allow logon locally	Used for interactive logons that originate on the local machine
Deny logon over the network, Allow logon over the network	Used for logons that originate from a remote machine
Deny logon through Terminal Services, Allow logon through Terminal Services	Used for logons through a Terminal Server client
Deny logon as a service, Allow logon as a service	Used by the service control manager when starting a service in a particular user account
Deny logon as a batch job, Allow logon as a batch job	Used when performing a logon of type batch

Many programs store user-data in global locations. File system and registry entries are virtualized. Services are NEVER virtualized.

Slider Position	When administrative user not running with administrative rights....	Remarks
Highest position ("Always notify")	...attempts to change Windows settings, for example, use certain Control Panel applets	...attempts to install software, or run a program whose manifest calls for elevation, or uses Run As Administrator
Second position	UAC elevation prompt appears on the secure desktop	UAC elevation prompt appears on the secure desktop
Third position	UAC elevation occurs automatically with no prompt or notification	UAC elevation prompt appears on the secure desktop
Fourth position	UAC elevation occurs automatically with no prompt	UAC elevation prompt appears on the user's normal desktop
Fifth position	UAC elevation occurs automatically with no prompt	UAC elevation prompt appears on the user's normal desktop



S-1-5-21-1712426984-1618080182-1209977580-1108

Revision level	Domain identifier	Relative ID :Unique identifier within domain
Structure used by SID	Unique number for a domain	Any group or user created (not default) has an RID of 1000 or greater

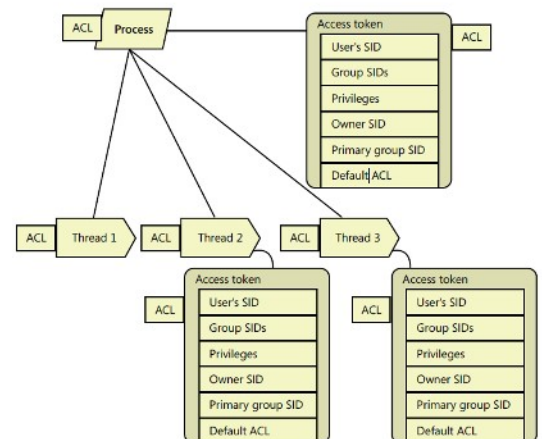


FIGURE 6-9 Process and thread security structures

TABLE 6-13 User Account Control Options

Slider Position	When administrative user not running with administrative rights...		Remarks
	...attempts to change Windows settings, for example, use certain Control Panel applets	...attempts to install software, or run a program whose manifest calls for elevation, or uses Run As Administrator	
Highest position ("Always notify")	UAC elevation prompt appears on the secure desktop	UAC elevation prompt appears on the secure desktop	This was the Windows Vista behavior
Second position	UAC elevation occurs automatically with no prompt or notification	UAC elevation prompt appears on the secure desktop	Windows 7 default setting
Third position	UAC elevation occurs automatically with no prompt or notification	UAC elevation prompt appears on the user's normal desktop	Not recommended
Lowest position ("Never notify")	UAC is turned off for administrative users	UAC is turned off for administrative users	Not recommended

Everything run in admin mode.

AppID:
Identify applications so admins can reference them in security policy decisions.
AppLocker: restrict exe, dll, com, scripts, etc from being run



FIGURE 6-9 Process and thread security structures

AD Recovery

Thursday, May 24, 2018 10:20 AM

Tombstone: When an object is deleted from AD, it is marked as deleted by setting *isDeleted* attribute to True. This strips most attributes, renames the object and sends it to the CN=Deleted Objects folder. It no longer shows up in MMC, LDAP. The deletion is then replicated to all DC's on the domain.

Lifetime of tombstone:

- Object deleted via LDAP or Security Accounts Manager (SAM) delete operation.
- Checks for:
 - The *isDeleted* attribute of the object is not already set to TRUE. (You can't delete an already deleted object)
 - The "private object" resource-specific security descriptor control flag is not set in the object's security descriptor. (This appears to be an undocumented "feature." The private object flag is bit 1 of the Sb21 byte of the security descriptor structure.)
 - The "disallow delete" bit (0x80000000) is not set in the systemFlags attribute of the object.
 - The *isCriticalSystemObject* attribute of the object is not set to TRUE.
 - The object's security descriptor grants the appropriate access rights to the user to delete the object. (More specifically, the user is allowed to delete the object itself and is allowed to delete a child from the parent object.)
 - The object is not a cross-reference object (objectClass = crossRef) for an existing naming context.
 - The object does not have any subordinate objects. (If the LDAP delete operation includes the LDAP "tree delete" control object id, Active Directory will automatically delete subordinate objects as well, unless they have the *isCriticalSystemObject* attribute set to TRUE. This prevents critical system objects from being deleted inadvertently as part of a tree deletion operation. You can of course delete these objects individually.)
 - The object is not a critical internal object (for instance, it isn't the nTDSDSA object for the DC or placeholder objects for ancestors of the NC heads).
- Active Directory must be in appropriate state (not renaming Domain)
- Object sent to Deleted Objects folder, where it resides for 60 days (default)
- DC performs garbage collection every 12 hours. Deletes any objects past tombstone lifetime

LDAP restore: <https://technet.microsoft.com/en-us/library/2007.09.tombstones.aspx>

Deleted objects are tombstoned after replication is full

Recovered objects will be versioned to a higher revision than tombstoned objects.

If deletion has not been replicated across all DCs:

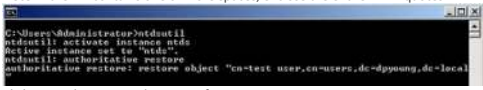
- recovery can be done by removing DC and rebooting in Directory Services Restore/Repair Mode (DSRM)
- Use *ntdsutil.exe* to mark undeleted object as authoritative
- Reboot DC and put back on network, this will cause the authoritative object to replace the "deleted" object

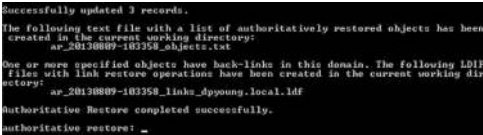
If deletion has been propagated:

- Locate DC with system-state backup prior to deletion
- Reboot into DSRM
- Restore system state without rebooting
- Mark objects as authoritative using *Ntdsutil.exe*
- Reboot DC

DSRM renders DC unavailable during operation.

Ntdsutil authoritative restore:

1. Determine the distinguished name (DN) of the object you wish to restore. If you are restoring a subtree of objects, such as an OU that contains user account objects, use the DN of the root of the subtree. For example, to restore a single user account named *jsmith* that exists in the *Sales* OU of a domain named *corp.mycompany.com*, use the DN *CN=jsmith,OU=Sales,DC=corp,DC=mycompany,DC=com*. To restore the entire *Sales* OU of the same domain, use the DN *OU=Sales,DC=corp,DC=mycompany,DC=com*.
2. At a command prompt, type **ntdsutil**.
3. Type **activate instance ntds**.
4. Type **authoritative restore**.
5. To mark a single object as authoritative, type **restore object <distinguished_name>**, where <distinguished_name> is the DN of the object as determined in step 1. To mark a subtree of objects as authoritative, type **restore subtree <distinguished_name>**, where <distinguished_name> is the DN of the subtree as determined in step 1. This will mark the container object and everything it contains as authoritative.
Note: If the DN contains one or more spaces, enclose the entire DN in quotes:

6. Click **Yes** in the message box to confirm.

7. Review the output. If it indicates that one or more objects have back-links in the domain, an .ldf file will be mentioned. Note the name of this file.

8. Type **quit** at each prompt until you have exited *Ntdsutil*.
9. Reboot the DC in normal mode or restart its AD DS service.
10. If an .ldf file was mentioned in step 7, copy this file to another DC in the domain and run it by typing **ldifde -i -k -f <filename>** at an elevated command prompt. This will ensure that the group memberships of all authoritatively restored objects remain intact.


From <http://www.dell.com/support/article/us/en/04/sln156828/windows-server-how-to-perform-an-authoritative-restore-of-active-directory-objects?lang=en>

ORACLE

Oracle DB basics

Friday, June 24, 2016 10:57 AM

Works on a client/server architecture, on a grid.

Datafile

Contain database data in forms of tables and indexes

Exclusive to one database

Are able to auto extend if database runs out of space

Tablespace: logical unit of datafiles

Once data from a datafile is read, it is stored in Oracle mem cache.

Control file

Every DB has a control file, which specifies the structure of the DB

Contains: Name, names and locations of datafiles/redo log files, timestamps

Multiple identical control files can be maintained

If the DB is altered, the control file is immediately modified by Oracle

Redo Log file

Made up of redo entries/records, which record all changes made to data on DB.

Multiple instances held on DB

Can be used to restore corrupt DB

ARCHIVELOG mode will archive redo logs

Parameter files

Hold config parameters for that instance and db

SPFILE: Server parameter file with initialization parameters

Alert and Trace log files

Server and background processes can write to trace files, which log internal errors

Can be used to tune apps and instances

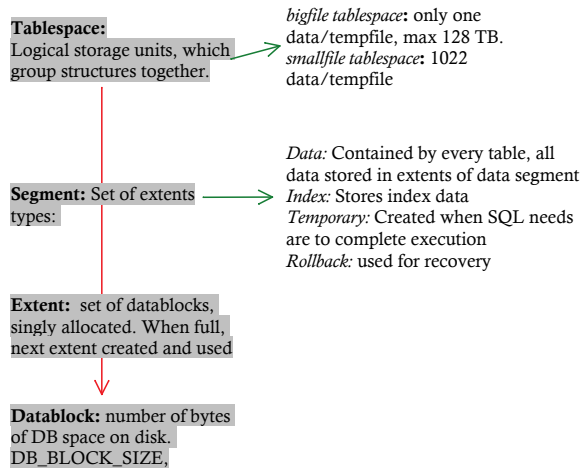
Alert file contains chrono log of messages/errors

Backup files

Backup files replace other files

Structure

Tuesday, June 28, 2016 7:34 AM



Schema: collection of database objects
Owned by DB user, with name of user

Object: structures that refer to data.
Made up of:

Tables: basic unit of storage. Rows/columns
Indexes: provide access path to table data
Views: alternate query, views of data. refer to data
Cluster: Groups of tables that share columns
Synonym: alias for any object, table, etc.

System Global Area (SGA): shared memory region that contains data info for one instance. Allocated when instance starts and deallocated when shut down. Should be as large as possible. Contains a *buffer cache* with all db buffers.

Program Global Area (PGA): info for server process

Oracle Data Dictionary: set of tables/views for read-only reference
Contains valid users, info about integrity constraints, amount of space allocated.

Oracle instance: DBS consists of Oracle DB and instance.
Combination of background processes and memory buffers. Memory structures are created to complete jobs:

Oracle 11g

Tuesday, May 23, 2017 9:27 AM

An instance must be created before a database.
An instance can only access one database

System Global Area (SGA): Shared memory structures

- Shared pool
- Database buffer cache
- Redo log buffer
- Large pool
- Java pool

Shared pool

Data used frequently, never written to disk.
Library cache: ready-to-go SQL statements. Syntax checking, etc.
Dictionary cache: collection of info about user and database objects
SQL result cache: Checks to see if data is already in memory
PL/SQL function result cache: stores result of a computation

Database buffer cache

Largest portion of SGA. Contains tables, indexes, system data
Stores database blocks (minimum amount of storage Oracle reads/writes)

Buffer cache state: state of blocks in cache:

Free: highest LRU, free to be used

Dirty: modified blocks, must be written to disk from buffer cache. Written when

Pinned: currently being accessed.

Redo Log Buffer:

Records every SQL statement that changes data. This is known as a redo entry. These are kept temporarily, and flushed out when:

- ✓ Every time there's a commit to data in the database
- ✓ Every three seconds
- ✓ When the redo buffer is 1/3 full
- ✓ Just before each dirty block is written to disk

Large pool

Not used by every database. Relieves shared pool of certain requirements.

- ✓ Oracle Recovery Manager
- ✓ Oracle Shared Server
- ✓ Parallel processing
- ✓ I/O-related server processes

Java pool

Rare, lets Java execute code out of SGA

Least recently Used (LRU): algorithm that determines which block has been used least recently.

Block: smallest unit to read/write by database. Every operation includes at least 1

Redo entry: statement and information pertaining to a change in data

- ✓ The database is issued a shutdown command.
- ✓ A full or partial *checkpoint* occurs — that's when the system periodically dumps all the dirty buffers to disk.
- ✓ A recovery time threshold, set by you, is met; the total number of dirty blocks causes an unacceptable recovery time.
- ✓ A free block is needed and none are found after a given amount of searching.
- ✓ Certain *data definition language* (DDL) commands. (DDL commands are SQL statements that define objects in a database. You find out more about DDL in Chapter 6.)
- ✓ Every three seconds.
- ✓ Other reasons. The algorithm is complex and we can't be certain with all the changes that occur with each software release.

FOR QUESTIONS, COMMENTS:
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